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27 **Country Differences in Nocturnal Sleep Variability: Observations**  
28 **from a Large-scale, Long-term Sleep Wearable Study**  
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## Abstract

### Study Objectives

Country or regional differences in sleep duration across countries are well-known, but few large-scale studies have evaluated sleep variability, either across the work week, or in terms of differences in weekday and weekend sleep. We capitalized on the increased use of wearable technology to characterise nocturnal sleep, weekday sleep variability and changes in sleep patterns at the weekend in a large, globally diverse sample.

### Methods

Using data from over 50 million night's sleep, sleep measures for ~220,000 wearable device users from 35 countries were calculated. Each person contributed an average of ~242 nights of data. Multiple regression was used to assess the impact country of residence had on a number of sleep measures, including sleep duration, weekday sleep variability, weekend sleep extension and social jetlag.

### Results

Nocturnal sleep duration was shorter in Asia than other regions. Furthermore, weekend sleep extension was longer in Europe and the USA than in Asia, and was only partially related to weekday sleep duration. There were also cross-country differences in social jetlag although the regional differences were less distinct than for weekend sleep extension. Variability in weekday sleep duration was greater, and sleep efficiency was lower, in Asia than other regions despite shorter nocturnal sleep. These regional differences may reflect under-appreciated effects of cultural differences.

### Conclusions

In addition to well-known regional differences in sleep duration, we provide clear evidence for cross-country differences in efficiency and variability; in particular, how weekends mitigate the effects of shorter weekday sleep (or not). Sleep norms around the world are subject to cultural influences and efforts to improve sleep habits should take these into consideration.

**Keywords:** Sleep Variability, Wearable, Culture, Social Jetlag, Weekend Sleep Extension, Weekday Sleep

## Introduction

Good sleep, encompassing dimensions of duration, timing, efficiency and regularity, is important for both our cognitive function<sup>1</sup> and our broader physical and psychological health<sup>2-5</sup>; yet many people from all parts of the world report that they do not get enough sleep<sup>6-10</sup>. The most well-studied feature of sleep is duration, and converging evidence from diverse epidemiology<sup>11</sup>, disease risk<sup>12</sup> and cognitive performance<sup>13,14</sup> studies indicate that, on average, the most positive outcomes are associated with ~7h of (self-reported) sleep (in working-age adults). However, beyond duration, conceptualization of what constitutes normative sleep, particularly in terms of how sleep changes in response to challenges to it, lacks detail. Furthermore, most sleep research, which determines how sleep is understood, has been conducted in the USA and Western Europe<sup>15</sup>. Differences in how people sleep around the world and how they cope with lack of sleep is an important area that merits attention.

Several cross-country differences in sleep are well-known; for example, nocturnal sleep in East Asia is shorter than that in Europe and the USA<sup>13,16-21</sup>, and there are significant differences in daytime napping across cultures<sup>22</sup> (although it is unknown how long these differences have existed for). However, a comprehensive understanding of how culture affects daily sleep habits, broader patterns of variability over longer periods (greater than two weeks) and differences between weekday and weekend sleep has yet to be reached. Evidence suggests that it is culture, rather than race or ethnicity that may underlie country differences in sleep. Cheung et al.<sup>23</sup> compared sleep across European-Canadian, Asian-Canadian and Japanese students, and showed that both groups of Canadian students had longer nocturnal sleep than Japanese students (by approximately 45-60 minutes). They also found that self-reported ideal sleep duration was shorter for Japanese students than both groups of Canadian students, suggesting the association was between sleep duration and cultural sleep beliefs, rather than ethnicity. In addition, most North American and Northern European adults consider monophasic sleep to be normal. In contrast, Al-Abri et al.<sup>24</sup> found that over 60% of Omani adults showed biphasic or polyphasic sleep. It is cultural practices, such as the observance of dawn prayers, after which some people may return to sleep, and day-time napping, which is more common in the Middle East<sup>25</sup>, that likely explain these differences in everyday sleep patterns.

Traditional cross-cultural (or cross-country) sleep research uses self-report questionnaire and survey data; however, this research often does not adequately differentiate between weekday and weekend sleep, and little long-term longitudinal data has been collected to investigate sleep variability. With the increasing adoption of reliable wearable devices that can adequately estimate sleep<sup>11</sup>, recent studies have been able to analyse objective sleep data on a larger scale and for longer durations than was previously possible<sup>26-28</sup>. Results from these studies are consistent with those obtained from surveys about regional sleep differences and have extended our understanding of sex differences in sleep and how sleep changes with age, both of which appear broadly similar across countries.

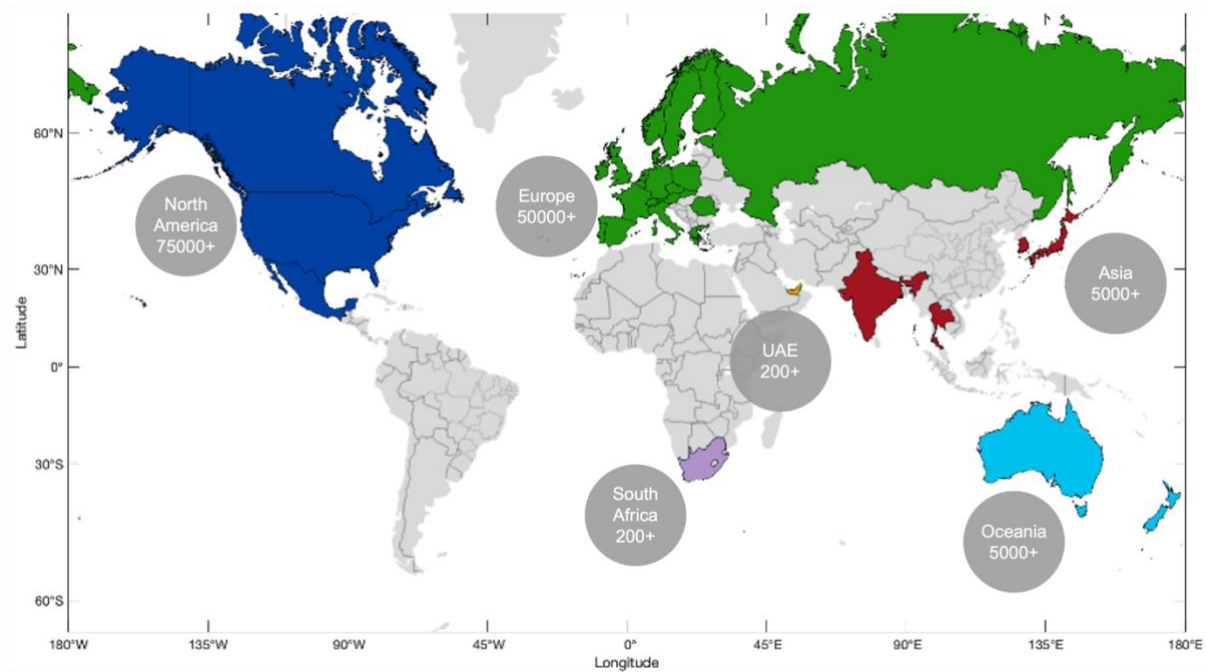
Sleep regularity is considered an important feature of healthy sleep<sup>3</sup>. Two independent reviews<sup>29,30</sup> and subsequent research<sup>31-33</sup> has shown that variable sleep can adversely affect both physical and mental health. However, despite the ability to collect long-term sleep

duration and timing information, sleep variability over the week, particularly across the work week, remains underexplored in cross-cultural studies. In order to assess sleep variability adequately, data needs to be collected over a sufficient duration. Research suggests that 3-5 nights of sleep are required to estimate a single week's variability accurately<sup>34</sup>; however, weekly sleep can vary significantly, with research suggesting that at least two months' data may be required to obtain stable estimates of weekly variability over an extended period<sup>35</sup>. In their review, Chaput et al.<sup>29</sup> found that only ~37% of the studies reviewed used objective measurements and most of these sampled sleep for only one week.

Most existing cross-country studies have focused on describing sleep variation as differences between weekday and weekend sleep, often in terms of 'social jetlag' (SJL). SJL was originally conceived as a measure of misalignment between social and biological time<sup>36</sup>, analogous to jetlag caused by travelling across time zones. However, because the original measure of SJL (the difference between mid-sleep time on work days and free days) is sensitive to both changes in sleep timing and asymmetric changes in sleep duration, it is now widely used as shorthand for describing the practice of both going to sleep later, and sleeping for longer, at the weekend (or on free days) than during the week (or on work days), thereby conflating changes in sleep timing with changes in sleep duration. This has sometimes resulted in SJL being misconstrued as a measure of catch-up sleep and as an indicator of sleep debt.

Both SJL and weekend catch-up sleep vary around the world<sup>26-28</sup>. Komada et al.<sup>37</sup> found that SJL in Japan was lower than in Europe, but higher than in the UK and Australia. Furthermore, Zhang et al.<sup>38</sup> found much lower levels of SJL in China, and that over 30% of the sample in fact reported negative SJL (that is, having an earlier mid-sleep time at the weekend than during the week). Sleep duration at the weekend is often, but not necessarily, longer than that during the week. This is often interpreted as catch-up sleep for sleep debt accumulated over the week, which would predict that weekend sleep extension should be greater in both individuals and countries where weekday sleep is shorter. Another feature distinguishing SJL and weekend sleep extension is that SJL is usually associated with negative health outcomes<sup>29,39</sup>, whereas catch-up sleep has been associated with neutral or more favourable health outcomes<sup>40,41</sup>. Comparing the effects of SJL and sleep debt, Okajima et al.<sup>42</sup> found the effect of sleep debt (reflecting sleep restriction) to be worse than SJL (reflecting degree of circadian misalignment). Currently, most of the literature on SJL and over 60% of that concerning sleep variability as a whole, is based on self-reported sleep, rather than objective measures<sup>29</sup>.

In this study, we used individual data collected from a validated consumer sleep wearable<sup>43-45</sup> (Oura Ring, Oura Health Oy, Oulu, Finland) from over 50 million nights of sleep in over 220,000 people from 35 countries (Fig. 1) to investigate sleep timing, duration and variability in different parts of the world. We focused on country differences in weekday variability in sleep duration and their relation to weekday/weekend sleep differences because prior work has already extensively characterized the effects of age and sex on sleep<sup>26-28</sup>. Our findings allow us to discuss potential cultural origins of cross-country differences and their importance in developing a clearer understanding of the forces that shape sleep around the world.



**Figure 1.** Global distribution of countries providing data for analysis. Numbers represent the minimum average number of users per week from each region (all countries have an average of more than 200 users per week across the analysis period). Colours indicating region are consistent with those used in subsequent charts. (South Africa is the only country contributing to Africa and the United Arab Emirates is the only country contributing to the Middle East region).

## Methods

### Oura Users, User Data and Inclusion Criteria

Data was collected by Oura Health Oy (Oulu, Finland) in accordance with their Terms and Conditions (<https://ouraring.com/terms-and-conditions>) and Privacy Policy (<https://ouraring.com/en/privacy-policy-oura-health>). Users electronically consented to the use of anonymized data for research purposes. The study was exempt from formal review by the National University of Singapore Institutional Review Board following their initial consideration of the protocol because the data contained no personal identifiers.

226,187 unique Oura Ring users contributed a total of 54,769,523 nights' data from the 52-week period from Monday 4<sup>th</sup> January 2021 to Sunday 2<sup>nd</sup> January 2022. On average, each user contributed data from 242 nights (minimum 90 nights; periods of transmeridian travel were excluded). The large number of nights ensured that we had a truly representative sample for each user's sleep, seasonal effects were averaged out, and that the effect of outlier nights was minimized (e.g., daylight savings changes, holidays, brief periods of illness). The mean user age was 44.40 (SD=12.48; range: 18-80); therefore, the majority of the users were working age adults between approximately 30 and 55 years old. The overall proportion of male users was 59.86%, ranging from 50.21% to 80.56%. See Supplementary Table S1 for full user demographic details and Supplementary Fig. S1 for age distributions.

Nightly sleep measures for each user were generated by the Oura App. For each night, sleep measures were derived from the user's longest sleep episode (with a minimum of 3 hours' time in bed). Daily measures were averaged to create estimates for both weekdays and weekends for each week of the analysis period. To be included in the weekday analysis, weeks had to have data from at least 4 out of the 5 work days. For the weekend analysis, data had to be present from both weekend nights. These weekly sleep measures were then averaged across weeks to create weekday and weekend estimates for the entire analysis period for each user. People at higher latitudes sleep longer in the winter than in the summer because of changes in photoperiod length<sup>46-48</sup>; therefore, we analysed data across a whole year to minimize the impact of seasonal variation. The average number of weekly users for each country was at least 200. This ensured reliable country-level estimates of sleep measures and minimized the impact of outlier users.

## Data Collection and Measurement Device

Data was collected using the 2<sup>nd</sup> Generation Oura Ring equipped with infrared photoplethysmography, and temperature and accelerometry sensors. Sleep was inferred using motion, heart rate variability and temperature, using proprietary algorithms<sup>49</sup>. Previous research<sup>43-45</sup> has evaluated the 2<sup>nd</sup> Generation Oura Ring against other research-grade tools in classifying sleep and wake states. Despite their limitations, data from wearable devices is particularly useful when collecting data from large samples or over an extended period, where polysomnography (PSG) is impractical<sup>11</sup>.

## Sleep and Sleep Variability Measures

*Basic Sleep Measures.* Sleep was characterized in terms of sleep onset time (SOT), wake time (WT), total sleep time (TST), sleep onset latency (SOL), wake after sleep onset (WASO), time-in-bed (TIB) and sleep efficiency (SE; calculated as  $(TST/TIB) \times 100$ , that is, the amount of time in bed that is actually spent asleep). The difference between TST and TIB is the sum of SOL and WASO. SOT is an objective marker of sleep onset that removes the uncertainty in determining what constitutes 'bedtime' which questionnaires assess<sup>50</sup>. Wearable devices infer TIB from changes in movement and heart rate when transitioning from being active to lying down in bed preparing for sleep. Occasionally, this can result in inaccuracies if the user is lying still but awake in bed; however, these effects are minimized in this study because estimates are averaged across a large number of days.

*Weekend/Weekday Differences in Sleep.* To assess how sleep changed from weekdays to weekends, the difference between weekday and weekend sleep measures was calculated. These differences were computed by subtracting individual users' weekday mean from the corresponding weekend mean. Friday and Saturday nights were considered weekend nights for all countries except the United Arab Emirates (where Thursday and Friday nights were considered weekend nights).

*Weekday Variability.* Variability in weekday sleep measures was calculated as the standard deviation over the four or five daily measures for that week; this ensured acceptable measures of weekly variability<sup>34</sup>. These weekly measures of variability were then averaged to produce an estimate of overall weekday variability across the whole year. While other

measures of variability can be computed, the standard deviation is sensitive to inter-day fluctuations in sleep metrics<sup>51</sup>.

## Data Modelling and Statistical Analysis

We analysed the data using MATLAB<sup>52</sup> and R version 4.2.2<sup>53</sup>. Linear modelling was used to test the associations between sleep measures and predictors variables. Where sleep variables had both weekday and weekend measures, we included sex, age (mean-centred), age<sup>2</sup>, country and day type (weekday or weekend) and their two-way interactions as predictors in the model. Where there was only one measure per week (e.g., the weekday/weekend difference in sleep duration), day type was omitted. Additional models included sleep variables as predictors. Two-way interactions were included to improve model fit but results are not reported. Basic models (including the day type predictor, and excluding any additional sleep variables as predictors) took the form of:

$$y = \beta_0 + \beta_1 \text{sex} + \beta_2 \text{age} + \beta_3 \text{age}^2 + \beta_4 \text{daytype} + \beta_5 \text{country} + \beta_6 \text{sex} * \text{age} + \beta_7 \text{sex} * \text{age}^2 + \beta_8 \text{sex} * \text{daytype} + \beta_9 \text{sex} * \text{country} + \beta_{10} \text{age} * \text{age}^2 + \beta_{11} \text{age} * \text{daytype} + \beta_{12} \text{age} * \text{country} + \beta_{13} \text{age}^2 * \text{daytype} + \beta_{14} \text{age}^2 * \text{country} + \beta_{15} \text{daytype} * \text{country} + \varepsilon$$

Full model specification and output for all models can be found in the Section 3 of the Supplementary Material. Where correlations are reported, Pearson's correlation coefficient is used for individual user-level data and Spearman's coefficient for country-level data because it is more robust to lower sample size and outlying data.

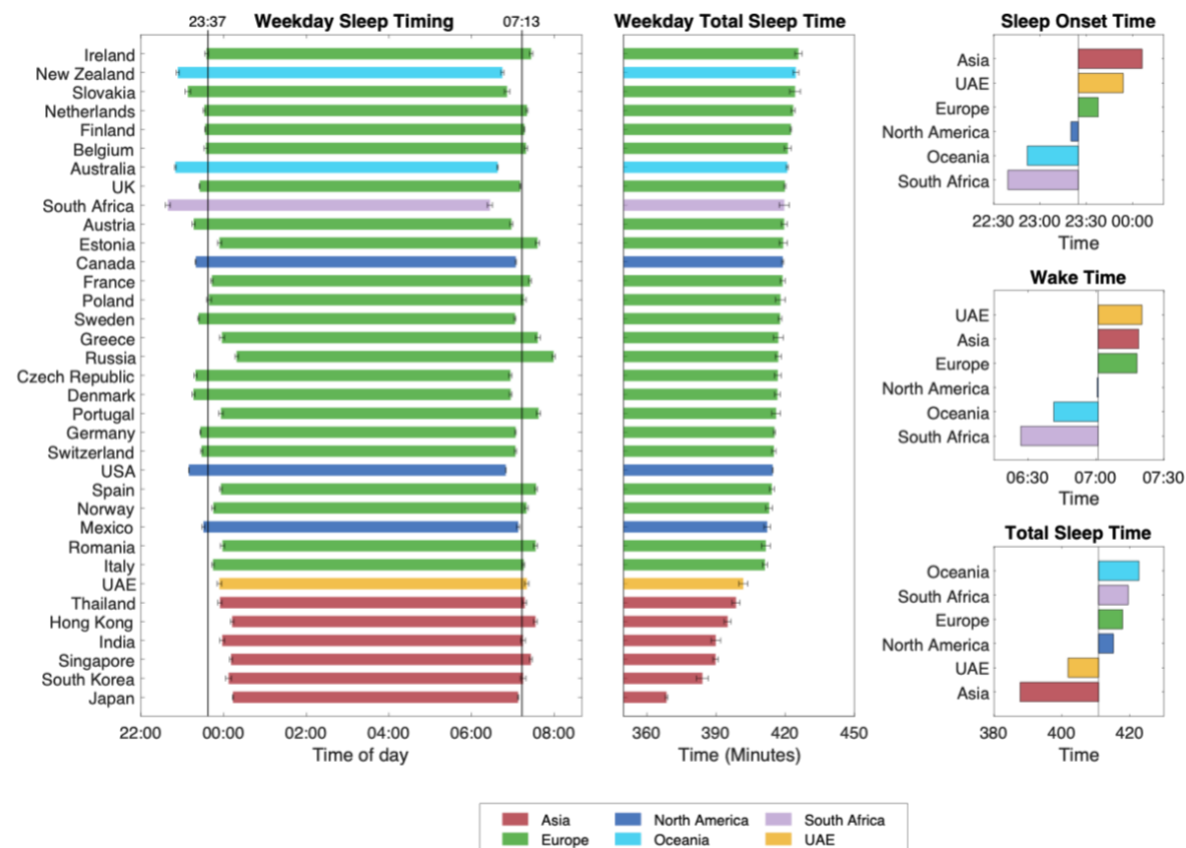
## Results

Reported sleep measures were estimated from linear models and statistical significance was tested using ANOVA. We also report effect sizes ( $\eta_p^2$ ) which, unlike p-values, are not influenced by sample size, as an indication of the strength with which predictors influence the response variable. Global differences in weekly sleep are characterized first to provide context for subsequent sleep variability analysis.

### Weekday Sleep

Consistent with previous research, we found that people in Asia slept less than those in other parts of the world by up to about 45 minutes (Fig. 2, centre panel). People in Asia also tended to fall asleep later than those in Europe, who in turn fell asleep later than those in North America and Oceania. A similar pattern is also seen for waking up; that is, people in Asia and Europe woke up later than those in North America and Oceania (although the difference between people in Asia and Europe was smaller; Fig. 2, left panel). Model estimated weekday TST had a very high correlation with uncorrected weekday TST means ( $\rho=0.95$ ,  $p<0.001$ ).





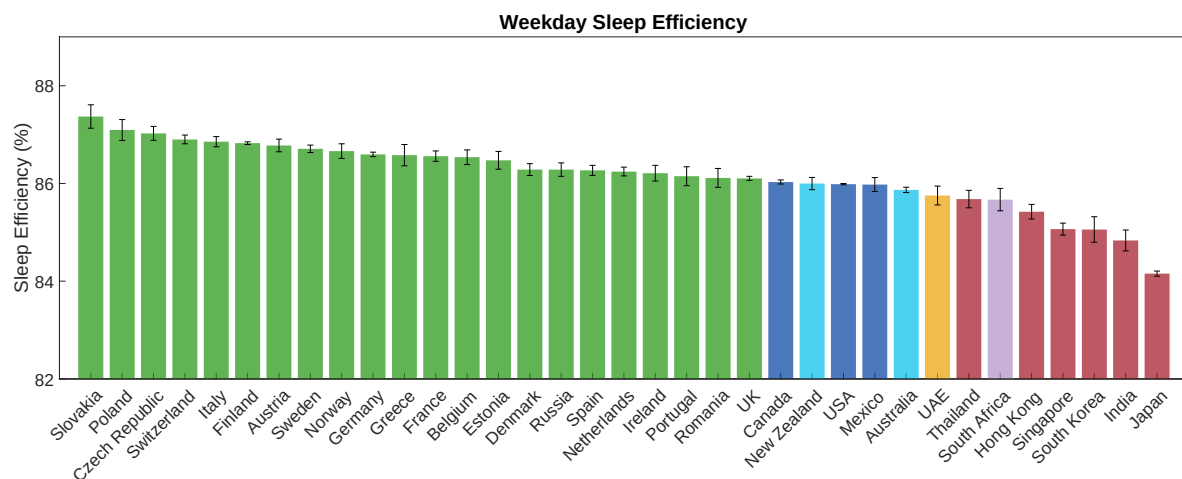
**Figure 2.** Sleep duration and timing across countries and global regions. Error bars represent the SEM. **Left Panel.** Weekday sleep onset time (SOT) and wake time (WT). Countries are ordered by total sleep time (TST). Vertical lines indicate the average SOT and WT across countries. **Centre Panel.** Weekday TST across countries. **Right Panels.** Regional means for SOT, WT and TST. Vertical lines indicate the unweighted mean across regions.

Results show significant effects for all variables apart from age<sup>2</sup>: sex ( $F(1, 452193)=39723$ ,  $p<.001$ ,  $\eta_p^2=0.081$ ), age ( $F(1, 452193)=464.1$ ,  $p<.001$ ,  $\eta_p^2=0.001$ ), age<sup>2</sup> ( $F(1, 452193)=0.030$ ,  $p=0.86$ ,  $\eta_p^2<0.0001$ ), day type ( $F(1, 452193)=19319$ ,  $p<.001$ ,  $\eta_p^2=0.041$ ) and country ( $F(34, 452193)=777.4$ ,  $p<.001$ ,  $\eta_p^2=0.055$ ). The effect size estimates show that country had a similar effect on weekday TST as sex and day type. The estimated effect of age is smaller than for previous studies<sup>13,26</sup> because most of the participants are in the working adult age range (30-55 years) during which sleep duration is relatively stable<sup>13</sup>. In addition to the differences in weekday TST across countries shown in Fig. 2, the results also show that women slept longer than men by approximately 25 minutes (women: 425 min. (95% CI: 424.5, 425.5 min.); men: 399.5 min. (95% CI: 399, 400 min.)).

The data also show that SE was influenced by country (Fig. 3). There were significant effects of sex ( $F(1, 452193)=29039$ ,  $p<.001$ ,  $\eta_p^2=0.060$ ), age ( $F(1, 452193)=10.04$ ,  $p=0.0016$ ,  $\eta_p^2<0.0001$ ), age<sup>2</sup> ( $F(1, 452193)=6994$ ,  $p<.001$ ,  $\eta_p^2=0.015$ ), day type ( $F(1, 452231)=512.6$ ,  $p<.001$ ,  $\eta_p^2=0.001$ ) and country ( $F(34, 452193)=212.59$ ,  $p<.001$ ,  $\eta_p^2=0.016$ ) on SE. However, the effect sizes suggest that sex and country had the largest impact on SE. The cross-country range in SE was 3.21%, from 87.37% (in Slovakia) to 84.16% (in Japan). In comparison, weekday SE in women (87.41%) was 2.47% higher than that in men (84.94%).

Experimental studies of sleep restriction show that lower sleep duration is usually associated with higher SE<sup>54,55</sup>. However, that is not the case here; shorter sleep is associated with lower SE. At the country level, those countries with the shortest sleep were also the ones with the lowest SE. This is reflected by a significant correlation between weekday TST and weekday SE ( $\rho=0.426$ ,  $p=0.011$ ). The same was also true at the individual user level; users who slept less also had lower SE ( $r=0.518$ ,  $p<0.001$ ). Including weekday TST in the model predicting weekday SE showed it had a very large effect size ( $\eta_p^2=0.248$ ), which was accompanied by a large decrease in the effect size of country ( $\eta_p^2$  reduced from 0.016 to 0.003). This indicates that the country level difference in SE reflects the aggregate effects of the sleep duration of its residents.

The range of weekday SOL across countries was 10.9-12.7 minutes, whereas the range of WASO was 39.1-48.0 minutes, suggesting that WASO contributed more than SOL to differences in SE. People in European countries showed the least WASO and had relatively longer sleep periods, resulting in higher SE, whereas those in Asian countries had average levels of WASO with shorter sleep periods, resulting in lower SE.

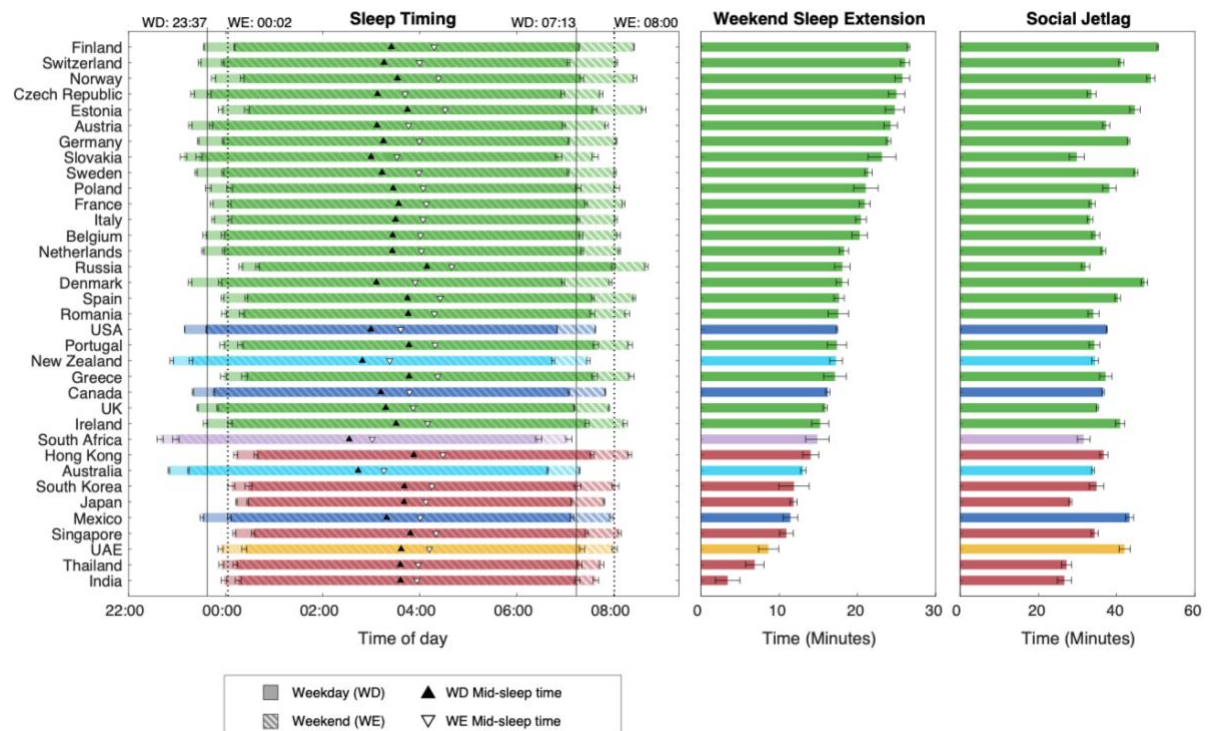


**Figure 3.** Weekday sleep efficiency (SE) across countries. Error bars represent the SEM.

### Weekend/Weekday Differences in Sleep

In general, people slept 5-25 minutes longer at the weekend than during the week, depending on their location; they fell asleep about 15-35 minutes later at the weekend and woke up about 30-60 minutes later (Fig. 4). In a model with weekday TST, sex, age, age<sup>2</sup> and country (and their two-way interactions) predicting weekend/weekday difference in TST, all predictors showed significant main effects: weekday TST ( $F(1, 226006)=14615.9$ ,  $p<.001$ ,  $\eta_p^2=0.060$ ), age ( $F(1, 226006)=909.7$ ,  $p<.001$ ,  $\eta_p^2=0.004$ ), age<sup>2</sup> ( $F(1, 226006)=3000.6$ ,  $p<.001$ ,  $\eta_p^2=0.013$ ), sex ( $F(1, 226006)=1410.4$ ,  $p<.001$ ,  $\eta_p^2=0.006$ ) and country ( $F(34, 226006)=143.56$ ,  $p<.001$ ,  $\eta_p^2=0.021$ ). The effect sizes suggest that weekday TST had the strongest effect on weekend TST difference and the negative regression coefficient ( $B=-0.158$  (95% CI: -0.174, -0.142),  $p<0.001$ ) indicates that shorter weekday TST is associated with longer weekend sleep extension. This may suggest that sleeping more on weekends may indeed reflect catch-up sleep. However, there was also an effect of country, indicating

that even after accounting for the effect of weekday TST, there are significant country effects on weekend sleep behaviour.

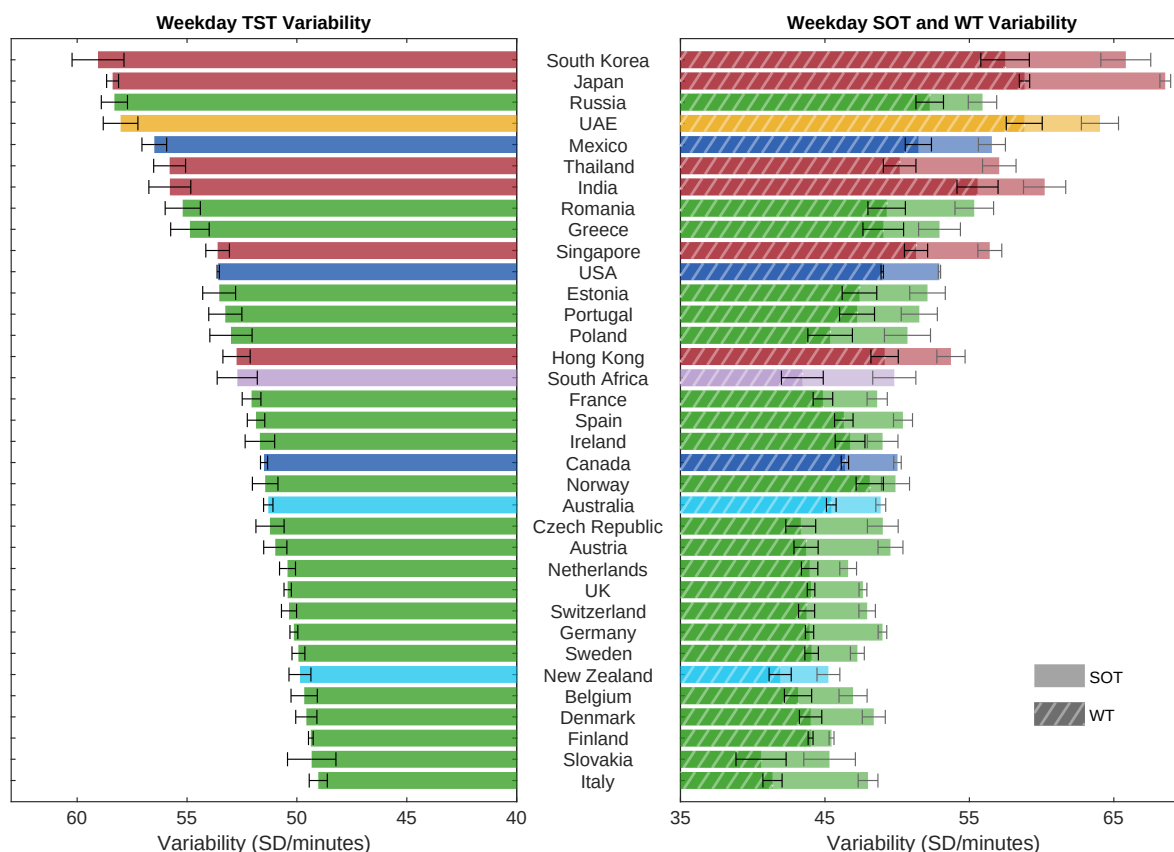


**Figure 4.** Weekday and weekend sleep timing, weekend sleep extension and SJL (ordered by weekend sleep extension). Error bars represent the SEM. **Left Panel.** Weekday (solid bars) and weekend (hatched bars) SOT and WT. Vertical lines represent mean weekday (solid line) and weekend (dotted line) SOT and WT. Filled triangles represent weekday mid-sleep time (MST), open triangles represent weekend MST. **Centre Panel.** Weekend sleep extension (difference between weekend TST and weekday TST). **Right Panel.** SJL (difference between weekend MST and weekday MST).

SJL is the difference between mid-sleep time (MST) on work days and free days. There was a positive correlation between SJL and weekend sleep extension ( $r=0.34$ ,  $p<0.001$ ) indicating that those who slept longer at the weekend also pushed their sleep back the most. However, these measures are not independent; changes in sleep duration also affect the measure of SJL. Therefore, in order to account for this, the model predicting SJL also included the weekday/weekend TST difference. This model showed significant effects for weekday/weekend difference in TST ( $F(1, 226006)=30790$ ,  $p<.001$ ,  $\eta_p^2=0.120$ ), age ( $F(1, 226006)=11415$ ,  $p<.001$ ,  $\eta_p^2=0.048$ ), age<sup>2</sup> ( $F(1, 226006)=14.61$ ,  $p<.001$ ,  $\eta_p^2<0.0001$ ), sex ( $F(1, 226006)=1055.11$ ,  $p<.001$ ,  $\eta_p^2=0.005$ ) and country ( $F(34, 226006)=229.09$ ,  $p<.001$ ,  $\eta_p^2=0.033$ ). While the effect size for weekday/weekend TST difference is the largest, there are also moderate effect sizes for age and country. This suggests that, consistent with prior studies, after accounting for weekend sleep extension, not only does SJL reduce with age<sup>26</sup>, but also that it varies across countries<sup>26,27</sup> (Fig. 4, right panel). While there are some countries that show both high (or low) weekend sleep extension and high (or low) SJL, the relationship between them is not consistent across countries, suggesting they are distinct constructs.

## Weekday Sleep Variability

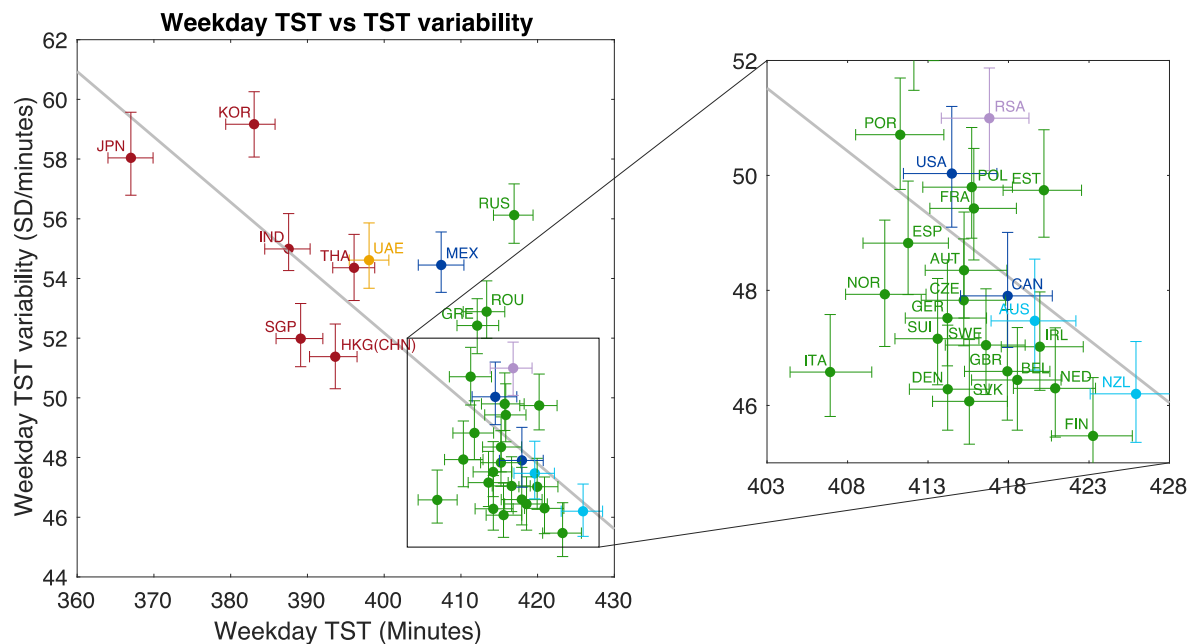
While there is often a difference in how people sleep at the weekend compared with during the week, there can also be significant variability in sleep timing and duration during the work week. Variability in SOT, WT and TST across weekdays for the different locations is shown in Fig. 5. In general, people in Asia displayed higher variability in both sleep timing and duration, whereas those in Europe and Oceania generally showed the least weekly variation, with North America in between. As might be expected, SOT variability was consistently higher than WT variability. When predicting weekday TST variability, there were significant effects for sex ( $F(1, 226044)=225.71, p<.001, \eta_p^2<0.001$ ), age ( $F(1, 226044)=1526.25, p<.001, \eta_p^2=0.007$ ), age<sup>2</sup> ( $F(1, 226044)=1724.79, p<.001, \eta_p^2=0.008$ ) and country ( $F(34, 226044)=179.32, p<.001, \eta_p^2=0.026$ ). The effect sizes show that the effect of country is much larger than the effects of age or sex.



**Figure 5.** Weekday variability in sleep onset time (SOT), wake time (WT) and total sleep time (TST). Countries are ordered by TST variability. Error bars represent the SEM. **Left Panel.** Weekday TST variability. **Right Panel.** SOT variability (solid bars) and WT variability (hatched bars).

Both at the individual and country level, weekday TST variability was significantly negatively correlated with weekday TST (individual:  $r=-0.22, p<0.001$ ; country:  $r=-0.71, p<0.001$ ), indicating that both within users, and across countries, higher variability in sleep duration was associated with shorter sleep. When including weekday TST in the model predicting weekday TST variability, weekday TST was the strongest predictor; however, country still exhibited a reasonable effect size ( $\eta_p^2=0.015$ ). Fig. 6 shows the relationship between

weekday TST and weekday TST variability at the country level. Median weekday TST and weekday TST variability are plotted with the scaled IRQ to illustrate distribution of both variables.

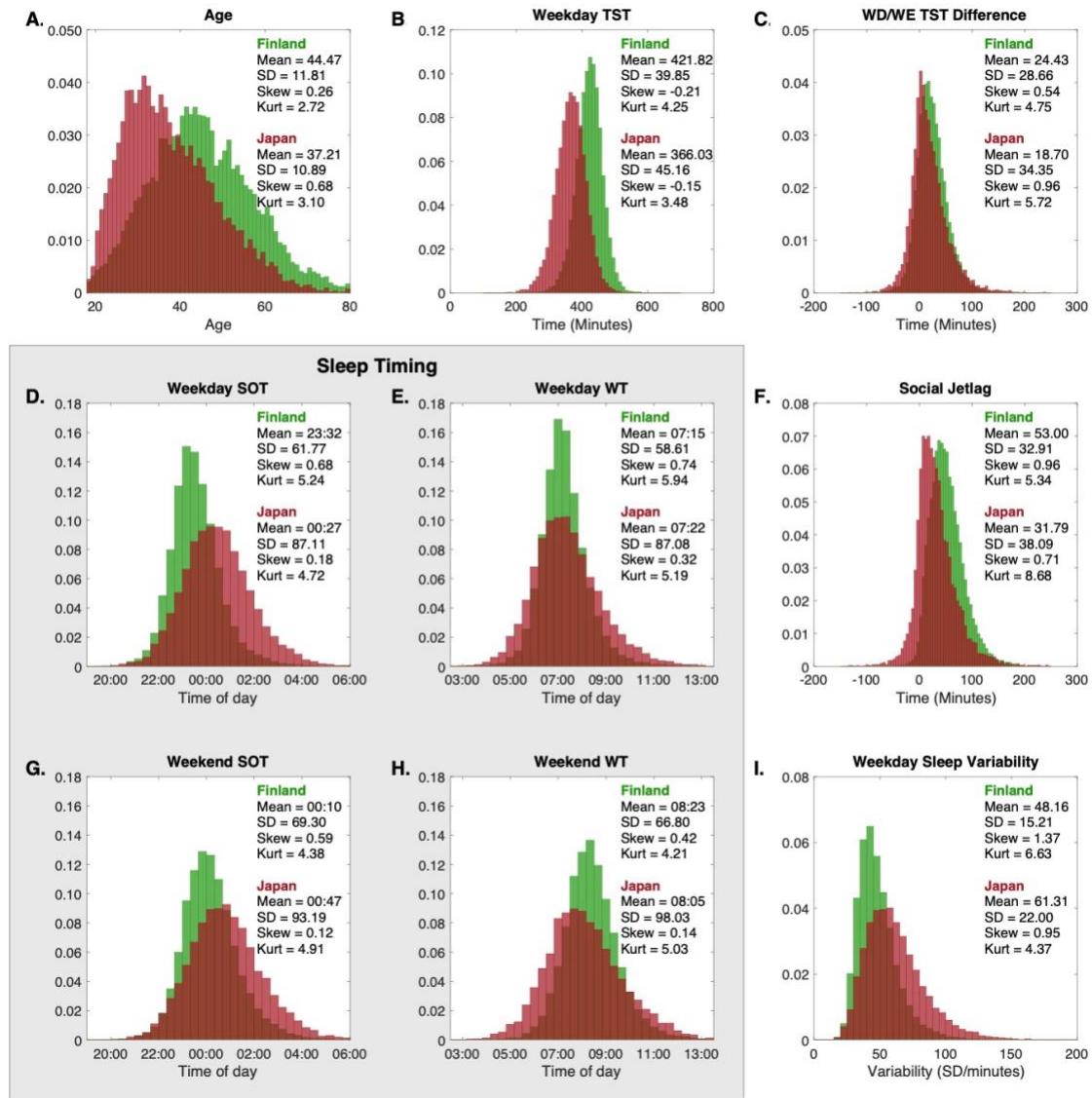


**Figure 6.** The relationship between median weekday TST and median weekday TST variability. Lower TST is associated with more variability in TST. Whiskers show scaled inter-quartile range (IRQ/10).

#### Illustrative Comparison: Finland vs Japan

Finland and Japan have both contrasting sleep patterns and a high number of users; therefore, it may be instructive to look more closely at the distributions of variables within these two countries (Fig. 7). Users in Japan are younger than those in Finland (Japan: 37.21 years; Finland: 44.47 years; Fig. 7, Panel A) and a higher proportion are male (Japan: 78.81%; Finland: 52.21%). The difference in weekday sleep duration is shown in Panel B. SOT and WT for both weekdays (Panels D and E) and weekends (Panels G and H) are displayed in the inset. Weekday SOT was later in Japan than Finland; however, weekday WT was the same, although there was greater variability in Japan. This presumably reflects the constraints imposed by the start of the workday. Weekend SOT was delayed more in Finland than in Japan and weekend WT in Finland was later than that in Japan resulting in greater weekend sleep extension and SJL in Finland (Fig. 7, Panels C and F). One notable feature of these distributions is that TST is not always longer at the weekend than during the week and that SJL is not always positive<sup>38,56</sup>. It is possible that these results reflect people for whom weekend days are not free days; nevertheless, it indicates that weekend sleep extension and (positive) SJL are not ubiquitous. Finally, Panel I shows not only that weekday TST variability was higher in Japan, but also that its variance was larger.





**Figure 7.** Distribution of age and the main sleep variables in Japan (red) and Finland (green). Mean, standard deviation, skew and kurtosis are shown for each distribution. **Panel A:** Age. **Panel B:** Weekly TST. **Panel C:** Weekday/Weekend TST difference (weekend sleep extension). **Panel D:** Weekday SOT. **Panel E:** Weekday WT. **Panel F:** Social jetlag. **Panel G:** Weekend SOT. **Panel H:** Weekend WT. **Panel I:** Weekday sleep variability.

## Discussion

The differences in nocturnal sleep across countries around the world found here mirror previous research showing shorter sleep duration and later sleep timing in Asia compared with Western Europe, North America and Oceania<sup>16,18,20</sup>. Additionally, shorter sleep was associated with lower sleep efficiency across countries. Weekend sleep extension, commonly thought of as catch-up sleep following sleep debt accumulated over the working week, was longest in Northern and Western European countries where sleep duration was also the longest. In contrast, parts of Asia and the Middle East, in particular Thailand, India and the UAE, showed the least weekend sleep extension. Social jetlag, which diminishes

with increasing age, was longer in Scandinavia and Northern Europe and generally shorter in Asia, but regional differences were not as pronounced as with weekend sleep extension. Finally, despite having shorter and later sleep, people in Asia also exhibited greater weekday sleep variability.

The homeostatic regulation of sleep suggests that for each person, there is an intrinsic sleep duration requirement that they endeavour to maintain across different durations of sleep opportunity. This explains why experimental sleep restriction is associated with higher sleep efficiency and sleep extension is associated with lower efficiency<sup>54,55,57</sup>. However, our findings show the opposite; despite sleeping less, people with shorter sleep also have less efficient sleep. This is consistent with Cheung et al.<sup>23</sup>, who found both shorter sleep duration and lower SE in Japanese students compared with Canadian ones. Similarly, in a study investigating sleep disturbances in middle aged and older adults of different ethnicities in the USA, Chen et al.<sup>58</sup> found that people of Chinese ethnicity, particularly the younger men in the cohort (54-64 years of age), showed a higher likelihood for both shorter sleep duration and lower sleep efficiency. A potential explanation for these results, and the apparent discrepancy with experimental sleep studies, may be that those factors which are likely to lead to short sleep (e.g., a sub-optimal sleep environment, caregiving, life stressors, illness) are also the ones likely to result in disrupted sleep and lower sleep efficiency. These factors are typically minimized or eliminated in lab-based sleep studies in which sleeping conditions are often ideal. This explanation is also consistent with a recent study<sup>59</sup> which found both shorter sleep duration and increased sleep variability were correlated with more self-reported stressful demands. Sleep variability was also associated with disrupted limbic network connectivity, thereby providing a potential mechanism for its negative effects.

Weekend sleep extension and social jetlag are related yet distinct phenomena. Weekend sleep extension is thought to reflect catch-up or recovery sleep following partial sleep deprivation during the week. However, our results only partly support this interpretation. While we found an association between shorter weekday sleep and longer weekend sleep extension, there was also a significant effect of country (particularly in Northern and Western Europe). Weekend sleep extension was greater in Europe than in Asia, even though weekday sleep duration was also longer in Europe. Sleep duration was similar between North America and Europe, yet sleep extension was still greater in Europe. This suggests that factors other than catch-up sleep influence weekend sleep extension. Putilov<sup>60</sup> has also argued that weekend sleep extension does not reflect catch-up sleep. Analysing data from studies investigating changes in sleep patterns during movement restrictions imposed by governments during the COVID pandemic, he showed that while weekday sleep duration increased during COVID restrictions, there was no associated decrease in weekend sleep duration, which would be expected with less sleep debt being accumulated during the week. Similar to the findings on weekend sleep extension, we also show that SJL differs across countries. SJL is often thought to reflect a shift to a more natural or preferred sleep timing, typically sleeping and waking later than workday schedules allow. While the evidence that later chronotypes suffer from greater SJL and related health consequences<sup>56,61,62</sup> supports this argument, our data suggests that country also has a significant effect on SJL. The present results also help reconcile the apparent contradiction between studies showing that SJL (an indicator of variable sleep timing) is associated with negative health consequences, whereas weekend sleep extension is usually

associated with health benefits<sup>29</sup> (although it is worth pointing out that weekend catch-up sleep is no substitute for good, regular sleep throughout the week<sup>63-67</sup>).

There are many socio-cultural factors that may affect sleep patterns that differ across country, including sleeping arrangements, child-care practices, and evening meal times and leisure activity. However, because work is such an integral part of our everyday lives, both the direct impact of work itself and the broader differences in work culture (in particular, the distinction between work and non-work time) are likely to play a considerable role in the cross-country differences in sleep reported here. Several time use surveys have shown a strong relationship between long work hours and short sleep<sup>68-70</sup> and Akerstedt et al.<sup>71</sup> found that both work demands and the inability to stop thinking about work were strongly associated with sleep disturbances. East Asian cultures are known for working long hours<sup>72,73</sup>; a recent example of this is the '996' work culture in China where employees are expected to work from 9am to 9pm six days a week<sup>74</sup>. In turn, long work hours are tied to work centrality; that is, the degree to which work plays a role in one's life and identity. This is more strongly expressed in people living in Asia than those in the USA and Europe<sup>75,76</sup>. While not directly related to employment work, an illustrative example comes from a recent survey of Singaporean adolescents which revealed that longer time spent studying or doing homework after school was associated with shorter time in bed for sleep<sup>77</sup>. Furthermore, approximately 35% of these students reported that completion of homework was the dominant prerequisite for going to bed on school nights, superseding feeling sleepy or tired. This indicates that people in cultures where emphasis is put on work and study may be willing to sacrifice their sleep to get work completed.

In addition to the direct effects of work, how work culture has evolved in different parts of the world may also influence sleep. The development of the weekend in Western Europe<sup>78</sup> and the efforts of labour and trade unions to limit working hours and to provide labour protection have resulted in a sharp distinction between work and non-work time in Europe and, to a lesser extent, the USA and Canada<sup>79</sup>. In these regions, weekends are regarded as a time for socializing with family and friends (often until late at night); relaxed days and weekend lie-ins are common<sup>78,80</sup>. While weekends are observed in most industrial nations, in countries without a history of strong labour protection laws, and where loss of employment is a more significant threat to livelihood because of limited social welfare, work is often longer and more integrated with other aspects of life. Therefore, cultural norms concerning 'work time' and 'non-work time' may result in less distinct weekday and weekend behaviours, leading to the differences we report in weekend sleep extension and SJL.

Given different work practices and cultural attitudes towards work (among other cultural differences), it is likely that because people in Asia sleep later, shorter and less well, they catch up on sleep *when they can* during the week, especially when the contrast in time use between workdays and free days is less distinct than among Europeans. This ongoing modulation of sleep may result in increased weekday sleep variability compared with Europeans whose weekday sleep is more likely to be sufficient.

Genetics are an important endogenous modulator of sleep, and determine 40-45% of the variance in adult sleep duration<sup>81</sup>. While this affects interindividual differences in sleep



duration and timing, there is no evidence to indicate that, at the population level, the need for sleep differs across ethnicity or countries. A systematic evaluation of vigilance and declarative memory across nocturnal sleep periods ranging from 5-9 hours, indicated that sleep restriction impairs cognitive performance in Singaporean adolescents in much the same way as it does those in Western countries<sup>82-84</sup>. Although students reported sleeping 6.5 hours a night on average, when such a duration was offered on successive nights in a controlled setting, poorer memory performance and a cumulative decline in vigilance were observed. In contrast, 8 hours of total sleep opportunity, whether presented exclusively nocturnally or in combination with a mid-afternoon nap, preserved vigilance and memory performance relative to a 9-hour sleep opportunity. Similar effects are also seen in studies of mortality and disease risk. A recent large-scale analysis of mortality in Japan, China, Singapore and Korea<sup>85</sup> has shown that a sleep duration of 7 hours is associated with the lowest mortality risk, with both shorter and longer sleep result in elevated risk; a pattern similar to that found in studies conducted elsewhere<sup>86-88</sup>. Furthermore, a study investigating the relationship between sleep duration and cognitive decline in large samples from the UK and China found that cognitive performance was best with 7 hours sleep in both cohorts<sup>89</sup>.

A strength of the current work is that because data was collected over an entire year, seasonal effects<sup>46-48</sup> (caused by varying intensity and duration of light exposure) on sleep duration and timing were minimized. Higher latitude has also been linked to longer sleep duration<sup>13,90</sup> but there no is compelling biological reason for why this might be so. One possibility is that the higher temperature in the tropics is unfavourable to nocturnal sleep. However, if temperature were an issue, it would not account for why more northerly Asian countries (such as Japan and South Korea) share similar sleep patterns with equatorial ones like Singapore. Conversely, major cities in Japan and South Korea are at approximately the same distance from the equator (35-38°) as major Australian cities, where sleep patterns are considerably different. Therefore, we believe that a significant portion of the variation in sleep patterns seen across countries is likely to be cultural in origin.

The present work focuses on differences in sleep patterns between several Asian countries and those primarily populated by people of European ancestry. Many regions of the world are not represented at all (Central and South America, in particular) and others are significantly under-represented, including the Middle East, where monophasic sleep occurs in a minority, and Africa (with representation only from South Africa; see Section 4 of the Supplementary Material for a comparison of global coverage for several large-scale sleep studies). Additionally, our sample is not representative of the wider population. People who use sleep/activity trackers tend to have higher socioeconomic status; they are also younger, more health-conscious (and healthy) and more technologically adept than the general population<sup>91</sup>. However, these limitations also apply to many other sleep studies, particularly those involving wearable devices (or relying on other potentially expensive technology) and requiring the time, interest and cooperation of participants<sup>92</sup>. A more representative characterization of global sleep habits requires sleep/activity trackers to achieve the ubiquity of mobile phones and a reduction in digital health inequality<sup>93</sup>.

Other limitations come from the wearables themselves. While sleep trackers continue to improve, most trackers have difficulty differentiating short periods of stationariness from

sleep, resulting in limited ability to detect naps and to accurately determine sleep onset<sup>94</sup>. Although Oura rings can presently detect naps of 15 minutes duration or longer, this feature has not yet been evaluated against polysomnography (PSG). Other consumer sleep trackers either do not disclose their ability to detect naps or only detect them if they longer than an hour. Conservatively, our analyses were restricted to the primary sleep episode during each 24hr period. While naps are relatively uncommon in the USA and Northern Europe, they are more prevalent in other parts of the world that have a siesta culture or are more tolerant of daytime sleeping<sup>22,95</sup>. As technology and algorithms improve, future research will be able to determine the extent to which people with shorter nocturnal sleep take naps during the day, and what effect these naps have on nocturnal sleep and its variability.

Another factor potentially influencing this study was COVID-19. Some of the data was collected when people in many countries in the world were under some degree of movement restrictions. During this period, sleep tended to be longer, later, and less variable than analogous timeframes in earlier years, although there were regional differences<sup>96-100</sup>. The reduction in both the weekday-weekend differences in sleep and weekday variability<sup>101</sup> suggest that the country level effects described here likely underestimate the differences in sleep patterns both before and after the COVID pandemic.

While we provide a reasonably comprehensive characterization of several different features of sleep, there are several ways future studies could build on our research. It would be helpful to collect additional information about users to investigate how these interact with socio-cultural factors affecting sleep. Furthermore, combining sleep data with longitudinal cognitive, affective and physical health outcomes would provide stronger causal evidence and help uncover the functional consequences of different sleep habits. Because sleep is intrinsically linked with so many aspects of our lives, a full understanding of the reciprocal relationship sleep has with our health and well-being<sup>102</sup> requires multidimensional data from a wide variety of sources.

This study, by its nature, has a broad scope, offering an overview of sleep characteristics across wide global regions, and some potential explanations for differences between them. However, there are often significant cultural differences between neighbouring countries, and within sub-populations of the same country. We hope this study inspires more targeted research into specific cultural factors that might affect sleep, and to what extent different populations within those cultures are affected. This will help us understand the specific mechanisms by which socio-cultural factors affect both our sleep and our waking lives.

The results presented here extend our understanding of how sleep varies geographically and how it is modulated over the week. The present findings are consistent with other research<sup>24,103-105</sup> demonstrating that what is considered ‘normal sleep’ differs around the world, and that it should not be defined solely according to norms or ideals derived from Western sleep research<sup>106-108</sup>. In addition to the Western origins of most sleep research, there is also a pronounced lack of diversity among editors and editorial board members in specialist sleep journals (Table S4 in Supplementary Material). In most of the top ten sleep journals, over 90% of editorial board members work in Western institutions. Together, these points demonstrate that social and cultural factors should be given more consideration when interpreting results from sleep studies. For example, large differences

651 between weekday and weekend sleep duration may not simply reflect homeostatic sleep  
652 drive following sleep restriction, but result from cultural differences in behaviour at the  
653 weekend. We suggest that understanding the influence of culture on sleep is at least as  
654 important as studying how it is influenced by age, sex and circadian biology.

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## Preprint Repositories

A preprint version of this manuscript can be found on the PsyArXiv repository (<https://psyarxiv.com/8ahsu/>).

## Data Availability

The dataset reported in this study is the property of Oura Health Oy and is not publicly available. However, access to anonymized, privacy-protected data can be obtained by writing to M.W.L.C. for further information.

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Supplementary Material for:

Country Differences in Nocturnal Sleep Variability: Observations  
from a Large-scale, Long-term Sleep Wearable Study

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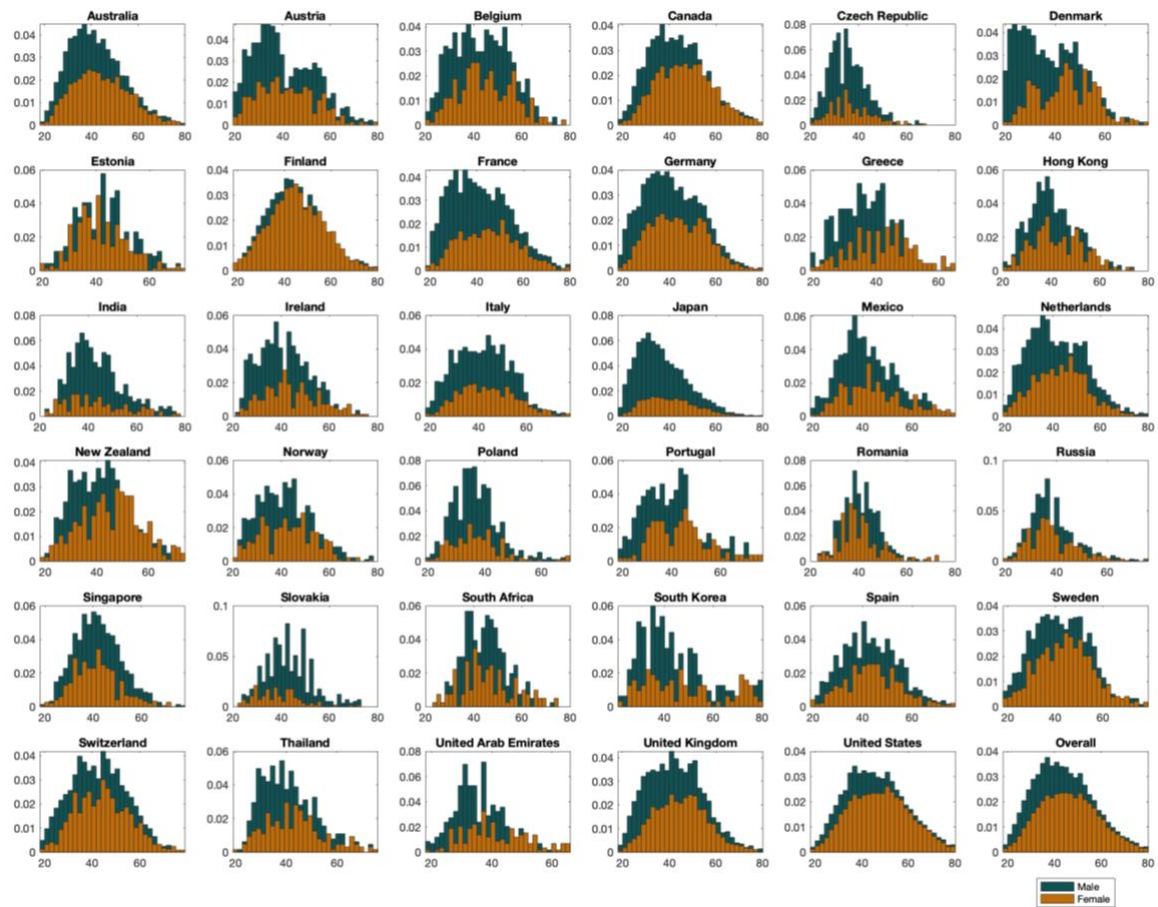
## User and Demographic Data

The average number of weekly users from which data was analysed is shown in Table S1. Owing to Oura's intellectual property restrictions we are unable to publish the exact number of users in each country. The number of nights, the mean age of the users from each country and the proportion of male users is also shown.

| Global Region | Country           | Mean Weekly Users | Mean Number of Nights (SD) | Mean Age (SD) | Proportion Male (%) |
|---------------|-------------------|-------------------|----------------------------|---------------|---------------------|
| Africa        | South Africa      | 200+              | 254.03 (78.78)             | 44.57 (10.81) | 67.90               |
| Asia          | Hong Kong (China) | 500+              | 246.91 (77.67)             | 42.71 (10.67) | 64.48               |
|               | India             | 200+              | 230.67 (81.03)             | 41.80 (12.24) | 77.86               |
|               | Japan             | 5000+             | 242.06 (78.54)             | 37.21 (10.89) | 78.81               |
|               | Singapore         | 1000+             | 243.49 (79.65)             | 42.02 (9.84)  | 67.87               |
|               | South Korea       | 200+              | 248.98 (77.92)             | 45.75 (14.82) | 69.62               |
|               | Thailand          | 200+              | 245.20 (78.94)             | 42.39 (11.31) | 66.56               |
| Europe        | Austria           | 500+              | 245.42 (76.97)             | 39.62 (12.47) | 68.24               |
|               | Belgium           | 500+              | 244.35 (78.90)             | 43.07 (12.03) | 67.15               |
|               | Czech Republic    | 1000+             | 247.46 (76.82)             | 36.99 (9.35)  | 77.70               |
|               | Denmark           | 500+              | 242.10 (79.54)             | 39.71 (12.81) | 68.21               |
|               | Estonia           | 200+              | 258.24 (75.31)             | 40.85 (9.61)  | 58.82               |
|               | Finland           | 10000+            | 271.12 (73.63)             | 44.47 (11.81) | 52.21               |
|               | France            | 1000+             | 239.29 (79.49)             | 42.35 (12.74) | 69.79               |
|               | Germany           | 5000+             | 243.34 (76.89)             | 41.59 (12.06) | 64.18               |
|               | Greece            | 200+              | 238.13 (77.96)             | 41.66 (10.68) | 67.10               |
|               | Ireland           | 500+              | 241.66 (76.95)             | 40.97 (11.54) | 70.61               |
|               | Italy             | 1000+             | 244.53 (78.36)             | 42.64 (11.77) | 71.09               |
|               | Netherlands       | 1000+             | 247.28 (75.10)             | 41.57 (11.74) | 66.08               |
|               | Norway            | 500+              | 247.37 (76.75)             | 41.07 (11.47) | 65.00               |
|               | Poland            | 200+              | 250.16 (78.35)             | 36.74 (8.63)  | 74.15               |
|               | Portugal          | 200+              | 242.91 (75.31)             | 42.87 (12.00) | 66.54               |
|               | Romania           | 200+              | 242.26 (78.08)             | 39.64 (8.97)  | 64.27               |
|               | Russia            | 500+              | 236.88 (78.64)             | 37.53 (8.95)  | 62.28               |
|               | Slovakia          | 200+              | 261.98 (74.69)             | 37.80 (9.17)  | 80.56               |
|               | Spain             | 1000+             | 243.33 (77.67)             | 43.49 (11.58) | 66.10               |
|               | Sweden            | 1000+             | 250.12 (77.55)             | 43.11 (11.81) | 60.16               |
|               | Switzerland       | 1000+             | 242.20 (76.47)             | 43.31 (11.78) | 62.44               |
|               | UK                | 5000+             | 243.92 (77.66)             | 42.94 (11.69) | 64.81               |
| Middle East   | UAE               | 200+              | 231.90 (81.02)             | 41.84 (10.44) | 68.54               |
| North         | Canada            | 5000+             | 243.22 (78.35)             | 44.43 (12.14) | 59.79               |
|               | Mexico            | 500+              | 231.80 (81.57)             | 42.66 (11.85) | 68.96               |
|               | USA               | 50000+            | 234.65 (77.31)             | 46.28 (12.74) | 56.89               |
| Oceania       | Australia         | 1000+             | 243.47 (77.62)             | 42.37 (11.70) | 62.98               |
|               | New Zealand       | 500+              | 248.98 (76.20)             | 44.98 (11.80) | 59.31               |

**Table S1.** Mean number of weekly users and demographic data for each country included in the analysis. 200+: 200-500 users; 500+: 500-1000 users; 1000+: 1000-5000 users; 5000+: 5000-10000 users; 10000+: 10000-50000 users; 50000+: >50000 users.

Figure S1 shows the age distribution for each country by sex and the overall sample.



**Figure S1.** Age distribution for males and females in each country (and the overall sample).

## Full Sleep Measure Data

Sleep measure means and standard deviations are shown in Tables S2a-f.

| Country           | SOTwd (SD)       | WTwd (SD)        | MSTwd (SD)       |
|-------------------|------------------|------------------|------------------|
| South Africa      | 22:37:42 (58.59) | 06:20:56 (54.77) | 02:29:11 (53.31) |
| Hong Kong (China) | 00:11:29 (84.99) | 07:28:46 (79.13) | 03:50:00 (78.81) |
| India             | 00:12:20 (86.42) | 07:25:26 (89.71) | 03:48:45 (85.49) |
| Japan             | 00:27:50 (87.11) | 07:22:04 (87.08) | 03:54:53 (83.92) |
| Singapore         | 00:12:47 (82.59) | 07:25:45 (74.78) | 03:49:09 (75.32) |
| South Korea       | 00:05:02 (91.44) | 07:09:59 (87.50) | 03:37:27 (86.61) |
| Thailand          | 23:56:45 (85.36) | 07:18:01 (81.32) | 03:37:17 (80.34) |
| Austria           | 23:24:33 (67.07) | 07:03:46 (63.80) | 03:14:01 (62.38) |
| Belgium           | 23:37:22 (61.14) | 07:19:05 (58.05) | 03:28:05 (55.82) |
| Czech Republic    | 23:29:12 (66.32) | 07:05:13 (63.43) | 03:17:05 (61.98) |
| Denmark           | 23:22:45 (60.12) | 07:03:06 (59.96) | 03:12:48 (56.67) |
| Estonia           | 23:56:59 (65.40) | 07:37:43 (60.69) | 03:47:10 (59.85) |
| Finland           | 23:32:35 (61.77) | 07:15:24 (58.61) | 03:23:51 (56.77) |
| France            | 23:48:14 (64.11) | 07:27:27 (62.34) | 03:37:42 (59.75) |
| Germany           | 23:31:21 (65.88) | 07:07:32 (64.29) | 03:19:20 (61.99) |
| Greece            | 00:00:02 (78.33) | 07:35:04 (74.54) | 03:47:28 (73.76) |
| Ireland           | 23:39:14 (64.75) | 07:26:59 (63.10) | 03:32:58 (60.90) |
| Italy             | 23:50:14 (62.00) | 07:17:33 (59.63) | 03:33:46 (57.61) |
| Netherlands       | 23:34:43 (62.84) | 07:21:16 (59.11) | 03:27:52 (57.61) |
| Norway            | 23:49:52 (69.01) | 07:23:53 (68.84) | 03:36:43 (65.40) |
| Poland            | 23:45:52 (67.90) | 07:20:36 (63.34) | 03:33:08 (62.52) |
| Portugal          | 00:01:29 (73.58) | 07:39:07 (70.99) | 03:50:14 (68.75) |
| Romania           | 00:05:50 (75.21) | 07:39:21 (71.51) | 03:52:30 (70.32) |
| Russia            | 00:33:19 (76.81) | 08:12:24 (76.66) | 04:22:44 (74.11) |
| Slovakia          | 23:20:38 (62.44) | 06:56:05 (55.89) | 03:08:16 (56.05) |
| Spain             | 23:59:04 (63.10) | 07:33:24 (61.51) | 03:46:09 (58.89) |
| Sweden            | 23:25:54 (64.35) | 07:04:02 (62.30) | 03:14:51 (60.02) |
| Switzerland       | 23:29:44 (61.91) | 07:03:19 (59.40) | 03:16:25 (57.33) |
| UK                | 23:27:20 (67.59) | 07:10:12 (64.69) | 03:18:38 (62.86) |
| UAE               | 23:57:14 (87.18) | 07:20:17 (87.39) | 03:38:36 (84.69) |
| Canada            | 23:20:30 (75.11) | 07:02:59 (70.80) | 03:11:38 (69.48) |
| Mexico            | 23:36:02 (67.97) | 07:08:12 (64.85) | 03:22:00 (62.56) |
| USA               | 23:07:47 (76.95) | 06:46:35 (73.64) | 02:57:05 (71.80) |
| Australia         | 22:52:22 (70.86) | 06:38:04 (66.62) | 02:45:05 (65.61) |
| New Zealand       | 22:52:50 (63.52) | 06:42:30 (56.23) | 02:47:31 (56.68) |

**Table S2a.** Mean (SD) Weekday Sleep onset timing: SOT, Wake time: WT and Mid sleep time: MST for each country (in clock time, SD in minutes).



1066

| Country           | SOTwe (SD)       | WTwe (SD)        | MSTwe (SD)       |
|-------------------|------------------|------------------|------------------|
| South Africa      | 22:57:43 (66.54) | 06:59:38 (63.31) | 02:58:35 (61.71) |
| Hong Kong (China) | 00:38:22 (87.79) | 08:17:25 (87.08) | 04:27:46 (84.01) |
| India             | 00:32:43 (90.23) | 07:50:30 (96.50) | 04:11:35 (90.83) |
| Japan             | 00:47:31 (93.19) | 08:05:56 (98.03) | 04:26:41 (92.25) |
| Singapore         | 00:38:03 (87.31) | 08:09:07 (85.63) | 04:23:28 (83.24) |
| South Korea       | 00:26:16 (98.92) | 07:55:42 (98.75) | 04:10:55 (95.84) |
| Thailand          | 00:16:04 (89.45) | 07:47:05 (89.04) | 04:01:30 (86.47) |
| Austria           | 23:53:39 (74.98) | 07:58:41 (69.66) | 03:55:59 (69.28) |
| Belgium           | 00:01:32 (64.41) | 08:06:42 (63.86) | 04:03:57 (60.23) |
| Czech Republic    | 23:55:37 (71.84) | 07:57:15 (68.91) | 03:56:18 (67.59) |
| Denmark           | 00:03:41 (71.19) | 08:03:46 (68.38) | 04:03:39 (66.82) |
| Estonia           | 00:31:52 (72.16) | 08:41:25 (67.90) | 04:36:31 (66.86) |
| Finland           | 00:10:20 (69.30) | 08:23:41 (66.80) | 04:16:51 (64.72) |
| France            | 00:12:18 (71.94) | 08:15:19 (70.77) | 04:13:38 (67.99) |
| Germany           | 00:04:18 (72.56) | 08:08:11 (70.22) | 04:06:05 (68.08) |
| Greece            | 00:28:25 (80.71) | 08:22:41 (77.64) | 04:25:24 (76.23) |
| Ireland           | 00:12:38 (74.11) | 08:16:31 (70.52) | 04:14:29 (69.29) |
| Italy             | 00:13:22 (68.01) | 08:05:56 (65.41) | 04:09:32 (63.52) |
| Netherlands       | 00:02:50 (70.84) | 08:09:22 (68.31) | 04:05:56 (66.05) |
| Norway            | 00:28:05 (77.79) | 08:32:06 (77.77) | 04:29:59 (74.68) |
| Poland            | 00:18:07 (75.21) | 08:12:50 (72.68) | 04:15:17 (71.11) |
| Portugal          | 00:25:24 (79.25) | 08:23:02 (77.65) | 04:24:02 (75.32) |
| Romania           | 00:31:04 (79.02) | 08:26:00 (75.20) | 04:28:21 (74.04) |
| Russia            | 00:58:36 (82.10) | 08:57:05 (80.03) | 04:57:45 (78.47) |
| Slovakia          | 23:45:15 (68.01) | 07:45:21 (62.06) | 03:45:11 (61.95) |
| Spain             | 00:29:57 (70.48) | 08:25:58 (69.82) | 04:27:52 (66.81) |
| Sweden            | 00:00:04 (73.02) | 08:03:33 (70.66) | 04:01:40 (68.76) |
| Switzerland       | 23:59:33 (68.72) | 08:03:56 (66.24) | 04:01:33 (64.20) |
| UK                | 23:53:30 (73.66) | 07:54:56 (71.69) | 03:54:04 (69.39) |
| UAE               | 00:28:49 (92.79) | 08:03:20 (97.40) | 04:16:01 (92.34) |
| Canada            | 23:46:59 (80.64) | 07:48:26 (75.90) | 03:47:34 (74.91) |
| Mexico            | 00:11:58 (76.08) | 07:59:00 (73.11) | 04:05:17 (70.87) |
| USA               | 23:33:27 (81.75) | 07:33:02 (78.32) | 03:33:08 (76.53) |
| Australia         | 23:18:24 (75.91) | 07:19:05 (73.30) | 03:18:34 (71.36) |
| New Zealand       | 23:16:36 (69.64) | 07:26:30 (62.12) | 03:21:24 (62.30) |

**Table S2b.** Mean (SD) Weekend SOT, WT and MST for each country (in clock time, SD in minutes).

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| Country           | TSTwd (SD)     | SOLwd (SD)   | SEwd/% (SD)  | WASOwd (SD)   |
|-------------------|----------------|--------------|--------------|---------------|
| South Africa      | 414.17 (39.94) | 11.79 (4.77) | 85.32 (4.47) | 48.83 (20.13) |
| Hong Kong (China) | 391.55 (43.72) | 11.35 (3.73) | 85.18 (4.61) | 45.47 (21.58) |
| India             | 384.68 (43.67) | 12.02 (3.84) | 84.26 (4.70) | 48.15 (19.95) |
| Japan             | 366.03 (45.16) | 12.87 (4.48) | 83.36 (5.10) | 48.07 (22.00) |
| Singapore         | 386.87 (45.83) | 11.77 (3.89) | 84.82 (4.81) | 45.86 (21.10) |
| South Korea       | 378.63 (45.77) | 12.44 (3.81) | 84.30 (4.89) | 46.21 (20.50) |
| Thailand          | 396.20 (42.68) | 11.63 (4.15) | 85.32 (4.33) | 44.88 (20.49) |
| Austria           | 415.47 (39.26) | 12.03 (4.38) | 86.25 (4.24) | 43.45 (19.75) |
| Belgium           | 416.49 (42.05) | 11.26 (4.11) | 86.08 (4.31) | 44.92 (19.75) |
| Czech Republic    | 413.97 (38.55) | 11.51 (4.24) | 86.52 (4.12) | 41.79 (19.08) |
| Denmark           | 414.01 (39.70) | 12.13 (4.43) | 85.70 (4.59) | 46.07 (21.58) |
| Estonia           | 418.59 (40.92) | 12.18 (4.51) | 86.53 (4.51) | 41.79 (20.20) |
| Finland           | 421.82 (39.85) | 12.27 (4.51) | 86.85 (4.29) | 40.70 (20.21) |
| France            | 413.85 (40.89) | 11.59 (4.33) | 85.96 (4.42) | 45.09 (20.86) |
| Germany           | 412.55 (39.33) | 11.63 (4.20) | 86.25 (4.30) | 43.41 (19.94) |
| Greece            | 411.78 (40.93) | 11.80 (4.40) | 86.12 (4.25) | 43.06 (19.11) |
| Ireland           | 420.04 (39.26) | 11.87 (4.53) | 85.62 (4.29) | 47.42 (20.49) |
| Italy             | 405.74 (39.95) | 11.68 (4.07) | 86.32 (4.21) | 41.33 (18.51) |
| Netherlands       | 420.10 (40.02) | 11.55 (4.10) | 85.87 (4.33) | 46.21 (20.68) |
| Norway            | 410.78 (40.76) | 11.80 (4.28) | 86.24 (4.19) | 42.90 (20.38) |
| Poland            | 413.25 (40.44) | 11.82 (4.47) | 86.50 (4.02) | 41.25 (18.06) |
| Portugal          | 411.27 (43.75) | 11.06 (3.86) | 85.75 (4.59) | 46.23 (21.79) |
| Romania           | 409.75 (41.83) | 12.84 (4.75) | 85.88 (4.32) | 43.61 (19.52) |
| Russia            | 415.94 (39.18) | 12.19 (4.07) | 86.15 (4.14) | 42.88 (19.33) |
| Slovakia          | 415.13 (37.49) | 11.72 (4.03) | 86.81 (3.95) | 40.15 (18.43) |
| Spain             | 409.99 (40.35) | 12.07 (4.27) | 85.92 (4.35) | 44.16 (20.05) |
| Sweden            | 415.52 (39.72) | 11.74 (4.36) | 86.48 (4.22) | 42.36 (19.51) |
| Switzerland       | 412.00 (40.05) | 11.57 (4.38) | 86.63 (4.33) | 41.36 (19.62) |
| UK                | 416.28 (41.81) | 11.62 (4.13) | 85.76 (4.61) | 46.30 (21.46) |
| UAE               | 397.48 (42.79) | 11.66 (3.99) | 85.39 (4.40) | 45.26 (19.72) |
| Canada            | 416.03 (44.37) | 11.78 (4.54) | 85.81 (4.68) | 46.20 (21.82) |
| Mexico            | 406.02 (44.49) | 11.78 (4.06) | 85.49 (4.37) | 45.88 (19.82) |
| USA               | 412.39 (45.21) | 11.52 (4.34) | 85.79 (4.63) | 46.20 (21.36) |
| Australia         | 417.71 (42.05) | 11.53 (4.18) | 85.59 (4.60) | 47.71 (21.43) |
| New Zealand       | 422.50 (40.78) | 11.72 (4.27) | 85.84 (4.55) | 46.86 (21.29) |

**Table S2c.** Mean (SD) Weekday TST, SOL, SE and WASO for each country (in minutes, except SE which is a percentage).

1079

| Country           | TSTwe (SD)     | SOLwe (SD)   | SEwe/% (SD)  | WASOwe (SD)   |
|-------------------|----------------|--------------|--------------|---------------|
| South Africa      | 428.52 (41.70) | 11.61 (4.67) | 85.01 (4.50) | 53.22 (21.06) |
| Hong Kong (China) | 409.15 (46.47) | 11.35 (4.12) | 84.93 (4.71) | 49.64 (22.52) |
| India             | 388.25 (45.64) | 11.81 (3.91) | 84.17 (4.72) | 49.49 (20.22) |
| Japan             | 384.74 (48.90) | 12.81 (4.59) | 82.95 (5.14) | 53.60 (23.99) |
| Singapore         | 401.87 (47.88) | 11.59 (3.96) | 84.71 (4.82) | 48.95 (21.68) |
| South Korea       | 396.68 (47.87) | 12.47 (3.70) | 83.76 (4.85) | 52.63 (22.86) |
| Thailand          | 404.04 (44.50) | 11.70 (4.11) | 85.09 (4.46) | 46.82 (20.70) |
| Austria           | 436.67 (42.40) | 11.87 (4.50) | 85.98 (4.21) | 47.98 (20.33) |
| Belgium           | 435.18 (44.77) | 11.17 (4.29) | 85.74 (4.29) | 49.66 (20.50) |
| Czech Republic    | 435.30 (40.94) | 11.26 (4.23) | 86.28 (4.09) | 46.08 (19.49) |
| Denmark           | 429.21 (41.65) | 11.85 (4.28) | 85.28 (4.50) | 50.73 (21.57) |
| Estonia           | 441.84 (44.59) | 11.77 (4.32) | 86.15 (4.58) | 47.48 (21.50) |
| Finland           | 446.24 (42.75) | 11.75 (4.33) | 86.43 (4.18) | 46.78 (20.83) |
| France            | 433.29 (43.94) | 11.42 (4.40) | 85.70 (4.43) | 49.37 (21.56) |
| Germany           | 434.92 (43.22) | 11.39 (4.20) | 85.93 (4.29) | 48.64 (21.01) |
| Greece            | 427.48 (45.15) | 11.71 (4.64) | 85.90 (4.25) | 46.45 (19.30) |
| Ireland           | 432.17 (43.08) | 11.39 (4.24) | 85.26 (4.31) | 51.49 (21.00) |
| Italy             | 425.90 (42.40) | 11.54 (4.10) | 85.95 (4.21) | 46.39 (19.56) |
| Netherlands       | 435.61 (44.36) | 11.39 (4.24) | 85.51 (4.32) | 50.59 (21.14) |
| Norway            | 435.29 (42.97) | 11.56 (4.41) | 85.90 (4.20) | 48.52 (21.51) |
| Poland            | 428.95 (41.45) | 11.64 (4.67) | 86.17 (3.99) | 45.39 (18.55) |
| Portugal          | 427.14 (44.87) | 10.92 (3.90) | 85.48 (4.55) | 50.14 (21.99) |
| Romania           | 426.91 (42.70) | 12.63 (4.72) | 85.61 (4.24) | 47.66 (20.18) |
| Russia            | 432.15 (41.15) | 11.88 (4.17) | 86.00 (4.16) | 46.16 (20.11) |
| Slovakia          | 435.25 (40.21) | 11.43 (4.01) | 86.50 (3.88) | 44.62 (18.66) |
| Spain             | 427.06 (43.57) | 11.95 (4.46) | 85.54 (4.37) | 48.76 (20.45) |
| Sweden            | 435.64 (41.86) | 11.61 (4.47) | 86.09 (4.15) | 47.56 (20.04) |
| Switzerland       | 437.08 (43.18) | 11.43 (4.45) | 86.29 (4.32) | 46.95 (20.61) |
| UK                | 430.54 (44.32) | 11.38 (4.23) | 85.42 (4.60) | 50.59 (22.05) |
| UAE               | 405.99 (44.83) | 11.36 (3.98) | 85.11 (4.43) | 48.41 (20.83) |
| Canada            | 430.90 (45.91) | 11.65 (4.62) | 85.53 (4.65) | 50.28 (22.44) |
| Mexico            | 416.90 (46.28) | 11.48 (4.21) | 85.21 (4.29) | 49.84 (20.44) |
| USA               | 429.01 (47.18) | 11.44 (4.46) | 85.55 (4.60) | 50.37 (22.17) |
| Australia         | 428.81 (44.32) | 11.55 (4.36) | 85.23 (4.57) | 51.53 (21.91) |
| New Zealand       | 438.20 (45.25) | 11.68 (4.27) | 85.46 (4.49) | 51.41 (21.67) |

**Table S2d.** Mean (SD) Weekend TST, SOL, SE and WASO for each country (in minutes, except SE which is a percentage).

1086

| Country           | SOTdiff (SD)  | WTdiff (SD)   | MSTdiff (SD)  | TSTdiff (SD)  |
|-------------------|---------------|---------------|---------------|---------------|
| South Africa      | 20.01 (28.44) | 38.70 (32.87) | 29.39 (26.95) | 14.36 (25.03) |
| Hong Kong (China) | 26.89 (31.37) | 48.65 (44.18) | 37.78 (34.12) | 17.60 (29.81) |
| India             | 20.38 (26.12) | 25.07 (31.20) | 22.83 (26.56) | 3.58 (19.78)  |
| Japan             | 19.67 (32.89) | 43.88 (51.58) | 31.79 (38.09) | 18.70 (34.35) |
| Singapore         | 25.27 (29.01) | 43.36 (39.25) | 34.32 (30.42) | 15.00 (28.68) |
| South Korea       | 21.23 (27.77) | 45.72 (50.34) | 33.47 (35.25) | 18.05 (33.42) |
| Thailand          | 19.33 (26.36) | 29.08 (35.19) | 24.21 (27.73) | 7.84 (24.83)  |
| Austria           | 29.10 (32.53) | 54.92 (40.39) | 41.97 (33.14) | 21.19 (27.01) |
| Belgium           | 24.16 (25.55) | 47.61 (35.70) | 35.87 (27.22) | 18.68 (26.03) |
| Czech Republic    | 26.41 (27.97) | 52.05 (37.08) | 39.22 (28.87) | 21.33 (27.99) |
| Denmark           | 40.93 (35.41) | 60.67 (40.71) | 50.85 (34.91) | 15.19 (26.76) |
| Estonia           | 34.88 (34.65) | 63.69 (40.45) | 49.35 (33.73) | 23.25 (29.49) |
| Finland           | 37.74 (31.72) | 68.29 (41.25) | 53.00 (32.91) | 24.43 (28.66) |
| France            | 24.07 (27.19) | 47.85 (37.72) | 35.93 (28.80) | 19.44 (27.50) |
| Germany           | 32.95 (31.03) | 60.65 (40.16) | 46.75 (31.89) | 22.37 (28.37) |
| Greece            | 28.38 (27.28) | 47.62 (35.76) | 37.94 (27.69) | 15.70 (26.87) |
| Ireland           | 33.41 (34.13) | 49.53 (39.04) | 41.53 (33.44) | 12.13 (26.72) |
| Italy             | 23.15 (26.21) | 48.38 (36.23) | 35.76 (27.64) | 20.16 (26.46) |
| Netherlands       | 28.12 (30.24) | 48.10 (36.32) | 38.07 (29.79) | 15.51 (26.31) |
| Norway            | 38.21 (34.85) | 68.22 (41.83) | 53.28 (34.64) | 24.51 (29.48) |
| Poland            | 32.24 (29.28) | 52.24 (39.10) | 42.15 (30.23) | 15.69 (29.48) |
| Portugal          | 23.91 (27.93) | 43.92 (35.98) | 33.80 (28.41) | 15.88 (25.32) |
| Romania           | 25.23 (27.35) | 46.65 (37.47) | 35.85 (29.01) | 17.17 (26.28) |
| Russia            | 25.27 (27.23) | 44.69 (36.40) | 35.01 (28.30) | 16.22 (26.29) |
| Slovakia          | 24.62 (26.62) | 49.27 (34.24) | 36.92 (27.02) | 20.12 (24.75) |
| Spain             | 30.88 (30.22) | 52.56 (38.53) | 41.73 (30.67) | 17.08 (27.91) |
| Sweden            | 34.17 (30.69) | 59.51 (38.23) | 46.83 (31.13) | 20.12 (26.37) |
| Switzerland       | 29.81 (30.04) | 60.60 (39.70) | 45.14 (31.40) | 25.08 (28.17) |
| UK                | 26.16 (29.44) | 44.75 (35.77) | 35.43 (29.15) | 14.25 (25.73) |
| UAE               | 31.60 (32.00) | 43.05 (44.88) | 37.43 (35.64) | 8.51 (27.00)  |
| Canada            | 26.47 (30.26) | 45.45 (38.31) | 35.95 (30.92) | 14.87 (26.14) |
| Mexico            | 35.94 (33.77) | 50.81 (39.27) | 43.28 (32.83) | 10.88 (27.61) |
| USA               | 25.66 (31.06) | 46.46 (41.31) | 36.06 (32.67) | 16.62 (27.99) |
| Australia         | 26.03 (30.05) | 41.01 (36.67) | 33.48 (29.86) | 11.09 (26.12) |
| New Zealand       | 23.76 (26.05) | 44.00 (33.25) | 33.88 (25.57) | 15.70 (26.41) |

**Table S2e.** Mean (SD) weekend-weekday differences in SOT, WT, MST (SJL) and TST (weekend sleep extension) (in minutes).

1093

| Country           | TSTwd Var (SD) | SOTwd Var (SD) | WTwd Var (SD) |
|-------------------|----------------|----------------|---------------|
| South Africa      | 52.09 (14.94)  | 50.20 (20.58)  | 43.16 (16.56) |
| Hong Kong (China) | 53.31 (16.81)  | 53.69 (24.69)  | 48.83 (24.51) |
| India             | 57.80 (16.16)  | 62.08 (28.90)  | 57.89 (28.83) |
| Japan             | 61.31 (22.00)  | 71.33 (43.22)  | 62.55 (44.44) |
| Singapore         | 54.41 (17.05)  | 55.67 (26.93)  | 50.21 (27.31) |
| South Korea       | 61.39 (18.40)  | 66.69 (29.23)  | 58.06 (28.20) |
| Thailand          | 56.15 (16.89)  | 57.61 (25.40)  | 50.32 (23.95) |
| Austria           | 50.56 (14.81)  | 50.60 (22.54)  | 45.73 (22.56) |
| Belgium           | 48.58 (15.62)  | 46.97 (22.92)  | 43.37 (22.80) |
| Czech Republic    | 50.57 (16.06)  | 49.73 (22.79)  | 43.96 (22.40) |
| Denmark           | 49.16 (14.81)  | 49.20 (22.66)  | 45.13 (23.12) |
| Estonia           | 52.01 (15.73)  | 50.65 (22.61)  | 45.99 (20.85) |
| Finland           | 48.16 (15.21)  | 45.24 (22.38)  | 43.39 (22.87) |
| France            | 51.37 (15.26)  | 49.28 (20.58)  | 45.37 (19.97) |
| Germany           | 49.85 (15.42)  | 49.60 (23.62)  | 44.73 (24.11) |
| Greece            | 53.68 (15.76)  | 53.40 (21.30)  | 49.13 (21.03) |
| Ireland           | 50.12 (17.22)  | 48.94 (31.31)  | 46.51 (29.10) |
| Italy             | 48.78 (14.38)  | 48.51 (22.29)  | 41.76 (21.63) |
| Netherlands       | 48.94 (15.83)  | 46.91 (23.76)  | 43.85 (22.88) |
| Norway            | 51.56 (17.78)  | 50.76 (26.73)  | 49.20 (28.27) |
| Poland            | 51.68 (15.13)  | 50.97 (20.13)  | 45.59 (20.39) |
| Portugal          | 52.42 (15.34)  | 51.68 (21.64)  | 47.00 (22.14) |
| Romania           | 55.11 (15.50)  | 56.23 (25.77)  | 49.65 (25.08) |
| Russia            | 57.71 (16.06)  | 56.48 (23.54)  | 52.94 (23.29) |
| Slovakia          | 48.66 (14.83)  | 47.29 (22.73)  | 41.41 (20.85) |
| Spain             | 50.99 (15.49)  | 50.28 (22.18)  | 46.21 (22.97) |
| Sweden            | 49.29 (15.58)  | 47.49 (22.62)  | 43.99 (22.51) |
| Switzerland       | 49.60 (15.14)  | 47.89 (21.67)  | 43.42 (21.54) |
| UK                | 49.36 (16.12)  | 47.83 (25.42)  | 44.14 (24.55) |
| UAE               | 57.98 (18.32)  | 64.46 (31.90)  | 58.78 (32.13) |
| Canada            | 50.45 (16.36)  | 50.15 (26.96)  | 46.16 (26.00) |
| Mexico            | 56.34 (15.97)  | 57.88 (28.47)  | 52.64 (27.40) |
| USA               | 52.81 (17.37)  | 52.64 (28.25)  | 48.42 (27.15) |
| Australia         | 50.13 (16.32)  | 49.15 (26.94)  | 45.59 (25.60) |
| New Zealand       | 48.36 (14.89)  | 45.27 (21.70)  | 41.67 (20.63) |

**Table S2f.** Mean (SD) weekday variability in TST, SOT and WT (in minutes).

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## Linear Models

### 1. Total Sleep Time

TST ~ 1 + sex + age + age<sup>2</sup> + daytype + country + ...  
sex:age + sex:age<sup>2</sup> + sex:daytype + sex:country + ...  
age:age<sup>2</sup> + age:daytype + age:country + ...  
age<sup>2</sup>:daytype + age<sup>2</sup>:country + daytype:country

**Table S3.1a.** Unstandardised Regression Coefficients for TST

| Characteristic               | B     | 95% CI <sup>1</sup> | p-value |
|------------------------------|-------|---------------------|---------|
| <b>Sex</b>                   |       |                     |         |
| Female                       | —     | —                   |         |
| Male                         | -26   | -28, -24            | <0.001  |
| <b>Age</b>                   | -0.38 | -0.44, -0.31        | <0.001  |
| <b>Age<sup>2</sup></b>       | 0.00  | 0.00, 0.01          | 0.037   |
| <b>DayType</b>               |       |                     |         |
| Weekday                      | —     | —                   |         |
| Weekend                      | 14    | 13, 16              | <0.001  |
| <b>Country</b>               |       |                     |         |
| Australia                    | —     | —                   |         |
| Austria                      | -0.74 | -4.7, 3.2           | 0.7     |
| Belgium                      | 0.14  | -4.3, 4.6           | >0.9    |
| Canada                       | -0.04 | -1.9, 1.8           | >0.9    |
| Czech Republic               | -5.2  | -9.3, -1.2          | 0.012   |
| Denmark                      | -7.0  | -11, -3.3           | <0.001  |
| Estonia                      | -0.26 | -5.0, 4.5           | >0.9    |
| Finland                      | 2.1   | 0.45, 3.7           | 0.013   |
| France                       | 0.10  | -3.3, 3.5           | >0.9    |
| Germany                      | -5.3  | -7.3, -3.3          | <0.001  |
| Greece                       | 0.48  | -5.5, 6.5           | 0.9     |
| Hong Kong                    | -25   | -29, -20            | <0.001  |
| India                        | -28   | -34, -21            | <0.001  |
| Ireland                      | 6.1   | 1.3, 11             | 0.014   |
| Italy                        | -8.8  | -12, -5.5           | <0.001  |
| Japan                        | -54   | -56, -52            | <0.001  |
| Mexico                       | -5.1  | -9.5, -0.68         | 0.024   |
| Netherlands                  | 5.0   | 2.1, 7.8            | <0.001  |
| New Zealand                  | 1.9   | -1.8, 5.6           | 0.3     |
| Norway                       | -9.3  | -14, -5.0           | <0.001  |
| Poland                       | -1.1  | -6.7, 4.6           | 0.7     |
| Portugal                     | 1.3   | -4.3, 6.8           | 0.7     |
| Romania                      | -7.6  | -13, -2.5           | 0.003   |
| Russia                       | -5.6  | -9.3, -1.9          | 0.003   |
| Singapore                    | -29   | -33, -25            | <0.001  |
| Slovakia                     | 4.4   | -2.3, 11            | 0.2     |
| South Africa                 | 0.31  | -6.2, 6.8           | >0.9    |
| South Korea                  | -34   | -42, -26            | <0.001  |
| Spain                        | -6.4  | -9.6, -3.3          | <0.001  |
| Sweden                       | -4.0  | -6.5, -1.4          | 0.002   |
| Switzerland                  | -2.0  | -4.9, 0.82          | 0.2     |
| Thailand                     | -27   | -32, -21            | <0.001  |
| United Arab Emirates         | -19   | -25, -14            | <0.001  |
| United Kingdom               | -0.90 | -2.8, 0.99          | 0.3     |
| United States                | -4.7  | -6.3, -3.2          | <0.001  |
| <b>Sex * Age</b>             |       |                     |         |
| Male * Age                   | 0.28  | 0.26, 0.30          | <0.001  |
| <b>Sex * Age<sup>2</sup></b> |       |                     |         |
| Male * Age <sup>2</sup>      | 0.01  | 0.01, 0.01          | <0.001  |

| Characteristic               | B     | 95% CI <sup>1</sup> | p-value |
|------------------------------|-------|---------------------|---------|
| <b>Sex *</b>                 |       |                     |         |
| Male * Weekend               | -0.11 | -0.62, 0.41         | 0.7     |
| <b>Sex * Country</b>         |       |                     |         |
| Male * Austria               | -1.0  | -4.9, 2.8           | 0.6     |
| Male * Belgium               | -2.3  | -6.8, 2.2           | 0.3     |
| Male * Canada                | -2.9  | -4.8, -1.0          | 0.002   |
| Male * Czech Republic        | 2.4   | -1.5, 6.2           | 0.2     |
| Male * Denmark               | 1.2   | -2.6, 5.0           | 0.5     |
| Male * Estonia               | 1.4   | -3.4, 6.3           | 0.6     |
| Male * Finland               | -0.18 | -1.8, 1.5           | 0.8     |
| Male * France                | -1.8  | -5.2, 1.6           | 0.3     |
| Male * Germany               | 0.60  | -1.3, 2.6           | 0.5     |
| Male * Greece                | -8.1  | -14, -2.1           | 0.009   |
| Male * Hong Kong             | -0.54 | -4.9, 3.8           | 0.8     |
| Male * India                 | 1.9   | -4.6, 8.3           | 0.6     |
| Male * Ireland               | -2.6  | -7.4, 2.2           | 0.3     |
| Male * Italy                 | -3.0  | -6.3, 0.25          | 0.071   |
| Male * Japan                 | 2.6   | 0.51, 4.6           | 0.015   |
| Male * Mexico                | -7.0  | -11, -2.7           | 0.001   |
| Male * Netherlands           | -0.98 | -3.9, 1.9           | 0.5     |
| Male * New Zealand           | 4.1   | 0.34, 7.8           | 0.033   |
| Male * Norway                | 2.2   | -2.1, 6.6           | 0.3     |
| Male * Poland                | -5.8  | -11, -0.51          | 0.032   |
| Male * Portugal              | -8.2  | -14, -2.6           | 0.004   |
| Male * Romania               | -4.6  | -9.7, 0.46          | 0.074   |
| Male * Russia                | 3.5   | -0.09, 7.0          | 0.056   |
| Male * Singapore             | 2.1   | -1.5, 5.7           | 0.3     |
| Male * Slovakia              | -2.3  | -8.7, 4.1           | 0.5     |
| Male * South Africa          | -3.1  | -9.5, 3.4           | 0.4     |
| Male * South Korea           | 1.9   | -5.5, 9.4           | 0.6     |
| Male * Spain                 | -1.7  | -4.9, 1.5           | 0.3     |
| Male * Sweden                | 0.60  | -2.0, 3.2           | 0.6     |
| Male * Switzerland           | -3.0  | -5.8, -0.08         | 0.044   |
| Male * Thailand              | 6.6   | 1.4, 12             | 0.013   |
| Male * United Arab Emirates  | -2.2  | -7.6, 3.3           | 0.4     |
| Male * United Kingdom        | 0.64  | -1.2, 2.5           | 0.5     |
| Male * United States         | -2.5  | -4.0, -0.98         | 0.001   |
| <b>Age * Age<sup>2</sup></b> | 0.00  | 0.00, 0.00          | <0.001  |
| <b>Age * DayType</b>         |       |                     |         |
| Age * Weekend                | 0.19  | 0.16, 0.21          | <0.001  |
| <b>Age * Country</b>         |       |                     |         |
| Age * Austria                | -0.01 | -0.16, 0.15         | >0.9    |
| Age * Belgium                | -0.09 | -0.27, 0.08         | 0.3     |
| Age * Canada                 | 0.00  | -0.07, 0.08         | >0.9    |
| Age * Czech Republic         | -0.38 | -0.57, -0.19        | <0.001  |
| Age * Denmark                | -0.17 | -0.32, -0.02        | 0.022   |
| Age * Estonia                | -0.23 | -0.48, 0.02         | 0.068   |
| Age * Finland                | -0.12 | -0.19, -0.05        | <0.001  |
| Age * France                 | -0.07 | -0.20, 0.05         | 0.3     |
| Age * Germany                | -0.16 | -0.24, -0.08        | <0.001  |
| Age * Greece                 | -0.12 | -0.40, 0.15         | 0.4     |
| Age * Hong Kong              | 0.01  | -0.18, 0.20         | >0.9    |
| Age * India                  | -0.32 | -0.54, -0.09        | 0.007   |
| Age * Ireland                | 0.01  | -0.18, 0.21         | 0.9     |
| Age * Italy                  | -0.29 | -0.42, -0.16        | <0.001  |
| Age * Japan                  | -0.43 | -0.51, -0.34        | <0.001  |
| Age * Mexico                 | 0.07  | -0.11, 0.24         | 0.5     |
| Age * Netherlands            | -0.18 | -0.30, -0.06        | 0.002   |
| Age * New Zealand            | 0.04  | -0.12, 0.20         | 0.6     |
| Age * Norway                 | -0.29 | -0.48, -0.11        | 0.002   |
| Age * Poland                 | -0.29 | -0.58, 0.00         | 0.050   |
| Age * Portugal               | -0.22 | -0.45, 0.01         | 0.060   |

| Characteristic                          | B     | 95% CI <sup>1</sup> | p-value |
|-----------------------------------------|-------|---------------------|---------|
| Age * Romania                           | -0.50 | -0.77, -0.24        | <0.001  |
| Age * Russia                            | -0.13 | -0.32, 0.07         | 0.2     |
| Age * Singapore                         | -0.08 | -0.24, 0.09         | 0.4     |
| Age * Slovakia                          | 0.17  | -0.14, 0.48         | 0.3     |
| Age * South Africa                      | -0.25 | -0.56, 0.05         | 0.10    |
| Age * South Korea                       | -0.14 | -0.45, 0.17         | 0.4     |
| Age * Spain                             | -0.22 | -0.35, -0.09        | 0.001   |
| Age * Sweden                            | -0.25 | -0.35, -0.14        | <0.001  |
| Age * Switzerland                       | -0.20 | -0.32, -0.08        | <0.001  |
| Age * Thailand                          | -0.31 | -0.53, -0.09        | 0.006   |
| Age * United Arab Emirates              | -0.40 | -0.64, -0.16        | 0.001   |
| Age * United Kingdom                    | -0.03 | -0.11, 0.05         | 0.5     |
| Age * United States                     | -0.05 | -0.12, 0.01         | 0.11    |
| <b>Age<sup>2</sup> * DayType</b>        |       |                     |         |
| Age <sup>2</sup> * Weekend              | -0.02 | -0.02, -0.02        | <0.001  |
| <b>Age<sup>2</sup> * Country</b>        |       |                     |         |
| Age <sup>2</sup> * Austria              | 0.00  | -0.01, 0.01         | >0.9    |
| Age <sup>2</sup> * Belgium              | 0.01  | 0.00, 0.02          | 0.2     |
| Age <sup>2</sup> * Canada               | 0.00  | -0.01, 0.00         | 0.3     |
| Age <sup>2</sup> * Czech Republic       | 0.00  | -0.01, 0.01         | >0.9    |
| Age <sup>2</sup> * Denmark              | 0.01  | 0.00, 0.02          | 0.004   |
| Age <sup>2</sup> * Estonia              | -0.01 | -0.03, 0.00         | 0.12    |
| Age <sup>2</sup> * Finland              | 0.00  | -0.01, 0.00         | 0.3     |
| Age <sup>2</sup> * France               | -0.01 | -0.02, 0.00         | 0.071   |
| Age <sup>2</sup> * Germany              | 0.00  | -0.01, 0.00         | 0.2     |
| Age <sup>2</sup> * Greece               | 0.00  | -0.02, 0.02         | 0.8     |
| Age <sup>2</sup> * Hong Kong            | -0.01 | -0.02, 0.01         | 0.4     |
| Age <sup>2</sup> * India                | -0.03 | -0.04, -0.01        | <0.001  |
| Age <sup>2</sup> * Ireland              | 0.00  | -0.01, 0.01         | >0.9    |
| Age <sup>2</sup> * Italy                | 0.00  | 0.00, 0.01          | 0.3     |
| Age <sup>2</sup> * Japan                | 0.01  | 0.00, 0.01          | 0.058   |
| Age <sup>2</sup> * Mexico               | 0.00  | -0.01, 0.01         | >0.9    |
| Age <sup>2</sup> * Netherlands          | -0.01 | -0.02, 0.00         | 0.004   |
| Age <sup>2</sup> * New Zealand          | 0.00  | -0.01, 0.01         | 0.8     |
| Age <sup>2</sup> * Norway               | 0.00  | -0.01, 0.01         | 0.7     |
| Age <sup>2</sup> * Poland               | 0.01  | -0.01, 0.02         | 0.5     |
| Age <sup>2</sup> * Portugal             | -0.01 | -0.03, 0.00         | 0.074   |
| Age <sup>2</sup> * Romania              | 0.00  | -0.01, 0.02         | 0.7     |
| Age <sup>2</sup> * Russia               | 0.00  | -0.01, 0.01         | >0.9    |
| Age <sup>2</sup> * Singapore            | -0.02 | -0.03, -0.01        | 0.001   |
| Age <sup>2</sup> * Slovakia             | 0.00  | -0.02, 0.02         | >0.9    |
| Age <sup>2</sup> * South Africa         | 0.00  | -0.02, 0.02         | >0.9    |
| Age <sup>2</sup> * South Korea          | -0.02 | -0.04, -0.01        | 0.002   |
| Age <sup>2</sup> * Spain                | 0.00  | 0.00, 0.01          | 0.4     |
| Age <sup>2</sup> * Sweden               | 0.00  | 0.00, 0.01          | 0.3     |
| Age <sup>2</sup> * Switzerland          | -0.01 | -0.02, -0.01        | <0.001  |
| Age <sup>2</sup> * Thailand             | 0.01  | -0.01, 0.02         | 0.3     |
| Age <sup>2</sup> * United Arab Emirates | 0.01  | -0.01, 0.03         | 0.3     |
| Age <sup>2</sup> * United Kingdom       | 0.00  | -0.01, 0.00         | 0.3     |
| Age <sup>2</sup> * United States        | 0.00  | -0.01, 0.00         | 0.3     |
| <b>Weekend * country</b>                |       |                     |         |
| Weekend * Austria                       | 11    | 7.7, 15             | <0.001  |
| Weekend * Belgium                       | 7.6   | 3.4, 12             | <0.001  |
| Weekend * Canada                        | 3.5   | 1.7, 5.3            | <0.001  |
| Weekend * Czech Republic                | 11    | 8.0, 15             | <0.001  |
| Weekend * Denmark                       | 5.5   | 2.0, 8.9            | 0.002   |
| Weekend * Estonia                       | 12    | 7.0, 16             | <0.001  |
| Weekend * Finland                       | 13    | 11, 15              | <0.001  |
| Weekend * France                        | 8.8   | 5.7, 12             | <0.001  |
| Weekend * Germany                       | 12    | 9.8, 14             | <0.001  |
| Weekend * Greece                        | 4.4   | -1.3, 10            | 0.13    |
| Weekend * Hong Kong                     | 6.0   | 1.8, 10             | 0.005   |



| Characteristic                 | B     | 95% CI <sup>1</sup> | p-value |
|--------------------------------|-------|---------------------|---------|
| Weekend * India                | -7.1  | -13, -1.7           | 0.010   |
| Weekend * Ireland              | 1.4   | -3.0, 5.7           | 0.5     |
| Weekend * Italy                | 9.0   | 6.0, 12             | <0.001  |
| Weekend * Japan                | 9.1   | 7.3, 11             | <0.001  |
| Weekend * Mexico               | -0.22 | -4.2, 3.8           | >0.9    |
| Weekend * Netherlands          | 4.7   | 1.9, 7.4            | <0.001  |
| Weekend * New Zealand          | 4.1   | 0.46, 7.7           | 0.027   |
| Weekend * Norway               | 14    | 9.6, 18             | <0.001  |
| Weekend * Poland               | 5.5   | 0.83, 10            | 0.021   |
| Weekend * Portugal             | 4.8   | -0.47, 10           | 0.075   |
| Weekend * Romania              | 5.8   | 1.0, 11             | 0.018   |
| Weekend * Russia               | 5.8   | 2.3, 9.2            | <0.001  |
| Weekend * Singapore            | 3.2   | -0.15, 6.6          | 0.061   |
| Weekend * Slovakia             | 9.6   | 4.5, 15             | <0.001  |
| Weekend * South Africa         | 2.4   | -3.7, 8.4           | 0.4     |
| Weekend * South Korea          | 7.9   | 1.1, 15             | 0.023   |
| Weekend * Spain                | 5.7   | 2.6, 8.7            | <0.001  |
| Weekend * Sweden               | 8.9   | 6.4, 11             | <0.001  |
| Weekend * Switzerland          | 14    | 11, 17              | <0.001  |
| Weekend * Thailand             | -3.4  | -8.3, 1.4           | 0.2     |
| Weekend * United Arab Emirates | -3.0  | -8.1, 2.1           | 0.3     |
| Weekend * United Kingdom       | 3.0   | 1.2, 4.8            | 0.001   |
| Weekend * United States        | 5.3   | 3.8, 6.7            | <0.001  |

<sup>1</sup>CI = Confidence Interval

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**Table S3.1b.** ANOVA table with effect sizes - TST

| Factor                     | Sum of squares | Df      | F value    | p     | $\eta_p^2$ |
|----------------------------|----------------|---------|------------|-------|------------|
| Sex                        | 72,361,600.53  | 1       | 39,723.406 | 0.000 | 0.081      |
| Age                        | 845,336.18     | 1       | 464.053    | 0.000 | 0.001      |
| Age <sup>2</sup>           | 55.24          | 1       | 0.030      | 0.862 | 0.000      |
| DayType                    | 35,193,632.89  | 1       | 19,319.791 | 0.000 | 0.041      |
| Country                    | 48,153,388.94  | 34      | 777.475    | 0.000 | 0.055      |
| Sex x Age                  | 1,114,700.95   | 1       | 611.923    | 0.000 | 0.001      |
| Sex x Age <sup>2</sup>     | 242,029.35     | 1       | 132.864    | 0.000 | 0.000      |
| Sex x DayType              | 300.66         | 1       | 0.165      | 0.685 | 0.000      |
| Sex x Country              | 334,339.24     | 34      | 5.398      | 0.000 | 0.000      |
| Age x Age <sup>2</sup>     | 198,175.09     | 1       | 108.790    | 0.000 | 0.000      |
| Age x DayType              | 538,650.31     | 1       | 295.696    | 0.000 | 0.001      |
| Age x Country              | 515,193.38     | 34      | 8.318      | 0.000 | 0.001      |
| Age <sup>2</sup> x DayType | 1,480,826.43   | 1       | 812.910    | 0.000 | 0.002      |
| Age <sup>2</sup> x Country | 193,360.25     | 34      | 3.122      | 0.000 | 0.000      |
| DayType x Country          | 1,361,434.30   | 34      | 21.981     | 0.000 | 0.002      |
| Residuals                  | 823,731,199.14 | 452,193 |            |       |            |

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## 2. Sleep Efficiency

Sleep efficiency (TST/TIB, that is, the amount of time in bed that is actually spent asleep). The difference between TST and TIB is the sum of SOL and WASO. Wearables infer TIB from change in movement and heart rate when transitioning from being active to lying down in bed preparing for sleep. On occasion, this can be tricked if one lies still but awake in bed, such as while reading an e-device. This is another advantage of the long-duration data used here.

SE ~ 1 + sex + age + age<sup>2</sup> + daytype + country + ...  
sex:age + sex:age<sup>2</sup> + sex:daytype + sex:country + ...  
age:age<sup>2</sup> + age:daytype + age:country + ...  
age<sup>2</sup>:daytype + age<sup>2</sup>:country + daytype:country

**Table S3.2a.** Unstandardised Regression Coefficients for SE

| Characteristic         | B     | 95% CI <sup>1</sup> | p-value |
|------------------------|-------|---------------------|---------|
| <b>Sex</b>             |       |                     |         |
| Female                 | —     | —                   |         |
| Male                   | -2.2  | -2.4, -2.1          | <0.001  |
| <b>Age</b>             | -0.02 | -0.03, -0.01        | <0.001  |
| <b>Age<sup>2</sup></b> | 0.00  | 0.00, 0.00          | <0.001  |
| <b>DayType</b>         |       |                     |         |
| Weekday                | —     | —                   |         |
| Weekend                | -0.47 | -0.62, -0.32        | <0.001  |
| <b>Country</b>         |       |                     |         |
| Australia              | —     | —                   |         |
| Austria                | 0.63  | 0.23, 1.0           | 0.002   |
| Belgium                | 0.50  | 0.04, 0.96          | 0.032   |
| Canada                 | 0.27  | 0.08, 0.46          | 0.006   |
| Czech Republic         | 0.93  | 0.51, 1.3           | <0.001  |
| Denmark                | -0.03 | -0.40, 0.34         | 0.9     |
| Estonia                | 0.87  | 0.39, 1.4           | <0.001  |
| Finland                | 1.1   | 0.90, 1.2           | <0.001  |
| France                 | 0.79  | 0.44, 1.1           | <0.001  |
| Germany                | 0.69  | 0.49, 0.89          | <0.001  |
| Greece                 | 0.58  | -0.03, 1.2          | 0.064   |
| Hong Kong              | -0.24 | -0.68, 0.21         | 0.3     |
| India                  | -1.3  | -2.0, -0.64         | <0.001  |
| Ireland                | 0.64  | 0.15, 1.1           | 0.011   |
| Italy                  | 0.89  | 0.54, 1.2           | <0.001  |
| Japan                  | -2.0  | -2.2, -1.8          | <0.001  |
| Mexico                 | 0.22  | -0.23, 0.67         | 0.3     |
| Netherlands            | 0.44  | 0.15, 0.74          | 0.003   |
| New Zealand            | 0.10  | -0.27, 0.48         | 0.6     |
| Norway                 | 0.91  | 0.47, 1.4           | <0.001  |
| Poland                 | 1.1   | 0.52, 1.7           | <0.001  |
| Portugal               | 0.55  | -0.02, 1.1          | 0.057   |
| Romania                | -0.05 | -0.57, 0.47         | 0.9     |
| Russia                 | 0.34  | -0.03, 0.72         | 0.073   |
| Singapore              | -0.53 | -0.90, -0.16        | 0.005   |
| Slovakia               | 1.5   | 0.77, 2.1           | <0.001  |
| South Africa           | -0.23 | -0.90, 0.44         | 0.5     |
| South Korea            | -0.64 | -1.5, 0.17          | 0.12    |
| Spain                  | 0.50  | 0.17, 0.83          | 0.003   |
| Sweden                 | 0.90  | 0.64, 1.2           | <0.001  |
| Switzerland            | 1.0   | 0.74, 1.3           | <0.001  |

| Characteristic               | B     | 95% CI <sup>1</sup> | p-value |
|------------------------------|-------|---------------------|---------|
| Thailand                     | -0.36 | -0.90, 0.17         | 0.2     |
| United Arab Emirates         | 0.03  | -0.54, 0.60         | >0.9    |
| United Kingdom               | 0.26  | 0.07, 0.45          | 0.008   |
| United States                | 0.03  | -0.13, 0.18         | 0.7     |
| <b>Sex x Age<sup>2</sup></b> |       |                     |         |
| Male * Age                   | 0.03  | 0.03, 0.03          | <0.001  |
| <b>Sex * Age<sup>2</sup></b> |       |                     |         |
| Male * Age <sup>2</sup>      | 0.00  | 0.00, 0.00          | <0.001  |
| <b>Sex * DayType</b>         |       |                     |         |
| Male * Weekend               | 0.08  | 0.03, 0.14          | 0.002   |
| <b>Sex * Country</b>         |       |                     |         |
| Male * Austria               | 0.43  | 0.03, 0.82          | 0.035   |
| Male * Belgium               | -0.10 | -0.56, 0.36         | 0.7     |
| Male * Canada                | -0.01 | -0.20, 0.18         | >0.9    |
| Male * Czech Republic        | 0.42  | 0.03, 0.82          | 0.037   |
| Male * Denmark               | 0.29  | -0.10, 0.67         | 0.14    |
| Male * Estonia               | -0.31 | -0.80, 0.19         | 0.2     |
| Male * Finland               | 0.11  | -0.05, 0.28         | 0.2     |
| Male * France                | -0.22 | -0.56, 0.13         | 0.2     |
| Male * Germany               | 0.09  | -0.11, 0.29         | 0.4     |
| Male * Greece                | -0.35 | -0.97, 0.27         | 0.3     |
| Male * Hong Kong             | -0.34 | -0.78, 0.10         | 0.13    |
| Male * India                 | 0.60  | -0.06, 1.3          | 0.074   |
| Male * Ireland               | -0.31 | -0.80, 0.18         | 0.2     |
| Male * Italy                 | -0.32 | -0.66, 0.02         | 0.064   |
| Male * Japan                 | 0.11  | -0.10, 0.32         | 0.3     |
| Male * Mexico                | -0.32 | -0.76, 0.12         | 0.2     |
| Male * Netherlands           | 0.00  | -0.30, 0.29         | >0.9    |
| Male * New Zealand           | 0.03  | -0.35, 0.41         | 0.9     |
| Male * Norway                | -0.34 | -0.79, 0.10         | 0.13    |
| Male * Poland                | -0.13 | -0.67, 0.41         | 0.6     |
| Male * Portugal              | -0.27 | -0.85, 0.30         | 0.3     |
| Male * Romania               | 0.15  | -0.37, 0.67         | 0.6     |
| Male * Russia                | 0.18  | -0.18, 0.55         | 0.3     |
| Male * Singapore             | -0.25 | -0.62, 0.11         | 0.2     |
| Male * Slovakia              | 0.20  | -0.46, 0.86         | 0.5     |
| Male * South Africa          | -0.19 | -0.85, 0.48         | 0.6     |
| Male * South Korea           | -0.35 | -1.1, 0.42          | 0.4     |
| Male * Spain                 | -0.21 | -0.54, 0.12         | 0.2     |
| Male * SWeekendden           | -0.14 | -0.41, 0.12         | 0.3     |
| Male * Switzerland           | -0.15 | -0.45, 0.14         | 0.3     |
| Male * Thailand              | -0.43 | -0.96, 0.10         | 0.11    |
| Male * United Arab Emirates  | -0.48 | -1.0, 0.08          | 0.094   |
| Male * United Kingdom        | -0.08 | -0.28, 0.11         | 0.4     |
| Male * United States         | 0.08  | -0.08, 0.23         | 0.3     |
| <b>Age * Age<sup>2</sup></b> | 0.00  | 0.00, 0.00          | <0.001  |
| <b>Age * DayType</b>         |       |                     |         |
| Age * Weekend                | 0.00  | 0.00, 0.00          | 0.8     |
| <b>Age * Country</b>         |       |                     |         |
| Age * Austria                | 0.01  | 0.00, 0.03          | 0.2     |
| Age * Belgium                | 0.01  | 0.00, 0.03          | 0.12    |
| Age * Canada                 | -0.01 | -0.01, 0.00         | 0.15    |
| Age * Czech Republic         | -0.01 | -0.03, 0.01         | 0.5     |
| Age * Denmark                | 0.01  | 0.00, 0.03          | 0.087   |
| Age * Estonia                | -0.03 | -0.05, 0.00         | 0.031   |
| Age * Finland                | 0.00  | -0.01, 0.00         | 0.3     |
| Age * France                 | 0.01  | -0.01, 0.02         | 0.3     |
| Age * Germany                | 0.00  | -0.01, 0.01         | 0.6     |
| Age * Greece                 | 0.01  | -0.02, 0.04         | 0.5     |
| Age * Hong Kong              | 0.00  | -0.02, 0.02         | 0.7     |
| Age * India                  | 0.02  | 0.00, 0.04          | 0.072   |
| Age * Ireland                | 0.01  | -0.01, 0.03         | 0.3     |

| Characteristic                          | B     | 95% CI <sup>1</sup> | p-value |
|-----------------------------------------|-------|---------------------|---------|
| Age * Italy                             | -0.02 | -0.03, 0.00         | 0.012   |
| Age * Japan                             | 0.00  | -0.01, 0.01         | 0.6     |
| Age * Mexico                            | 0.00  | -0.02, 0.01         | 0.6     |
| Age * Netherlands                       | 0.00  | -0.01, 0.01         | >0.9    |
| Age * New Zealand                       | 0.00  | -0.01, 0.02         | 0.8     |
| Age * Norway                            | 0.01  | -0.01, 0.03         | 0.5     |
| Age * Poland                            | 0.00  | -0.03, 0.03         | >0.9    |
| Age * Portugal                          | 0.00  | -0.03, 0.02         | 0.7     |
| Age * Romania                           | 0.00  | -0.03, 0.03         | >0.9    |
| Age * Russia                            | -0.01 | -0.03, 0.01         | 0.4     |
| Age * Singapore                         | 0.01  | -0.01, 0.02         | 0.5     |
| Age * Slovakia                          | -0.01 | -0.04, 0.02         | 0.6     |
| Age * South Africa                      | 0.00  | -0.03, 0.03         | 0.9     |
| Age * South Korea                       | -0.03 | -0.06, 0.00         | 0.044   |
| Age * Spain                             | -0.01 | -0.02, 0.01         | 0.3     |
| Age * Sweden                            | 0.00  | -0.01, 0.01         | 0.6     |
| Age * Switzerland                       | -0.01 | -0.02, 0.00         | 0.2     |
| Age * Thailand                          | -0.04 | -0.07, -0.02        | <0.001  |
| Age * United Arab Emirates              | -0.02 | -0.05, 0.00         | 0.10    |
| Age * United Kingdom                    | 0.01  | 0.00, 0.01          | 0.10    |
| Age * United States                     | 0.00  | -0.01, 0.00         | 0.4     |
| <b>Age<sup>2</sup> * DayType</b>        |       |                     |         |
| Age <sup>2</sup> * Weekend              | 0.00  | 0.00, 0.00          | <0.001  |
| <b>Age<sup>2</sup> * Country</b>        |       |                     |         |
| Age <sup>2</sup> * Austria              | 0.00  | 0.00, 0.00          | 0.4     |
| Age <sup>2</sup> * Belgium              | 0.00  | 0.00, 0.00          | 0.031   |
| Age <sup>2</sup> * Canada               | 0.00  | 0.00, 0.00          | 0.017   |
| Age <sup>2</sup> * Czech Republic       | 0.00  | 0.00, 0.00          | 0.9     |
| Age <sup>2</sup> * Denmark              | 0.00  | 0.00, 0.00          | <0.001  |
| Age <sup>2</sup> * Estonia              | 0.00  | 0.00, 0.00          | 0.4     |
| Age <sup>2</sup> * Finland              | 0.00  | 0.00, 0.00          | <0.001  |
| Age <sup>2</sup> * France               | 0.00  | 0.00, 0.00          | >0.9    |
| Age <sup>2</sup> * Germany              | 0.00  | 0.00, 0.00          | 0.8     |
| Age <sup>2</sup> * Greece               | 0.00  | 0.00, 0.00          | 0.045   |
| Age <sup>2</sup> * Hong Kong            | 0.00  | 0.00, 0.00          | 0.7     |
| Age <sup>2</sup> * India                | 0.00  | 0.00, 0.00          | 0.9     |
| Age <sup>2</sup> * Ireland              | 0.00  | 0.00, 0.00          | 0.2     |
| Age <sup>2</sup> * Italy                | 0.00  | 0.00, 0.00          | <0.001  |
| Age <sup>2</sup> * Japan                | 0.00  | 0.00, 0.00          | <0.001  |
| Age <sup>2</sup> * Mexico               | 0.00  | 0.00, 0.00          | 0.6     |
| Age <sup>2</sup> * Netherlands          | 0.00  | 0.00, 0.00          | 0.3     |
| Age <sup>2</sup> * New Zealand          | 0.00  | 0.00, 0.00          | >0.9    |
| Age <sup>2</sup> * Norway               | 0.00  | 0.00, 0.00          | 0.6     |
| Age <sup>2</sup> * Poland               | 0.00  | 0.00, 0.00          | 0.2     |
| Age <sup>2</sup> * Portugal             | 0.00  | 0.00, 0.00          | 0.2     |
| Age <sup>2</sup> * Romania              | 0.00  | 0.00, 0.00          | 0.12    |
| Age <sup>2</sup> * Russia               | 0.00  | 0.00, 0.00          | 0.8     |
| Age <sup>2</sup> * Singapore            | 0.00  | 0.00, 0.00          | 0.13    |
| Age <sup>2</sup> * Slovakia             | 0.00  | 0.00, 0.00          | 0.7     |
| Age <sup>2</sup> * South Africa         | 0.00  | 0.00, 0.00          | 0.4     |
| Age <sup>2</sup> * South Korea          | 0.00  | 0.00, 0.00          | >0.9    |
| Age <sup>2</sup> * Spain                | 0.00  | 0.00, 0.00          | >0.9    |
| Age <sup>2</sup> * SWeekendden          | 0.00  | 0.00, 0.00          | 0.8     |
| Age <sup>2</sup> * Switzerland          | 0.00  | 0.00, 0.00          | 0.3     |
| Age <sup>2</sup> * Thailand             | 0.00  | 0.00, 0.00          | 0.001   |
| Age <sup>2</sup> * United Arab Emirates | 0.00  | 0.00, 0.00          | 0.5     |
| Age <sup>2</sup> * United Kingdom       | 0.00  | 0.00, 0.00          | 0.7     |
| Age <sup>2</sup> * United States        | 0.00  | 0.00, 0.00          | 0.13    |
| <b>DayType * Country</b>                |       |                     |         |
| Weekend * Austria                       | 0.08  | -0.29, 0.45         | 0.7     |
| Weekend * Belgium                       | 0.01  | -0.41, 0.44         | >0.9    |
| Weekend * Canada                        | 0.08  | -0.10, 0.27         | 0.4     |

| Characteristic                 | B     | 95% CI <sup>1</sup> | p-value |
|--------------------------------|-------|---------------------|---------|
| Weekend * Czech Republic       | 0.11  | -0.22, 0.45         | 0.5     |
| Weekend * Denmark              | -0.07 | -0.42, 0.28         | 0.7     |
| Weekend * Estonia              | 0.00  | -0.49, 0.48         | >0.9    |
| Weekend * Finland              | -0.04 | -0.21, 0.12         | 0.6     |
| Weekend * France               | 0.10  | -0.22, 0.42         | 0.5     |
| Weekend * Germany              | 0.04  | -0.15, 0.23         | 0.7     |
| Weekend * Greece               | 0.15  | -0.43, 0.73         | 0.6     |
| Weekend * Hong Kong            | 0.12  | -0.30, 0.55         | 0.6     |
| Weekend * India                | 0.25  | -0.30, 0.81         | 0.4     |
| Weekend * Ireland              | 0.00  | -0.45, 0.44         | >0.9    |
| Weekend * Italy                | -0.01 | -0.32, 0.30         | >0.9    |
| Weekend * Japan                | -0.07 | -0.26, 0.11         | 0.4     |
| Weekend * Mexico               | 0.08  | -0.33, 0.49         | 0.7     |
| Weekend * Netherlands          | 0.01  | -0.27, 0.29         | >0.9    |
| Weekend * New Zealand          | 0.00  | -0.37, 0.37         | >0.9    |
| Weekend * Norway               | 0.03  | -0.39, 0.45         | >0.9    |
| Weekend * Poland               | 0.03  | -0.44, 0.51         | 0.9     |
| Weekend * Portugal             | 0.09  | -0.45, 0.63         | 0.7     |
| Weekend * Romania              | 0.11  | -0.39, 0.60         | 0.7     |
| Weekend * Russia               | 0.22  | -0.13, 0.57         | 0.2     |
| Weekend * Singapore            | 0.27  | -0.08, 0.61         | 0.13    |
| Weekend * Slovakia             | 0.05  | -0.48, 0.57         | 0.9     |
| Weekend * South Africa         | 0.06  | -0.55, 0.68         | 0.8     |
| Weekend * South Korea          | -0.20 | -0.90, 0.49         | 0.6     |
| Weekend * Spain                | -0.01 | -0.32, 0.30         | >0.9    |
| Weekend * Sweden               | -0.02 | -0.27, 0.23         | 0.9     |
| Weekend * Switzerland          | 0.03  | -0.26, 0.31         | 0.9     |
| Weekend * Thailand             | 0.14  | -0.36, 0.64         | 0.6     |
| Weekend * United Arab Emirates | 0.09  | -0.43, 0.61         | 0.7     |
| Weekend * United Kingdom       | 0.03  | -0.16, 0.21         | 0.8     |
| Weekend * United States        | 0.12  | -0.03, 0.27         | 0.11    |

<sup>1</sup>CI = Confidence Interval

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Table S3.2b. ANOVA table with effect sizes - SE

| Factor                     | Sum of squares | Df      | F value    | p     | $\eta_p^2$ |
|----------------------------|----------------|---------|------------|-------|------------|
| Sex                        | 552,775.59     | 1       | 29,039.035 | 0.000 | 0.060      |
| Age                        | 191.23         | 1       | 10.046     | 0.002 | 0.000      |
| Age <sup>2</sup>           | 133,125.82     | 1       | 6,993.517  | 0.000 | 0.015      |
| DayType                    | 9,756.72       | 1       | 512.551    | 0.000 | 0.001      |
| Country                    | 137,592.45     | 34      | 212.593    | 0.000 | 0.016      |
| Sex x Age                  | 13,823.11      | 1       | 726.171    | 0.000 | 0.002      |
| Sex x Age <sup>2</sup>     | 3,231.75       | 1       | 169.774    | 0.000 | 0.000      |
| Sex x DayType              | 181.90         | 1       | 9.556      | 0.002 | 0.000      |
| Sex x Country              | 1,398.36       | 34      | 2.161      | 0.000 | 0.000      |
| Age x Age <sup>2</sup>     | 939.81         | 1       | 49.371     | 0.000 | 0.000      |
| Age x DayType              | 0.85           | 1       | 0.045      | 0.833 | 0.000      |
| Age x Country              | 1,454.54       | 34      | 2.247      | 0.000 | 0.000      |
| Age <sup>2</sup> x DayType | 499.74         | 1       | 26.253     | 0.000 | 0.000      |
| Age <sup>2</sup> x Country | 5,175.31       | 34      | 7.996      | 0.000 | 0.001      |
| DayType x Country          | 593.36         | 34      | 0.917      | 0.607 | 0.000      |
| Residuals                  | 8,607,767.14   | 452,193 |            |       |            |

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1137 3. Sleep Efficiency (accounting for TST)  
 1138  
 1139 Model including Total Sleep Time (TST)  
 1140  
 1141  $SE \sim 1 + TST + sex + age + age^2 + daytype + country + \dots$   
 1142  $TST:sex + TST:age + TST:age^2 + TST:daytype + TST:country + \dots$   
 1143  $sex:age + sex:age^2 + sex:daytype + sex:country + \dots$   
 1144  $age:age^2 + age:daytype + age:country + \dots$   
 1145  $age^2:daytype + age^2:country + daytype:country$   
 1146  
 1147  
 1148

**Table S3.3a.** Unstandardised Regression Coefficients for SE (with TST)

| Characteristic               | B     | 95% CI <sup>1</sup> | p-value |
|------------------------------|-------|---------------------|---------|
| <b>TST</b>                   | 0.04  | 0.04, 0.05          | <0.001  |
| <b>Sex</b>                   |       |                     |         |
| Female                       | —     | —                   |         |
| Male                         | -6.9  | -7.1, -6.6          | <0.001  |
| <b>Age</b>                   | -0.02 | -0.03, -0.01        | <0.001  |
| <b>Age<sup>2</sup></b>       | -0.01 | -0.01, -0.01        | <0.001  |
| <b>DayType</b>               |       |                     |         |
| Weekday                      | —     | —                   |         |
| Weekend                      | -0.60 | -0.86, -0.34        | <0.001  |
| <b>Country</b>               |       |                     |         |
| Australia                    | —     | —                   |         |
| Austria                      | 3.6   | 1.8, 5.4            | <0.001  |
| Belgium                      | 3.3   | 1.3, 5.3            | 0.001   |
| Canada                       | 2.0   | 1.2, 2.8            | <0.001  |
| Czech Republic               | 3.0   | 1.4, 4.7            | <0.001  |
| Denmark                      | -1.5  | -3.1, 0.21          | 0.087   |
| Estonia                      | 0.75  | -1.5, 3.0           | 0.5     |
| Finland                      | 3.6   | 2.8, 4.4            | <0.001  |
| France                       | 4.2   | 2.7, 5.7            | <0.001  |
| Germany                      | 3.7   | 2.8, 4.6            | <0.001  |
| Greece                       | 2.3   | -0.45, 5.1          | 0.10    |
| Hong Kong                    | 5.8   | 4.0, 7.6            | <0.001  |
| India                        | -1.5  | -3.9, 0.81          | 0.2     |
| Ireland                      | 3.7   | 1.5, 5.9            | <0.001  |
| Italy                        | 2.5   | 1.0, 4.0            | <0.001  |
| Japan                        | 1.9   | 1.1, 2.7            | <0.001  |
| Mexico                       | 4.5   | 2.7, 6.4            | <0.001  |
| Netherlands                  | 3.2   | 1.9, 4.6            | <0.001  |
| New Zealand                  | -0.85 | -2.6, 0.86          | 0.3     |
| Norway                       | 8.2   | 6.2, 10             | <0.001  |
| Poland                       | 5.5   | 3.1, 7.8            | <0.001  |
| Portugal                     | 3.2   | 0.71, 5.7           | 0.012   |
| Romania                      | 1.6   | -0.73, 4.0          | 0.2     |
| Russia                       | 3.9   | 2.1, 5.6            | <0.001  |
| Singapore                    | 2.2   | 0.81, 3.6           | 0.002   |
| Slovakia                     | 6.3   | 3.5, 9.0            | <0.001  |
| South Africa                 | -2.5  | -5.6, 0.53          | 0.10    |
| South Korea                  | 2.6   | -0.13, 5.4          | 0.062   |
| Spain                        | 2.8   | 1.3, 4.3            | <0.001  |
| Sweden                       | 5.0   | 3.8, 6.2            | <0.001  |
| Switzerland                  | 3.2   | 1.9, 4.6            | <0.001  |
| Thailand                     | 6.1   | 3.9, 8.2            | <0.001  |
| United Arab Emirates         | 4.5   | 2.2, 6.8            | <0.001  |
| United Kingdom               | 0.53  | -0.33, 1.4          | 0.2     |
| United States                | 2.6   | 1.9, 3.3            | <0.001  |
| <b>TST * Sex</b>             |       |                     |         |
| TST * Male                   | 0.01  | 0.01, 0.01          | <0.001  |
| <b>TST * Age</b>             | 0.00  | 0.00, 0.00          | <0.001  |
| <b>TST * Age<sup>2</sup></b> | 0.00  | 0.00, 0.00          | <0.001  |
| <b>TST * DayType</b>         |       |                     |         |
| TST * Weekend                | 0.00  | 0.00, 0.00          | <0.001  |

| Characteristic               | B     | 95% CI <sup>1</sup> | p-value |
|------------------------------|-------|---------------------|---------|
| <b>TST * Country</b>         |       |                     |         |
| TST * Austria                | -0.01 | -0.01, 0.00         | 0.001   |
| TST * Belgium                | -0.01 | -0.01, 0.00         | 0.005   |
| TST * Canada                 | 0.00  | -0.01, 0.00         | <0.001  |
| TST * Czech Republic         | 0.00  | -0.01, 0.00         | 0.029   |
| TST * Denmark                | 0.00  | 0.00, 0.01          | 0.032   |
| TST * Estonia                | 0.00  | 0.00, 0.01          | 0.9     |
| TST * Finland                | -0.01 | -0.01, 0.00         | <0.001  |
| TST * France                 | -0.01 | -0.01, 0.00         | <0.001  |
| TST * Germany                | -0.01 | -0.01, 0.00         | <0.001  |
| TST * Greece                 | 0.00  | -0.01, 0.00         | 0.2     |
| TST * Hong Kong              | -0.01 | -0.02, -0.01        | <0.001  |
| TST * India                  | 0.00  | 0.00, 0.01          | 0.2     |
| TST * Ireland                | -0.01 | -0.01, 0.00         | 0.002   |
| TST * Italy                  | 0.00  | -0.01, 0.00         | 0.11    |
| TST * Japan                  | 0.00  | -0.01, 0.00         | <0.001  |
| TST * Mexico                 | -0.01 | -0.01, -0.01        | <0.001  |
| TST * Netherlands            | -0.01 | -0.01, 0.00         | <0.001  |
| TST * New Zealand            | 0.00  | 0.00, 0.01          | 0.3     |
| TST * Norway                 | -0.02 | -0.02, -0.01        | <0.001  |
| TST * Poland                 | -0.01 | -0.02, 0.00         | <0.001  |
| TST * Portugal               | -0.01 | -0.01, 0.00         | 0.031   |
| TST * Romania                | 0.00  | -0.01, 0.00         | 0.3     |
| TST * Russia                 | -0.01 | -0.01, 0.00         | <0.001  |
| TST * Singapore              | 0.00  | -0.01, 0.00         | 0.035   |
| TST * Slovakia               | -0.01 | -0.02, -0.01        | <0.001  |
| TST * South Africa           | 0.01  | 0.00, 0.01          | 0.14    |
| TST * South Korea            | 0.00  | -0.01, 0.00         | 0.2     |
| TST * Spain                  | 0.00  | -0.01, 0.00         | 0.006   |
| TST * Sweden                 | -0.01 | -0.01, -0.01        | <0.001  |
| TST * Switzerland            | 0.00  | -0.01, 0.00         | 0.002   |
| TST * Thailand               | -0.01 | -0.02, -0.01        | <0.001  |
| TST * United Arab Emirates   | -0.01 | -0.01, 0.00         | 0.002   |
| TST * United Kingdom         | 0.00  | 0.00, 0.00          | 0.6     |
| TST * United States          | -0.01 | -0.01, 0.00         | <0.001  |
| <b>Sex * Age</b>             |       |                     |         |
| Male * Age                   | 0.02  | 0.02, 0.02          | <0.001  |
| <b>Sex * Age<sup>2</sup></b> |       |                     |         |
| Male * Age <sup>2</sup>      | 0.00  | 0.00, 0.00          | <0.001  |
| <b>Sex * DayType</b>         |       |                     |         |
| Male * Weekend               | -0.20 | -0.25, -0.15        | <0.001  |
| <b>Sex * Country</b>         |       |                     |         |
| Male * Austria               | 0.25  | -0.11, 0.61         | 0.2     |
| Male * Belgium               | -0.21 | -0.63, 0.20         | 0.3     |
| Male * Canada                | 0.03  | -0.14, 0.20         | 0.7     |
| Male * Czech Republic        | 0.15  | -0.21, 0.50         | 0.4     |
| Male * Denmark               | 0.35  | 0.00, 0.70          | 0.050   |
| Male * Estonia               | -0.43 | -0.88, 0.01         | 0.057   |
| Male * Finland               | -0.14 | -0.29, 0.01         | 0.072   |
| Male * France                | -0.36 | -0.68, -0.05        | 0.023   |
| Male * Germany               | -0.10 | -0.28, 0.08         | 0.3     |
| Male * Greece                | -0.04 | -0.61, 0.54         | >0.9    |
| Male * Hong Kong             | -0.30 | -0.70, 0.10         | 0.14    |
| Male * India                 | 1.1   | 0.50, 1.7           | <0.001  |
| Male * Ireland               | -0.45 | -0.90, -0.01        | 0.047   |
| Male * Italy                 | -0.16 | -0.47, 0.14         | 0.3     |
| Male * Japan                 | 0.55  | 0.36, 0.74          | <0.001  |
| Male * Mexico                | -0.12 | -0.53, 0.29         | 0.6     |
| Male * Netherlands           | -0.21 | -0.48, 0.06         | 0.13    |
| Male * New Zealand           | -0.23 | -0.57, 0.11         | 0.2     |
| Male * Norway                | -0.81 | -1.2, -0.41         | <0.001  |
| Male * Poland                | -0.16 | -0.66, 0.34         | 0.5     |
| Male * Portugal              | -0.03 | -0.56, 0.50         | >0.9    |
| Male * Romania               | 0.36  | -0.12, 0.84         | 0.14    |
| Male * Russia                | -0.16 | -0.49, 0.17         | 0.3     |
| Male * Singapore             | -0.04 | -0.37, 0.29         | 0.8     |
| Male * Slovakia              | -0.09 | -0.69, 0.51         | 0.8     |
| Male * South Africa          | 0.14  | -0.46, 0.75         | 0.6     |
| Male * South Korea           | -0.06 | -0.74, 0.62         | 0.9     |
| Male * Spain                 | -0.19 | -0.49, 0.11         | 0.2     |

| Characteristic                    | B     | 95% CI <sup>1</sup> | p-value |
|-----------------------------------|-------|---------------------|---------|
| Male * Sweden                     | -0.42 | -0.65, -0.18        | <0.001  |
| Male * Switzerland                | -0.15 | -0.42, 0.12         | 0.3     |
| Male * Thailand                   | -0.68 | -1.2, -0.21         | 0.005   |
| Male * United Arab Emirates       | -0.32 | -0.83, 0.19         | 0.2     |
| Male * United Kingdom             | -0.14 | -0.32, 0.03         | 0.11    |
| Male * United States              | 0.11  | -0.03, 0.25         | 0.12    |
| Age * Age <sup>2</sup>            | 0.00  | 0.00, 0.00          | 0.002   |
| Age * DayType                     |       |                     |         |
| Age * Weekend                     | -0.01 | -0.01, -0.01        | <0.001  |
| Age * Country                     |       |                     |         |
| Age * Austria                     | 0.01  | 0.00, 0.02          | 0.13    |
| Age * Belgium                     | 0.02  | 0.00, 0.03          | 0.032   |
| Age * Canada                      | -0.01 | -0.01, 0.00         | 0.069   |
| Age * Czech Republic              | 0.01  | 0.00, 0.03          | 0.11    |
| Age * Denmark                     | 0.02  | 0.01, 0.04          | <0.001  |
| Age * Estonia                     | -0.01 | -0.04, 0.01         | 0.3     |
| Age * Finland                     | 0.00  | 0.00, 0.01          | 0.7     |
| Age * France                      | 0.01  | 0.00, 0.02          | 0.060   |
| Age * Germany                     | 0.01  | 0.00, 0.01          | 0.12    |
| Age * Greece                      | 0.02  | -0.01, 0.04         | 0.14    |
| Age * Hong Kong                   | 0.00  | -0.02, 0.01         | 0.6     |
| Age * India                       | 0.04  | 0.02, 0.06          | <0.001  |
| Age * Ireland                     | 0.01  | -0.01, 0.03         | 0.3     |
| Age * Italy                       | 0.00  | -0.01, 0.01         | 0.8     |
| Age * Japan                       | 0.03  | 0.02, 0.03          | <0.001  |
| Age * Mexico                      | -0.01 | -0.02, 0.01         | 0.3     |
| Age * Netherlands                 | 0.01  | 0.00, 0.02          | 0.11    |
| Age * New Zealand                 | 0.00  | -0.01, 0.01         | >0.9    |
| Age * Norway                      | 0.02  | 0.00, 0.04          | 0.021   |
| Age * Poland                      | 0.01  | -0.01, 0.04         | 0.3     |
| Age * Portugal                    | 0.01  | -0.01, 0.03         | 0.4     |
| Age * Romania                     | 0.03  | 0.00, 0.05          | 0.025   |
| Age * Russia                      | 0.00  | -0.02, 0.01         | 0.8     |
| Age * Singapore                   | 0.01  | 0.00, 0.03          | 0.2     |
| Age * Slovakia                    | -0.02 | -0.04, 0.01         | 0.3     |
| Age * South Africa                | 0.02  | -0.01, 0.04         | 0.3     |
| Age * South Korea                 | -0.02 | -0.05, 0.00         | 0.090   |
| Age * Spain                       | 0.00  | -0.01, 0.02         | 0.4     |
| Age * Sweden                      | 0.01  | 0.00, 0.02          | 0.003   |
| Age * Switzerland                 | 0.00  | -0.01, 0.01         | 0.7     |
| Age * Thailand                    | -0.03 | -0.05, -0.01        | 0.004   |
| Age * United Arab Emirates        | 0.00  | -0.02, 0.02         | >0.9    |
| Age * United Kingdom              | 0.01  | 0.00, 0.01          | 0.029   |
| Age * United States               | 0.00  | -0.01, 0.01         | >0.9    |
| Age <sup>2</sup> * DayType        |       |                     |         |
| Age <sup>2</sup> * Weekend        | 0.00  | 0.00, 0.00          | <0.001  |
| Age <sup>2</sup> * Country        |       |                     |         |
| Age <sup>2</sup> * Austria        | 0.00  | 0.00, 0.00          | 0.3     |
| Age <sup>2</sup> * Belgium        | 0.00  | 0.00, 0.00          | 0.12    |
| Age <sup>2</sup> * Canada         | 0.00  | 0.00, 0.00          | 0.033   |
| Age <sup>2</sup> * Czech Republic | 0.00  | 0.00, 0.00          | 0.7     |
| Age <sup>2</sup> * Denmark        | 0.00  | 0.00, 0.00          | 0.010   |
| Age <sup>2</sup> * Estonia        | 0.00  | 0.00, 0.00          | 0.7     |
| Age <sup>2</sup> * Finland        | 0.00  | 0.00, 0.00          | <0.001  |
| Age <sup>2</sup> * France         | 0.00  | 0.00, 0.00          | 0.3     |
| Age <sup>2</sup> * Germany        | 0.00  | 0.00, 0.00          | 0.6     |
| Age <sup>2</sup> * Greece         | 0.00  | 0.00, 0.00          | 0.012   |
| Age <sup>2</sup> * Hong Kong      | 0.00  | 0.00, 0.00          | 0.7     |
| Age <sup>2</sup> * India          | 0.00  | 0.00, 0.00          | 0.002   |
| Age <sup>2</sup> * Ireland        | 0.00  | 0.00, 0.00          | 0.2     |
| Age <sup>2</sup> * Italy          | 0.00  | 0.00, 0.00          | <0.001  |
| Age <sup>2</sup> * Japan          | 0.00  | 0.00, 0.00          | <0.001  |
| Age <sup>2</sup> * Mexico         | 0.00  | 0.00, 0.00          | 0.3     |
| Age <sup>2</sup> * Netherlands    | 0.00  | 0.00, 0.00          | 0.6     |
| Age <sup>2</sup> * New Zealand    | 0.00  | 0.00, 0.00          | >0.9    |
| Age <sup>2</sup> * Norway         | 0.00  | 0.00, 0.00          | 0.5     |
| Age <sup>2</sup> * Poland         | 0.00  | 0.00, 0.00          | 0.2     |
| Age <sup>2</sup> * Portugal       | 0.00  | 0.00, 0.00          | 0.7     |
| Age <sup>2</sup> * Romania        | 0.00  | 0.00, 0.00          | 0.10    |
| Age <sup>2</sup> * Russia         | 0.00  | 0.00, 0.00          | 0.8     |



| Characteristic                          | B     | 95% CI <sup>1</sup> | p-value |
|-----------------------------------------|-------|---------------------|---------|
| Age <sup>2</sup> * Singapore            | 0.00  | 0.00, 0.00          | 0.4     |
| Age <sup>2</sup> * Slovakia             | 0.00  | 0.00, 0.00          | 0.7     |
| Age <sup>2</sup> * South Africa         | 0.00  | 0.00, 0.00          | 0.3     |
| Age <sup>2</sup> * South Korea          | 0.00  | 0.00, 0.00          | 0.016   |
| Age <sup>2</sup> * Spain                | 0.00  | 0.00, 0.00          | 0.9     |
| Age <sup>2</sup> * Sweden               | 0.00  | 0.00, 0.00          | 0.8     |
| Age <sup>2</sup> * Switzerland          | 0.00  | 0.00, 0.00          | <0.001  |
| Age <sup>2</sup> * Thailand             | 0.00  | 0.00, 0.00          | <0.001  |
| Age <sup>2</sup> * United Arab Emirates | 0.00  | 0.00, 0.00          | 0.6     |
| Age <sup>2</sup> * United Kingdom       | 0.00  | 0.00, 0.00          | 0.3     |
| Age <sup>2</sup> * United States        | 0.00  | 0.00, 0.00          | 0.009   |
| <b>DayType * Country</b>                |       |                     |         |
| Weekend * Austria                       | -0.38 | -0.71, -0.05        | 0.023   |
| Weekend * Belgium                       | -0.28 | -0.66, 0.10         | 0.2     |
| Weekend * Canada                        | -0.05 | -0.21, 0.11         | 0.6     |
| Weekend * Czech Republic                | -0.40 | -0.70, -0.10        | 0.010   |
| Weekend * Denmark                       | -0.43 | -0.73, -0.12        | 0.007   |
| Weekend * Estonia                       | -0.64 | -1.1, -0.20         | 0.004   |
| Weekend * Finland                       | -0.58 | -0.73, -0.44        | <0.001  |
| Weekend * France                        | -0.23 | -0.51, 0.05         | 0.11    |
| Weekend * Germany                       | -0.45 | -0.62, -0.28        | <0.001  |
| Weekend * Greece                        | 0.00  | -0.52, 0.51         | >0.9    |
| Weekend * Hong Kong                     | -0.01 | -0.39, 0.36         | >0.9    |
| Weekend * India                         | 0.64  | 0.16, 1.1           | 0.009   |
| Weekend * Ireland                       | 0.03  | -0.36, 0.41         | 0.9     |
| Weekend * Italy                         | -0.45 | -0.73, -0.18        | 0.001   |
| Weekend * Japan                         | -0.54 | -0.70, -0.37        | <0.001  |
| Weekend * Mexico                        | 0.20  | -0.16, 0.55         | 0.3     |
| Weekend * Netherlands                   | -0.12 | -0.37, 0.13         | 0.3     |
| Weekend * New Zealand                   | -0.25 | -0.58, 0.08         | 0.13    |
| Weekend * Norway                        | -0.33 | -0.71, 0.05         | 0.088   |
| Weekend * Poland                        | -0.09 | -0.52, 0.33         | 0.7     |
| Weekend * Portugal                      | -0.07 | -0.54, 0.41         | 0.8     |
| Weekend * Romania                       | -0.16 | -0.60, 0.28         | 0.5     |
| Weekend * Russia                        | 0.03  | -0.28, 0.34         | 0.9     |
| Weekend * Singapore                     | 0.12  | -0.19, 0.42         | 0.5     |
| Weekend * Slovakia                      | -0.24 | -0.72, 0.23         | 0.3     |
| Weekend * South Africa                  | -0.14 | -0.69, 0.40         | 0.6     |
| Weekend * South Korea                   | -0.59 | -1.2, 0.02          | 0.059   |
| Weekend * Spain                         | -0.24 | -0.52, 0.03         | 0.082   |
| Weekend * Sweden                        | -0.32 | -0.55, -0.10        | 0.005   |
| Weekend * Switzerland                   | -0.60 | -0.86, -0.34        | <0.001  |
| Weekend * Thailand                      | 0.42  | -0.01, 0.86         | 0.057   |
| Weekend * United Arab Emirates          | 0.31  | -0.15, 0.76         | 0.2     |
| Weekend * United Kingdom                | -0.13 | -0.29, 0.04         | 0.13    |
| Weekend * United States                 | -0.08 | -0.21, 0.05         | 0.3     |

<sup>1</sup>CI = Confidence Interval

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**Table S3.3b.** ANOVA table with effect sizes – SE (with TST)

| Factor                     | Sum of squares | Df      | F value     | p     | $\eta_p^2$ |
|----------------------------|----------------|---------|-------------|-------|------------|
| TST                        | 2,055,413.70   | 1       | 142,896.428 | 0.000 | 0.240      |
| Sex                        | 95,106.37      | 1       | 6,611.983   | 0.000 | 0.014      |
| Age                        | 2,810.54       | 1       | 195.394     | 0.000 | 0.000      |
| Age <sup>2</sup>           | 135,417.46     | 1       | 9,414.490   | 0.000 | 0.020      |
| DayType                    | 142,129.29     | 1       | 9,881.109   | 0.000 | 0.021      |
| Country                    | 19,888.96      | 34      | 40.668      | 0.000 | 0.003      |
| TST x Sex                  | 39,311.02      | 1       | 2,732.980   | 0.000 | 0.006      |
| TST x Age                  | 218.20         | 1       | 15.170      | 0.000 | 0.000      |
| TST x Age <sup>2</sup>     | 2,622.42       | 1       | 182.316     | 0.000 | 0.000      |
| TST x DayType              | 218.43         | 1       | 15.186      | 0.000 | 0.000      |
| TST x Country              | 3,893.03       | 34      | 7.960       | 0.000 | 0.001      |
| Sex x Age                  | 5,472.01       | 1       | 380.425     | 0.000 | 0.001      |
| Sex x Age <sup>2</sup>     | 4,523.66       | 1       | 314.494     | 0.000 | 0.001      |
| Sex x DayType              | 907.74         | 1       | 63.108      | 0.000 | 0.000      |
| Sex x Country              | 3,311.29       | 34      | 6.771       | 0.000 | 0.001      |
| Age x Age <sup>2</sup>     | 138.15         | 1       | 9.605       | 0.002 | 0.000      |
| Age x DayType              | 1,742.58       | 1       | 121.147     | 0.000 | 0.000      |
| Age x Country              | 2,915.08       | 34      | 5.961       | 0.000 | 0.000      |
| Age <sup>2</sup> x DayType | 5,485.59       | 1       | 381.369     | 0.000 | 0.001      |
| Age <sup>2</sup> x Country | 5,004.55       | 34      | 10.233      | 0.000 | 0.001      |
| DayType x Country          | 4,723.52       | 34      | 9.658       | 0.000 | 0.001      |
| Residuals                  | 6,503,756.17   | 452,154 |             |       |            |

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1156 4. Weekend Sleep Extension

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1158 Model includes weekday TST.

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1160 WESE ~ 1 + TSTwd + sex + age + age<sup>2</sup> + country + ...

1161 TSTwd:sex + TSTwd:age + TSTwd:age<sup>2</sup> + TSTwd:country + ...

1162 sex:age + sex:age<sup>2</sup> + sex:country + ...

1163 age:age<sup>2</sup> + age:country + age<sup>2</sup>:country

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1166 **Table S3.4a.** Unstandardised Regression Coefficients for Weekend sleep extension (WESE).

| Characteristic                 | B     | 95% CI <sup>1</sup> | p-value |
|--------------------------------|-------|---------------------|---------|
| <b>TSTwd</b>                   | -0.16 | -0.17, -0.14        | <0.001  |
| <b>Sex</b>                     |       |                     |         |
| Female                         | —     | —                   |         |
| Male                           | -11   | -14, -8.8           | <0.001  |
| <b>Age</b>                     | -1.7  | -1.8, -1.6          | <0.001  |
| <b>Age<sup>2</sup></b>         | -0.01 | -0.02, 0.00         | 0.006   |
| <b>Country</b>                 |       |                     |         |
| Australia                      | —     | —                   |         |
| Austria                        | 20    | 1.2, 38             | 0.037   |
| Belgium                        | 1.9   | -19, 22             | 0.9     |
| Canada                         | 8.5   | -0.11, 17           | 0.053   |
| Czech Republic                 | 29    | 12, 46              | <0.001  |
| Denmark                        | 7.4   | -9.8, 25            | 0.4     |
| Estonia                        | 24    | 0.67, 47            | 0.044   |
| Finland                        | 38    | 30, 46              | <0.001  |
| France                         | 7.9   | -7.6, 23            | 0.3     |
| Germany                        | 21    | 12, 30              | <0.001  |
| Greece                         | 3.5   | -25, 32             | 0.8     |
| Hong Kong                      | 15    | -3.6, 33            | 0.11    |
| India                          | -44   | -68, -20            | <0.001  |
| Ireland                        | -4.1  | -27, 19             | 0.7     |
| Italy                          | 12    | -3.0, 27            | 0.12    |
| Japan                          | 18    | 10, 27              | <0.001  |
| Mexico                         | 8.4   | -10, 27             | 0.4     |
| Netherlands                    | -6.5  | -20, 7.3            | 0.4     |
| New Zealand                    | -12   | -29, 6.3            | 0.2     |
| Norway                         | 32    | 12, 53              | 0.002   |
| Poland                         | 53    | 29, 77              | <0.001  |
| Portugal                       | 14    | -11, 39             | 0.3     |
| Romania                        | 17    | -7.2, 41            | 0.2     |
| Russia                         | 13    | -4.4, 30            | 0.14    |
| Singapore                      | 1.3   | -13, 16             | 0.9     |
| Slovakia                       | 9.0   | -19, 37             | 0.5     |
| South Africa                   | 8.0   | -23, 39             | 0.6     |
| South Korea                    | 37    | 8.7, 65             | 0.010   |
| Spain                          | 11    | -3.8, 27            | 0.14    |
| Sweden                         | 13    | 0.96, 26            | 0.035   |
| Switzerland                    | 35    | 21, 49              | <0.001  |
| Thailand                       | -12   | -34, 9.5            | 0.3     |
| United Arab Emirates           | 3.1   | -20, 27             | 0.8     |
| United Kingdom                 | 0.22  | -8.5, 8.9           | >0.9    |
| United States                  | 16    | 9.3, 23             | <0.001  |
| <b>TSTwd * Sex</b>             |       |                     |         |
| TSTwd * Male                   | 0.02  | 0.01, 0.02          | <0.001  |
| <b>TSTwd * Age</b>             | 0.00  | 0.00, 0.00          | <0.001  |
| <b>TSTwd * Age<sup>2</sup></b> | 0.00  | 0.00, 0.00          | 0.5     |
| <b>TSTwd * Country</b>         |       |                     |         |

| Characteristic               | B     | 95% CI <sup>1</sup> | p-value |
|------------------------------|-------|---------------------|---------|
| TSTwd * Austria              | -0.02 | -0.06, 0.03         | 0.5     |
| TSTwd * Belgium              | 0.01  | -0.03, 0.06         | 0.5     |
| TSTwd * Canada               | -0.01 | -0.03, 0.01         | 0.3     |
| TSTwd * Czech Republic       | -0.04 | -0.08, 0.00         | 0.070   |
| TSTwd * Denmark              | -0.01 | -0.05, 0.03         | 0.6     |
| TSTwd * Estonia              | -0.02 | -0.08, 0.03         | 0.4     |
| TSTwd * Finland              | -0.05 | -0.07, -0.03        | <0.001  |
| TSTwd * France               | 0.00  | -0.03, 0.04         | 0.9     |
| TSTwd * Germany              | -0.02 | -0.04, 0.00         | 0.087   |
| TSTwd * Greece               | 0.01  | -0.05, 0.08         | 0.7     |
| TSTwd * Hong Kong            | -0.03 | -0.07, 0.01         | 0.2     |
| TSTwd * India                | 0.08  | 0.02, 0.14          | 0.005   |
| TSTwd * Ireland              | 0.01  | -0.04, 0.06         | 0.7     |
| TSTwd * Italy                | -0.01 | -0.04, 0.03         | 0.6     |
| TSTwd * Japan                | -0.04 | -0.06, -0.02        | <0.001  |
| TSTwd * Mexico               | -0.02 | -0.06, 0.02         | 0.3     |
| TSTwd * Netherlands          | 0.03  | 0.00, 0.07          | 0.032   |
| TSTwd * New Zealand          | 0.04  | 0.00, 0.08          | 0.069   |
| TSTwd * Norway               | -0.05 | -0.09, 0.00         | 0.049   |
| TSTwd * Poland               | -0.10 | -0.15, -0.05        | <0.001  |
| TSTwd * Portugal             | -0.02 | -0.07, 0.04         | 0.6     |
| TSTwd * Romania              | -0.02 | -0.08, 0.03         | 0.4     |
| TSTwd * Russia               | -0.02 | -0.06, 0.02         | 0.3     |
| TSTwd * Singapore            | -0.01 | -0.04, 0.03         | 0.6     |
| TSTwd * Slovakia             | 0.01  | -0.06, 0.07         | 0.8     |
| TSTwd * South Africa         | -0.01 | -0.08, 0.06         | 0.8     |
| TSTwd * South Korea          | -0.10 | -0.16, -0.03        | 0.006   |
| TSTwd * Spain                | -0.01 | -0.05, 0.02         | 0.5     |
| TSTwd * Sweden               | -0.01 | -0.04, 0.02         | 0.5     |
| TSTwd * Switzerland          | -0.05 | -0.08, -0.01        | 0.005   |
| TSTwd * Thailand             | 0.01  | -0.04, 0.06         | 0.7     |
| TSTwd * United Arab Emirates | -0.03 | -0.08, 0.03         | 0.4     |
| TSTwd * United Kingdom       | 0.01  | -0.01, 0.03         | 0.5     |
| TSTwd * United States        | -0.03 | -0.04, -0.01        | 0.002   |
| <b>Sex * Age</b>             |       |                     |         |
| Male * Age                   | 0.26  | 0.24, 0.29          | <0.001  |
| <b>Sex * Age<sup>2</sup></b> |       |                     |         |
| Male * Age <sup>2</sup>      | 0.00  | 0.00, 0.00          | <0.001  |
| <b>Sex * Country</b>         |       |                     |         |
| Male * Austria               | -2.0  | -5.6, 1.6           | 0.3     |
| Male * Belgium               | 0.43  | -3.7, 4.6           | 0.8     |
| Male * Canada                | -0.55 | -2.3, 1.2           | 0.5     |
| Male * Czech Republic        | -2.8  | -6.4, 0.75          | 0.12    |
| Male * Denmark               | 3.2   | -0.29, 6.7          | 0.072   |
| Male * Estonia               | -1.2  | -5.6, 3.3           | 0.6     |
| Male * Finland               | -1.4  | -3.0, 0.09          | 0.065   |
| Male * France                | 2.8   | -0.40, 5.9          | 0.087   |
| Male * Germany               | -1.3  | -3.1, 0.51          | 0.2     |
| Male * Greece                | -6.2  | -12, -0.46          | 0.034   |
| Male * Hong Kong             | -0.62 | -4.6, 3.4           | 0.8     |
| Male * India                 | -0.69 | -6.6, 5.2           | 0.8     |
| Male * Ireland               | 1.3   | -3.2, 5.8           | 0.6     |
| Male * Italy                 | 0.36  | -2.7, 3.5           | 0.8     |
| Male * Japan                 | -1.9  | -3.8, 0.02          | 0.052   |
| Male * Mexico                | -1.8  | -5.9, 2.3           | 0.4     |
| Male * Netherlands           | -2.0  | -4.7, 0.65          | 0.14    |
| Male * New Zealand           | -0.63 | -4.0, 2.8           | 0.7     |
| Male * Norway                | 1.1   | -2.9, 5.1           | 0.6     |
| Male * Poland                | -8.1  | -13, -3.2           | 0.001   |
| Male * Portugal              | -1.9  | -7.2, 3.4           | 0.5     |
| Male * Romania               | -1.4  | -6.2, 3.4           | 0.6     |
| Male * Russia                | 0.55  | -2.7, 3.8           | 0.7     |

| Characteristic                    | B     | 95% CI <sup>1</sup> | p-value |
|-----------------------------------|-------|---------------------|---------|
| Male * Singapore                  | 0.44  | -2.9, 3.7           | 0.8     |
| Male * Slovakia                   | -0.88 | -6.9, 5.1           | 0.8     |
| Male * South Africa               | 0.23  | -5.9, 6.3           | >0.9    |
| Male * South Korea                | 4.8   | -2.0, 12            | 0.2     |
| Male * Spain                      | -0.06 | -3.1, 2.9           | >0.9    |
| Male * Sweden                     | 1.6   | -0.73, 4.0          | 0.2     |
| Male * Switzerland                | -2.4  | -5.1, 0.32          | 0.085   |
| Male * Thailand                   | -2.3  | -7.0, 2.4           | 0.3     |
| Male * United Arab Emirates       | -2.8  | -7.9, 2.2           | 0.3     |
| Male * United Kingdom             | -0.66 | -2.4, 1.1           | 0.5     |
| Male * United States              | -1.6  | -3.0, -0.21         | 0.024   |
| <b>Age * Age<sup>2</sup></b>      | 0.00  | 0.00, 0.00          | <0.001  |
| <b>Age * Country</b>              |       |                     |         |
| Age * Austria                     | -0.05 | -0.19, 0.08         | 0.4     |
| Age * Belgium                     | -0.04 | -0.20, 0.11         | 0.6     |
| Age * Canada                      | 0.00  | -0.07, 0.07         | >0.9    |
| Age * Czech Republic              | -0.05 | -0.23, 0.12         | 0.5     |
| Age * Denmark                     | 0.12  | -0.01, 0.25         | 0.066   |
| Age * Estonia                     | 0.08  | -0.15, 0.30         | 0.5     |
| Age * Finland                     | 0.02  | -0.05, 0.08         | 0.6     |
| Age * France                      | -0.03 | -0.14, 0.08         | 0.6     |
| Age * Germany                     | -0.04 | -0.11, 0.03         | 0.3     |
| Age * Greece                      | -0.09 | -0.34, 0.15         | 0.4     |
| Age * Hong Kong                   | -0.12 | -0.29, 0.05         | 0.2     |
| Age * India                       | -0.16 | -0.36, 0.04         | 0.13    |
| Age * Ireland                     | 0.10  | -0.07, 0.27         | 0.2     |
| Age * Italy                       | 0.02  | -0.10, 0.13         | 0.8     |
| Age * Japan                       | -0.13 | -0.20, -0.05        | 0.001   |
| Age * Mexico                      | 0.01  | -0.15, 0.16         | >0.9    |
| Age * Netherlands                 | 0.02  | -0.08, 0.13         | 0.7     |
| Age * New Zealand                 | 0.09  | -0.05, 0.23         | 0.2     |
| Age * Norway                      | 0.05  | -0.12, 0.21         | 0.6     |
| Age * Poland                      | 0.03  | -0.23, 0.30         | 0.8     |
| Age * Portugal                    | 0.06  | -0.15, 0.26         | 0.6     |
| Age * Romania                     | -0.07 | -0.31, 0.17         | 0.6     |
| Age * Russia                      | -0.09 | -0.26, 0.09         | 0.3     |
| Age * Singapore                   | -0.24 | -0.39, -0.09        | 0.002   |
| Age * Slovakia                    | -0.01 | -0.28, 0.27         | >0.9    |
| Age * South Africa                | 0.10  | -0.17, 0.37         | 0.5     |
| Age * South Korea                 | -0.12 | -0.40, 0.15         | 0.4     |
| Age * Spain                       | 0.12  | 0.00, 0.23          | 0.055   |
| Age * Sweden                      | 0.06  | -0.03, 0.16         | 0.2     |
| Age * Switzerland                 | -0.05 | -0.15, 0.06         | 0.4     |
| Age * Thailand                    | -0.29 | -0.49, -0.09        | 0.004   |
| Age * United Arab Emirates        | -0.07 | -0.29, 0.15         | 0.5     |
| Age * United Kingdom              | -0.03 | -0.10, 0.04         | 0.3     |
| Age * United States               | -0.05 | -0.11, 0.00         | 0.056   |
| <b>Age<sup>2</sup> * Country</b>  |       |                     |         |
| Age <sup>2</sup> * Austria        | -0.01 | -0.02, 0.00         | 0.2     |
| Age <sup>2</sup> * Belgium        | -0.01 | -0.02, 0.00         | 0.3     |
| Age <sup>2</sup> * Canada         | 0.00  | -0.01, 0.00         | 0.040   |
| Age <sup>2</sup> * Czech Republic | -0.01 | -0.02, 0.00         | 0.2     |
| Age <sup>2</sup> * Denmark        | 0.00  | -0.01, 0.01         | >0.9    |
| Age <sup>2</sup> * Estonia        | -0.02 | -0.03, 0.00         | 0.026   |
| Age <sup>2</sup> * Finland        | -0.01 | -0.02, -0.01        | <0.001  |
| Age <sup>2</sup> * France         | -0.02 | -0.02, -0.01        | <0.001  |
| Age <sup>2</sup> * Germany        | -0.01 | -0.02, -0.01        | <0.001  |
| Age <sup>2</sup> * Greece         | -0.01 | -0.03, 0.00         | 0.14    |
| Age <sup>2</sup> * Hong Kong      | -0.01 | -0.02, 0.00         | 0.13    |
| Age <sup>2</sup> * India          | 0.01  | 0.00, 0.02          | 0.2     |
| Age <sup>2</sup> * Ireland        | 0.00  | -0.01, 0.02         | 0.4     |
| Age <sup>2</sup> * Italy          | -0.01 | -0.02, 0.00         | 0.020   |

| Characteristic                          | B     | 95% CI <sup>1</sup> | p-value |
|-----------------------------------------|-------|---------------------|---------|
| Age <sup>2</sup> * Japan                | -0.01 | -0.01, 0.00         | <0.001  |
| Age <sup>2</sup> * Mexico               | 0.00  | -0.01, 0.01         | 0.4     |
| Age <sup>2</sup> * Netherlands          | -0.01 | -0.02, 0.00         | 0.014   |
| Age <sup>2</sup> * New Zealand          | 0.00  | -0.01, 0.01         | 0.5     |
| Age <sup>2</sup> * Norway               | -0.01 | -0.02, 0.01         | 0.3     |
| Age <sup>2</sup> * Poland               | 0.00  | -0.02, 0.02         | >0.9    |
| Age <sup>2</sup> * Portugal             | -0.01 | -0.02, 0.00         | 0.052   |
| Age <sup>2</sup> * Romania              | -0.01 | -0.02, 0.01         | 0.3     |
| Age <sup>2</sup> * Russia               | 0.00  | -0.01, 0.01         | 0.6     |
| Age <sup>2</sup> * Singapore            | 0.00  | -0.01, 0.01         | 0.8     |
| Age <sup>2</sup> * Slovakia             | -0.01 | -0.03, 0.01         | 0.2     |
| Age <sup>2</sup> * South Africa         | -0.01 | -0.03, 0.00         | 0.2     |
| Age <sup>2</sup> * South Korea          | -0.01 | -0.02, 0.01         | 0.3     |
| Age <sup>2</sup> * Spain                | -0.01 | -0.02, 0.00         | 0.015   |
| Age <sup>2</sup> * Sweden               | -0.01 | -0.02, -0.01        | <0.001  |
| Age <sup>2</sup> * Switzerland          | -0.01 | -0.02, 0.00         | 0.004   |
| Age <sup>2</sup> * Thailand             | 0.01  | 0.00, 0.03          | 0.028   |
| Age <sup>2</sup> * United Arab Emirates | 0.03  | 0.01, 0.05          | <0.001  |
| Age <sup>2</sup> * United Kingdom       | 0.00  | -0.01, 0.00         | 0.6     |
| Age <sup>2</sup> * United States        | 0.00  | -0.01, 0.00         | 0.066   |

<sup>1</sup>CI = Confidence Interval

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**Table S3.4b.** ANOVA table with effect sizes - WESE

| Factor                           | Sum of squares | Df      | F value    | p     | $\eta_p^2$ |
|----------------------------------|----------------|---------|------------|-------|------------|
| TST (weekday)                    | 10,476,255.07  | 1       | 14,615.978 | 0.000 | 0.061      |
| Sex                              | 1,010,928.97   | 1       | 1,410.401  | 0.000 | 0.006      |
| Age                              | 652,099.77     | 1       | 909.779    | 0.000 | 0.004      |
| Age <sup>2</sup>                 | 2,150,773.69   | 1       | 3,000.658  | 0.000 | 0.013      |
| Country                          | 3,498,646.68   | 34      | 143.563    | 0.000 | 0.021      |
| TST (weekday) x Sex              | 29,957.21      | 1       | 41.795     | 0.000 | 0.000      |
| TST (weekday) x Age              | 968,177.15     | 1       | 1,350.755  | 0.000 | 0.006      |
| TST (weekday) x Age <sup>2</sup> | 384.91         | 1       | 0.537      | 0.464 | 0.000      |
| TST (weekday) x Country          | 118,860.84     | 34      | 4.877      | 0.000 | 0.001      |
| Sex x Age                        | 454,701.89     | 1       | 634.379    | 0.000 | 0.003      |
| Sex x Age <sup>2</sup>           | 9,196.89       | 1       | 12.831     | 0.000 | 0.000      |
| Sex x Country                    | 44,452.60      | 34      | 1.824      | 0.002 | 0.000      |
| Age x Age <sup>2</sup>           | 35,061.73      | 1       | 48.916     | 0.000 | 0.000      |
| Age x Country                    | 62,424.93      | 34      | 2.562      | 0.000 | 0.000      |
| Age <sup>2</sup> x Country       | 145,656.62     | 34      | 5.977      | 0.000 | 0.001      |
| Residuals                        | 161,993,711.24 | 226,006 |            |       |            |

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1172 5. Social Jetlag  
 1173  
 1174 Model includes weekend sleep extension.  
 1175  
 1176  $SJL \sim 1 + WESE + sex + age + age^2 + country + \dots$   
 1177  $WESE:sex + WESE:age + WESE:age^2 + WESE:country + \dots$   
 1178  $sex:age + sex:age^2 + sex:country + \dots$   
 1179  $age:age^2 + age:country + age^2:country$   
 1180  
 1181  
 1182

**Table S3.5a.** Unstandardised Regression Coefficients for SJL

| Characteristic                   | B     | 95% CI <sup>1</sup> | p-value |
|----------------------------------|-------|---------------------|---------|
| <b>TSTdiff</b>                   | 0.31  | 0.28, 0.34          | <0.001  |
| <b>Sex</b>                       |       |                     |         |
| Female                           | —     | —                   |         |
| Male                             | 4.6   | 3.2, 6.1            | <0.001  |
| <b>Age</b>                       | -0.45 | -0.51, -0.39        | <0.001  |
| <b>Age<sup>2</sup></b>           | 0.01  | 0.00, 0.01          | 0.001   |
| <b>Country</b>                   |       |                     |         |
| Australia                        | —     | —                   |         |
| Austria                          | 4.4   | 0.58, 8.2           | 0.024   |
| Belgium                          | 1.9   | -2.3, 6.1           | 0.4     |
| Canada                           | 2.7   | 1.0, 4.3            | 0.002   |
| Czech Republic                   | -0.61 | -4.5, 3.3           | 0.8     |
| Denmark                          | 12    | 8.8, 15             | <0.001  |
| Estonia                          | 16    | 12, 21              | <0.001  |
| Finland                          | 18    | 17, 20              | <0.001  |
| France                           | -1.1  | -4.3, 2.0           | 0.5     |
| Germany                          | 9.3   | 7.5, 11             | <0.001  |
| Greece                           | 5.5   | -0.20, 11           | 0.059   |
| Hong Kong                        | 1.5   | -2.4, 5.5           | 0.4     |
| India                            | -4.0  | -10, 2.0            | 0.2     |
| Ireland                          | 9.7   | 5.4, 14             | <0.001  |
| Italy                            | -0.91 | -4.1, 2.2           | 0.6     |
| Japan                            | -8.2  | -10, -6.3           | <0.001  |
| Mexico                           | 10    | 6.3, 14             | <0.001  |
| Netherlands                      | 3.3   | 0.63, 6.0           | 0.015   |
| New Zealand                      | 1.2   | -2.2, 4.5           | 0.5     |
| Norway                           | 17    | 13, 21              | <0.001  |
| Poland                           | -1.3  | -6.6, 4.0           | 0.6     |
| Portugal                         | 4.5   | -0.72, 9.7          | 0.092   |
| Romania                          | 1.8   | -2.9, 6.5           | 0.4     |
| Russia                           | -0.77 | -4.2, 2.6           | 0.7     |
| Singapore                        | 0.90  | -2.4, 4.2           | 0.6     |
| Slovakia                         | -5.6  | -12, 0.94           | 0.094   |
| South Africa                     | 1.7   | -4.2, 7.7           | 0.6     |
| South Korea                      | -2.4  | -9.6, 4.8           | 0.5     |
| Spain                            | 8.2   | 5.3, 11             | <0.001  |
| Sweden                           | 11    | 8.2, 13             | <0.001  |
| Switzerland                      | 3.7   | 0.94, 6.4           | 0.008   |
| Thailand                         | -6.8  | -11, -2.2           | 0.004   |
| United Arab Emirates             | 8.1   | 3.1, 13             | 0.001   |
| United Kingdom                   | 2.7   | 0.97, 4.3           | 0.002   |
| United States                    | 3.0   | 1.6, 4.4            | <0.001  |
| <b>TSTdiff * Sex</b>             |       |                     |         |
| TSTdiff * Male                   | -0.04 | -0.05, -0.03        | <0.001  |
| <b>TSTdiff * Age</b>             | 0.00  | 0.00, 0.00          | <0.001  |
| <b>TSTdiff * Age<sup>2</sup></b> | 0.00  | 0.00, 0.00          | <0.001  |
| <b>TSTdiff * Country</b>         |       |                     |         |

| Characteristic                 | B     | 95% CI <sup>1</sup> | p-value |
|--------------------------------|-------|---------------------|---------|
| TSTdiff * Austria              | 0.10  | 0.03, 0.17          | 0.003   |
| TSTdiff * Belgium              | 0.12  | 0.04, 0.20          | 0.004   |
| TSTdiff * Canada               | 0.05  | 0.02, 0.08          | 0.004   |
| TSTdiff * Czech Republic       | 0.08  | 0.02, 0.14          | 0.005   |
| TSTdiff * Denmark              | 0.04  | -0.03, 0.10         | 0.3     |
| TSTdiff * Estonia              | -0.07 | -0.15, 0.01         | 0.083   |
| TSTdiff * Finland              | 0.06  | 0.03, 0.09          | <0.001  |
| TSTdiff * France               | 0.13  | 0.07, 0.19          | <0.001  |
| TSTdiff * Germany              | 0.08  | 0.05, 0.11          | <0.001  |
| TSTdiff * Greece               | 0.05  | -0.06, 0.15         | 0.4     |
| TSTdiff * Hong Kong            | 0.16  | 0.09, 0.23          | <0.001  |
| TSTdiff * India                | 0.06  | -0.07, 0.19         | 0.4     |
| TSTdiff * Ireland              | -0.02 | -0.10, 0.06         | 0.6     |
| TSTdiff * Italy                | 0.11  | 0.05, 0.17          | <0.001  |
| TSTdiff * Japan                | 0.29  | 0.26, 0.32          | <0.001  |
| TSTdiff * Mexico               | -0.03 | -0.10, 0.04         | 0.4     |
| TSTdiff * Netherlands          | 0.01  | -0.04, 0.07         | 0.6     |
| TSTdiff * New Zealand          | -0.01 | -0.08, 0.06         | 0.7     |
| TSTdiff * Norway               | 0.01  | -0.06, 0.08         | 0.8     |
| TSTdiff * Poland               | 0.12  | 0.05, 0.20          | 0.002   |
| TSTdiff * Portugal             | 0.04  | -0.06, 0.15         | 0.4     |
| TSTdiff * Romania              | 0.12  | 0.02, 0.21          | 0.013   |
| TSTdiff * Russia               | 0.08  | 0.02, 0.15          | 0.011   |
| TSTdiff * Singapore            | 0.09  | 0.03, 0.15          | 0.002   |
| TSTdiff * Slovakia             | 0.07  | -0.03, 0.18         | 0.2     |
| TSTdiff * South Africa         | -0.06 | -0.18, 0.06         | 0.3     |
| TSTdiff * South Korea          | 0.32  | 0.22, 0.42          | <0.001  |
| TSTdiff * Spain                | 0.02  | -0.03, 0.08         | 0.4     |
| TSTdiff * Sweden               | 0.07  | 0.02, 0.12          | 0.004   |
| TSTdiff * Switzerland          | 0.11  | 0.06, 0.16          | <0.001  |
| TSTdiff * Thailand             | 0.11  | 0.02, 0.21          | 0.023   |
| TSTdiff * United Arab Emirates | 0.28  | 0.18, 0.37          | <0.001  |
| TSTdiff * United Kingdom       | -0.01 | -0.04, 0.02         | 0.6     |
| TSTdiff * United States        | 0.10  | 0.07, 0.12          | <0.001  |
| <b>Sex * Age</b>               |       |                     |         |
| Male * Age                     | -0.10 | -0.12, -0.07        | <0.001  |
| <b>Sex * Age<sup>2</sup></b>   |       |                     |         |
| Male * Age <sup>2</sup>        | 0.00  | 0.00, 0.00          | <0.001  |
| <b>Sex * Country</b>           |       |                     |         |
| Male * Austria                 | -3.0  | -6.8, 0.74          | 0.11    |
| Male * Belgium                 | -3.5  | -7.9, 0.82          | 0.11    |
| Male * Canada                  | 1.3   | -0.57, 3.1          | 0.2     |
| Male * Czech Republic          | 0.23  | -3.5, 4.0           | >0.9    |
| Male * Denmark                 | 0.31  | -3.4, 4.0           | 0.9     |
| Male * Estonia                 | -1.3  | -6.0, 3.4           | 0.6     |
| Male * Finland                 | -0.62 | -2.2, 1.0           | 0.5     |
| Male * France                  | -2.7  | -6.0, 0.55          | 0.10    |
| Male * Germany                 | -1.6  | -3.5, 0.27          | 0.092   |
| Male * Greece                  | -3.9  | -9.8, 2.1           | 0.2     |
| Male * Hong Kong               | -0.54 | -4.7, 3.7           | 0.8     |
| Male * India                   | -0.78 | -7.1, 5.5           | 0.8     |
| Male * Ireland                 | -3.4  | -8.1, 1.3           | 0.2     |
| Male * Italy                   | -1.2  | -4.4, 2.0           | 0.5     |
| Male * Japan                   | -2.3  | -4.3, -0.33         | 0.023   |
| Male * Mexico                  | -0.51 | -4.7, 3.7           | 0.8     |
| Male * Netherlands             | -0.39 | -3.2, 2.4           | 0.8     |
| Male * New Zealand             | -0.17 | -3.8, 3.4           | >0.9    |
| Male * Norway                  | -1.7  | -5.9, 2.5           | 0.4     |
| Male * Poland                  | 1.4   | -3.8, 6.6           | 0.6     |
| Male * Portugal                | -5.8  | -11, -0.37          | 0.036   |
| Male * Romania                 | -3.0  | -7.9, 1.9           | 0.2     |



| Characteristic                    | B     | 95% CI <sup>1</sup> | p-value |
|-----------------------------------|-------|---------------------|---------|
| Male * Russia                     | -0.62 | -4.1, 2.8           | 0.7     |
| Male * Singapore                  | -1.5  | -5.0, 2.0           | 0.4     |
| Male * Slovakia                   | 3.2   | -3.1, 9.5           | 0.3     |
| Male * South Africa               | -7.1  | -13, -0.82          | 0.027   |
| Male * South Korea                | -4.9  | -12, 2.4            | 0.2     |
| Male * Spain                      | 2.1   | -1.1, 5.2           | 0.2     |
| Male * Sweden                     | -0.68 | -3.2, 1.8           | 0.6     |
| Male * Switzerland                | -0.09 | -2.9, 2.7           | >0.9    |
| Male * Thailand                   | -0.24 | -5.3, 4.8           | >0.9    |
| Male * United Arab Emirates       | -6.7  | -12, -1.4           | 0.013   |
| Male * United Kingdom             | -0.32 | -2.2, 1.5           | 0.7     |
| Male * United States              | 0.15  | -1.3, 1.6           | 0.8     |
| <b>Age * Age<sup>2</sup></b>      | 0.00  | 0.00, 0.00          | <0.001  |
| <b>Age * Country</b>              |       |                     |         |
| Age * Austria                     | -0.01 | -0.16, 0.14         | >0.9    |
| Age * Belgium                     | 0.11  | -0.07, 0.28         | 0.2     |
| Age * Canada                      | 0.09  | 0.01, 0.16          | 0.024   |
| Age * Czech Republic              | 0.15  | -0.03, 0.34         | 0.11    |
| Age * Denmark                     | -0.03 | -0.17, 0.12         | 0.7     |
| Age * Estonia                     | -0.17 | -0.42, 0.07         | 0.2     |
| Age * Finland                     | -0.13 | -0.19, -0.06        | <0.001  |
| Age * France                      | 0.05  | -0.07, 0.17         | 0.4     |
| Age * Germany                     | 0.04  | -0.04, 0.12         | 0.3     |
| Age * Greece                      | 0.06  | -0.20, 0.33         | 0.6     |
| Age * Hong Kong                   | 0.05  | -0.13, 0.24         | 0.6     |
| Age * India                       | 0.40  | 0.17, 0.62          | <0.001  |
| Age * Ireland                     | 0.05  | -0.14, 0.24         | 0.6     |
| Age * Italy                       | 0.07  | -0.06, 0.20         | 0.3     |
| Age * Japan                       | 0.26  | 0.18, 0.34          | <0.001  |
| Age * Mexico                      | -0.21 | -0.37, -0.04        | 0.016   |
| Age * Netherlands                 | 0.07  | -0.04, 0.19         | 0.2     |
| Age * New Zealand                 | 0.07  | -0.08, 0.23         | 0.4     |
| Age * Norway                      | -0.02 | -0.20, 0.16         | 0.8     |
| Age * Poland                      | 0.03  | -0.25, 0.32         | 0.8     |
| Age * Portugal                    | 0.18  | -0.05, 0.40         | 0.13    |
| Age * Romania                     | 0.18  | -0.08, 0.44         | 0.2     |
| Age * Russia                      | 0.10  | -0.09, 0.29         | 0.3     |
| Age * Singapore                   | 0.23  | 0.06, 0.39          | 0.006   |
| Age * Slovakia                    | -0.04 | -0.34, 0.27         | 0.8     |
| Age * South Africa                | -0.15 | -0.45, 0.14         | 0.3     |
| Age * South Korea                 | 0.06  | -0.24, 0.36         | 0.7     |
| Age * Spain                       | 0.14  | 0.01, 0.27          | 0.033   |
| Age * Sweden                      | -0.01 | -0.12, 0.09         | 0.8     |
| Age * Switzerland                 | -0.14 | -0.25, -0.02        | 0.022   |
| Age * Thailand                    | 0.27  | 0.06, 0.49          | 0.013   |
| Age * United Arab Emirates        | 0.12  | -0.12, 0.36         | 0.3     |
| Age * United Kingdom              | 0.03  | -0.04, 0.11         | 0.4     |
| Age * United States               | 0.02  | -0.04, 0.08         | 0.5     |
| <b>Age<sup>2</sup> * Country</b>  |       |                     |         |
| Age <sup>2</sup> * Austria        | -0.01 | -0.02, 0.00         | 0.11    |
| Age <sup>2</sup> * Belgium        | -0.01 | -0.02, 0.00         | 0.11    |
| Age <sup>2</sup> * Canada         | -0.01 | -0.02, -0.01        | <0.001  |
| Age <sup>2</sup> * Czech Republic | -0.01 | -0.02, 0.00         | 0.15    |
| Age <sup>2</sup> * Denmark        | 0.00  | -0.01, 0.01         | 0.8     |
| Age <sup>2</sup> * Estonia        | -0.02 | -0.04, 0.00         | 0.013   |
| Age <sup>2</sup> * Finland        | -0.01 | -0.02, -0.01        | <0.001  |
| Age <sup>2</sup> * France         | 0.00  | -0.01, 0.01         | >0.9    |
| Age <sup>2</sup> * Germany        | -0.01 | -0.01, 0.00         | 0.055   |
| Age <sup>2</sup> * Greece         | -0.01 | -0.03, 0.01         | 0.4     |
| Age <sup>2</sup> * Hong Kong      | -0.01 | -0.02, 0.00         | 0.2     |
| Age <sup>2</sup> * India          | -0.02 | -0.04, -0.01        | <0.001  |
| Age <sup>2</sup> * Ireland        | 0.00  | -0.02, 0.01         | 0.5     |

| Characteristic                          | B     | 95% CI <sup>1</sup> | p-value |
|-----------------------------------------|-------|---------------------|---------|
| Age <sup>2</sup> * Italy                | -0.01 | -0.02, 0.00         | 0.084   |
| Age <sup>2</sup> * Japan                | -0.01 | -0.02, 0.00         | <0.001  |
| Age <sup>2</sup> * Mexico               | 0.00  | -0.01, 0.01         | 0.8     |
| Age <sup>2</sup> * Netherlands          | 0.00  | -0.01, 0.00         | 0.3     |
| Age <sup>2</sup> * New Zealand          | 0.00  | -0.01, 0.01         | 0.7     |
| Age <sup>2</sup> * Norway               | -0.01 | -0.02, 0.00         | 0.083   |
| Age <sup>2</sup> * Poland               | 0.02  | 0.00, 0.03          | 0.060   |
| Age <sup>2</sup> * Portugal             | -0.01 | -0.03, 0.00         | 0.077   |
| Age <sup>2</sup> * Romania              | -0.01 | -0.03, 0.00         | 0.090   |
| Age <sup>2</sup> * Russia               | -0.01 | -0.03, 0.00         | 0.025   |
| Age <sup>2</sup> * Singapore            | -0.01 | -0.02, 0.00         | 0.15    |
| Age <sup>2</sup> * Slovakia             | -0.01 | -0.03, 0.01         | 0.4     |
| Age <sup>2</sup> * South Africa         | 0.00  | -0.02, 0.02         | 0.7     |
| Age <sup>2</sup> * South Korea          | 0.00  | -0.01, 0.02         | 0.9     |
| Age <sup>2</sup> * Spain                | -0.02 | -0.03, -0.01        | <0.001  |
| Age <sup>2</sup> * Sweden               | 0.00  | -0.01, 0.00         | 0.4     |
| Age <sup>2</sup> * Switzerland          | 0.01  | 0.00, 0.02          | 0.010   |
| Age <sup>2</sup> * Thailand             | -0.01 | -0.03, 0.00         | 0.12    |
| Age <sup>2</sup> * United Arab Emirates | -0.01 | -0.03, 0.01         | 0.3     |
| Age <sup>2</sup> * United Kingdom       | -0.01 | -0.01, 0.00         | 0.003   |
| Age <sup>2</sup> * United States        | -0.01 | -0.01, 0.00         | <0.001  |

<sup>1</sup>CI = Confidence Interval

**Table S3.5b.** ANOVA table with effect sizes - SJL

| Factor                      | Sum of squares | Df      | F value    | p     | $\eta_p^2$ |
|-----------------------------|----------------|---------|------------|-------|------------|
| TST diff                    | 26,528,178.27  | 1       | 30,790.139 | 0.000 | 0.120      |
| Sex                         | 909,060.43     | 1       | 1,055.108  | 0.000 | 0.005      |
| Age                         | 9,834,549.65   | 1       | 11,414.547 | 0.000 | 0.048      |
| Age <sup>2</sup>            | 12,590.51      | 1       | 14.613     | 0.000 | 0.000      |
| Country                     | 6,710,966.72   | 34      | 229.092    | 0.000 | 0.033      |
| TST diff x Sex              | 66,256.51      | 1       | 76.901     | 0.000 | 0.000      |
| TST diff x Age              | 309,029.61     | 1       | 358.678    | 0.000 | 0.002      |
| TST diff x Age <sup>2</sup> | 81,053.53      | 1       | 94.075     | 0.000 | 0.000      |
| TST diff x Country          | 715,992.08     | 34      | 24.442     | 0.000 | 0.004      |
| Sex x Age                   | 65,852.82      | 1       | 76.433     | 0.000 | 0.000      |
| Sex x Age <sup>2</sup>      | 21,373.46      | 1       | 24.807     | 0.000 | 0.000      |
| Sex x Country               | 53,481.36      | 34      | 1.826      | 0.002 | 0.000      |
| Age x Age <sup>2</sup>      | 308,076.12     | 1       | 357.571    | 0.000 | 0.002      |
| Age x Country               | 189,934.24     | 34      | 6.484      | 0.000 | 0.001      |
| Age <sup>2</sup> x Country  | 124,186.28     | 34      | 4.239      | 0.000 | 0.001      |
| Residuals                   | 194,722,325.21 | 226,006 |            |       |            |

1189 6. Weekday TST Variability  
 1190  
 1191 TSTvar ~ 1 + sex + age + age<sup>2</sup> + country + ...  
 1192 sex:age + sex:age<sup>2</sup> + sex:country + ...  
 1193 age:age<sup>2</sup> + age:country + age<sup>2</sup>:country  
 1194  
 1195  
 1196

**Table S3.6a.** Unstandardised Regression Coefficients for TST Variability

| Characteristic               | B     | 95% CI <sup>1</sup> | p-value |
|------------------------------|-------|---------------------|---------|
| <b>Sex</b>                   |       |                     |         |
| Female                       | —     | —                   |         |
| Male                         | -0.03 | -0.86, 0.80         | >0.9    |
| <b>Age</b>                   | 0.00  | -0.04, 0.03         | 0.9     |
| <b>Age<sup>2</sup></b>       | 0.01  | 0.01, 0.01          | <0.001  |
| <b>Country</b>               |       |                     |         |
| Australia                    | —     | —                   |         |
| Austria                      | 0.52  | -1.4, 2.5           | 0.6     |
| Belgium                      | -0.49 | -2.7, 1.7           | 0.7     |
| Canada                       | -0.37 | -1.3, 0.54          | 0.4     |
| Czech Republic               | 0.77  | -1.3, 2.8           | 0.5     |
| Denmark                      | -0.40 | -2.2, 1.4           | 0.7     |
| Estonia                      | 0.56  | -1.7, 2.9           | 0.6     |
| Finland                      | -3.7  | -4.5, -2.9          | <0.001  |
| France                       | 1.3   | -0.39, 3.0          | 0.13    |
| Germany                      | -0.65 | -1.6, 0.31          | 0.2     |
| Greece                       | 5.1   | 2.1, 8.1            | <0.001  |
| Hong Kong                    | 1.8   | -0.33, 3.9          | 0.10    |
| India                        | 6.3   | 2.9, 9.7            | <0.001  |
| Ireland                      | 0.70  | -1.7, 3.1           | 0.6     |
| Italy                        | -0.74 | -2.4, 0.92          | 0.4     |
| Japan                        | 13    | 12, 14              | <0.001  |
| Mexico                       | 6.3   | 4.1, 8.5            | <0.001  |
| Netherlands                  | -0.63 | -2.0, 0.79          | 0.4     |
| New Zealand                  | -1.9  | -3.7, -0.16         | 0.033   |
| Norway                       | -0.50 | -2.6, 1.6           | 0.6     |
| Poland                       | 2.1   | -0.79, 4.9          | 0.2     |
| Portugal                     | 2.0   | -0.73, 4.7          | 0.2     |
| Romania                      | 3.3   | 0.84, 5.8           | 0.009   |
| Russia                       | 7.8   | 6.0, 9.6            | <0.001  |
| Singapore                    | 5.0   | 3.2, 6.8            | <0.001  |
| Slovakia                     | -3.5  | -7.0, -0.08         | 0.045   |
| South Africa                 | 1.9   | -1.4, 5.1           | 0.3     |
| South Korea                  | 10    | 6.2, 14             | <0.001  |
| Spain                        | 1.5   | -0.05, 3.1          | 0.057   |
| Sweden                       | -2.1  | -3.4, -0.91         | <0.001  |
| Switzerland                  | 0.52  | -0.88, 1.9          | 0.5     |
| Thailand                     | 7.2   | 4.6, 9.8            | <0.001  |
| United Arab Emirates         | 7.5   | 4.7, 10             | <0.001  |
| United Kingdom               | -0.61 | -1.5, 0.31          | 0.2     |
| United States                | 3.0   | 2.2, 3.7            | <0.001  |
| <b>Sex * Age</b>             |       |                     |         |
| Male * Age                   | 0.01  | -0.01, 0.02         | 0.3     |
| <b>Sex * Age<sup>2</sup></b> |       |                     |         |
| Male * Age <sup>2</sup>      | 0.00  | 0.00, 0.00          | <0.001  |
| <b>Sex * Country</b>         |       |                     |         |
| Male * Austria               | -0.69 | -2.8, 1.5           | 0.5     |
| Male * Belgium               | -0.48 | -3.0, 2.0           | 0.7     |
| Male * Canada                | 0.37  | -0.67, 1.4          | 0.5     |
| Male * Czech Republic        | -0.50 | -2.7, 1.7           | 0.6     |
| Male * Denmark               | -2.1  | -4.2, -0.03         | 0.046   |

| Characteristic              | B     | 95% CI <sup>1</sup> | p-value |
|-----------------------------|-------|---------------------|---------|
| Male * Estonia              | 1.8   | -0.85, 4.5          | 0.2     |
| Male * Finland              | 1.2   | 0.30, 2.1           | 0.009   |
| Male * France               | 0.49  | -1.4, 2.4           | 0.6     |
| Male * Germany              | 0.06  | -1.0, 1.1           | >0.9    |
| Male * Greece               | -1.2  | -4.6, 2.2           | 0.5     |
| Male * Hong Kong            | 0.92  | -1.5, 3.3           | 0.5     |
| Male * India                | 1.3   | -2.3, 4.9           | 0.5     |
| Male * Ireland              | -2.1  | -4.8, 0.58          | 0.12    |
| Male * Italy                | 0.01  | -1.8, 1.8           | >0.9    |
| Male * Japan                | -3.6  | -4.8, -2.5          | <0.001  |
| Male * Mexico               | 0.58  | -1.8, 3.0           | 0.6     |
| Male * Netherlands          | -1.4  | -3.0, 0.20          | 0.086   |
| Male * New Zealand          | 1.4   | -0.70, 3.4          | 0.2     |
| Male * Norway               | 0.81  | -1.6, 3.2           | 0.5     |
| Male * Poland               | -0.74 | -3.7, 2.2           | 0.6     |
| Male * Portugal             | -0.12 | -3.2, 3.0           | >0.9    |
| Male * Romania              | 1.6   | -1.2, 4.4           | 0.3     |
| Male * Russia               | -0.02 | -2.0, 1.9           | >0.9    |
| Male * Singapore            | -2.2  | -4.2, -0.20         | 0.031   |
| Male * Slovakia             | 2.9   | -0.66, 6.5          | 0.11    |
| Male * South Africa         | 1.9   | -1.7, 5.5           | 0.3     |
| Male * South Korea          | 0.19  | -4.0, 4.3           | >0.9    |
| Male * Spain                | -1.5  | -3.3, 0.30          | 0.10    |
| Male * Sweden               | 0.93  | -0.49, 2.4          | 0.2     |
| Male * Switzerland          | -1.3  | -2.9, 0.30          | 0.11    |
| Male * Thailand             | -1.8  | -4.7, 1.0           | 0.2     |
| Male * United Arab Emirates | -0.67 | -3.7, 2.4           | 0.7     |
| Male * United Kingdom       | -0.22 | -1.3, 0.83          | 0.7     |
| Male * United States        | -0.83 | -1.7, 0.02          | 0.054   |
| Age * Age <sup>2</sup>      | 0.00  | 0.00, 0.00          | <0.001  |
| <b>Age * Country</b>        |       |                     |         |
| Age * Austria               | 0.00  | -0.09, 0.08         | >0.9    |
| Age * Belgium               | -0.01 | -0.10, 0.09         | >0.9    |
| Age * Canada                | -0.05 | -0.09, 0.00         | 0.032   |
| Age * Czech Republic        | 0.01  | -0.10, 0.12         | 0.9     |
| Age * Denmark               | 0.00  | -0.08, 0.08         | >0.9    |
| Age * Estonia               | 0.08  | -0.06, 0.22         | 0.3     |
| Age * Finland               | -0.05 | -0.09, -0.01        | 0.010   |
| Age * France                | -0.04 | -0.11, 0.03         | 0.3     |
| Age * Germany               | -0.03 | -0.07, 0.02         | 0.2     |
| Age * Greece                | 0.09  | -0.06, 0.24         | 0.2     |
| Age * Hong Kong             | -0.10 | -0.21, 0.01         | 0.071   |
| Age * India                 | -0.29 | -0.42, -0.16        | <0.001  |
| Age * Ireland               | -0.03 | -0.14, 0.08         | 0.6     |
| Age * Italy                 | 0.01  | -0.06, 0.08         | 0.8     |
| Age * Japan                 | -0.22 | -0.27, -0.17        | <0.001  |
| Age * Mexico                | -0.04 | -0.14, 0.06         | 0.4     |
| Age * Netherlands           | 0.01  | -0.06, 0.08         | 0.8     |
| Age * New Zealand           | 0.01  | -0.08, 0.09         | >0.9    |
| Age * Norway                | -0.10 | -0.20, 0.00         | 0.048   |
| Age * Poland                | 0.06  | -0.10, 0.22         | 0.5     |
| Age * Portugal              | -0.01 | -0.13, 0.12         | >0.9    |
| Age * Romania               | -0.09 | -0.24, 0.06         | 0.2     |
| Age * Russia                | 0.02  | -0.09, 0.13         | 0.7     |
| Age * Singapore             | -0.11 | -0.20, -0.02        | 0.021   |
| Age * Slovakia              | 0.03  | -0.14, 0.20         | 0.7     |
| Age * South Africa          | 0.13  | -0.04, 0.30         | 0.13    |
| Age * South Korea           | -0.20 | -0.37, -0.03        | 0.024   |
| Age * Spain                 | -0.06 | -0.13, 0.01         | 0.11    |
| Age * Sweden                | -0.01 | -0.07, 0.05         | 0.7     |
| Age * Switzerland           | 0.03  | -0.04, 0.09         | 0.4     |
| Age * Thailand              | -0.14 | -0.26, -0.02        | 0.025   |

| Characteristic                          | B     | 95% CI <sup>1</sup> | p-value |
|-----------------------------------------|-------|---------------------|---------|
| Age * United Arab Emirates              | -0.03 | -0.16, 0.11         | 0.7     |
| Age * United Kingdom                    | -0.01 | -0.06, 0.03         | 0.5     |
| Age * United States                     | -0.07 | -0.11, -0.04        | <0.001  |
| <b>Age<sup>2</sup> * Country</b>        |       |                     |         |
| Age <sup>2</sup> * Austria              | 0.00  | -0.01, 0.01         | 0.9     |
| Age <sup>2</sup> * Belgium              | -0.01 | -0.01, 0.00         | 0.13    |
| Age <sup>2</sup> * Canada               | 0.00  | 0.00, 0.01          | 0.017   |
| Age <sup>2</sup> * Czech Republic       | 0.00  | -0.01, 0.01         | 0.9     |
| Age <sup>2</sup> * Denmark              | 0.00  | 0.00, 0.01          | 0.5     |
| Age <sup>2</sup> * Estonia              | 0.01  | 0.00, 0.02          | 0.2     |
| Age <sup>2</sup> * Finland              | 0.01  | 0.01, 0.01          | <0.001  |
| Age <sup>2</sup> * France               | 0.00  | -0.01, 0.00         | 0.11    |
| Age <sup>2</sup> * Germany              | 0.00  | 0.00, 0.00          | 0.7     |
| Age <sup>2</sup> * Greece               | 0.00  | -0.01, 0.01         | 0.5     |
| Age <sup>2</sup> * Hong Kong            | 0.01  | 0.00, 0.01          | 0.076   |
| Age <sup>2</sup> * India                | 0.00  | -0.01, 0.01         | 0.5     |
| Age <sup>2</sup> * Ireland              | 0.00  | 0.00, 0.01          | 0.3     |
| Age <sup>2</sup> * Italy                | 0.00  | -0.01, 0.00         | 0.094   |
| Age <sup>2</sup> * Japan                | 0.00  | -0.01, 0.00         | 0.027   |
| Age <sup>2</sup> * Mexico               | 0.00  | -0.01, 0.00         | 0.3     |
| Age <sup>2</sup> * Netherlands          | 0.00  | 0.00, 0.01          | 0.4     |
| Age <sup>2</sup> * New Zealand          | 0.00  | -0.01, 0.00         | 0.2     |
| Age <sup>2</sup> * Norway               | 0.01  | 0.00, 0.01          | 0.052   |
| Age <sup>2</sup> * Poland               | 0.00  | -0.01, 0.01         | 0.6     |
| Age <sup>2</sup> * Portugal             | 0.00  | 0.00, 0.01          | 0.4     |
| Age <sup>2</sup> * Romania              | 0.00  | -0.01, 0.01         | 0.4     |
| Age <sup>2</sup> * Russia               | 0.00  | -0.01, 0.01         | 0.7     |
| Age <sup>2</sup> * Singapore            | 0.01  | 0.00, 0.01          | 0.032   |
| Age <sup>2</sup> * Slovakia             | 0.00  | -0.01, 0.01         | 0.8     |
| Age <sup>2</sup> * South Africa         | -0.01 | -0.02, 0.00         | 0.2     |
| Age <sup>2</sup> * South Korea          | 0.01  | 0.00, 0.01          | 0.2     |
| Age <sup>2</sup> * Spain                | 0.00  | 0.00, 0.01          | 0.3     |
| Age <sup>2</sup> * Sweden               | 0.00  | 0.00, 0.01          | 0.013   |
| Age <sup>2</sup> * Switzerland          | 0.00  | -0.01, 0.00         | 0.5     |
| Age <sup>2</sup> * Thailand             | 0.00  | -0.01, 0.01         | 0.8     |
| Age <sup>2</sup> * United Arab Emirates | 0.01  | 0.00, 0.02          | 0.086   |
| Age <sup>2</sup> * United Kingdom       | 0.00  | 0.00, 0.00          | >0.9    |
| Age <sup>2</sup> * United States        | 0.00  | 0.00, 0.00          | 0.11    |

<sup>1</sup>CI = Confidence Interval

1197

1198

1199

**Table S3.6b.** ANOVA table with effect sizes – TST Variability

| Factor                     | Sum of squares | Df      | F value   | p     | $\eta_p^2$ |
|----------------------------|----------------|---------|-----------|-------|------------|
| Sex                        | 63,637.97      | 1       | 225.711   | 0.000 | 0.001      |
| Age                        | 430,317.68     | 1       | 1,526.252 | 0.000 | 0.007      |
| Age <sup>2</sup>           | 486,292.86     | 1       | 1,724.785 | 0.000 | 0.008      |
| Country                    | 1,718,969.67   | 34      | 179.319   | 0.000 | 0.026      |
| Sex x Age                  | 352.06         | 1       | 1.249     | 0.264 | 0.000      |
| Sex x Age <sup>2</sup>     | 30,094.83      | 1       | 106.740   | 0.000 | 0.000      |
| Sex x Country              | 53,300.84      | 34      | 5.560     | 0.000 | 0.001      |
| Age x Age <sup>2</sup>     | 59,798.96      | 1       | 212.095   | 0.000 | 0.001      |
| Age x Country              | 58,830.69      | 34      | 6.137     | 0.000 | 0.001      |
| Age <sup>2</sup> x Country | 55,098.24      | 34      | 5.748     | 0.000 | 0.001      |
| Residuals                  | 63,731,760.29  | 226,044 |           |       |            |

1200

1201

1202 7. Weekday sleep variability (accounting for TST)

1203

1204 Model including weekday TST.

1205

1206  $TSTvar \sim 1 + TSTwd + sex + age + age^2 + country + \dots$

1207  $TSTwd:sex + TSTwd:age + TSTwd:age^2 + TSTwd:country + \dots$

1208  $sex:age + sex:age^2 + sex:country + \dots$

1209  $age:age^2 + age:country + age^2:country$

1210

1211

1212

**Table S3.7a.** Unstandardised Regression Coefficients for TST Variability with TST

| Characteristic                 | B     | 95% CI <sup>1</sup> | p-value |
|--------------------------------|-------|---------------------|---------|
| <b>TSTwd</b>                   | -0.10 | -0.11, -0.09        | <0.001  |
| <b>Sex</b>                     |       |                     |         |
| Female                         | —     | —                   |         |
| Male                           | -8.3  | -10, -6.6           | <0.001  |
| <b>Age</b>                     | -0.28 | -0.34, -0.21        | <0.001  |
| <b>Age<sup>2</sup></b>         | 0.01  | 0.01, 0.01          | <0.001  |
| <b>Country</b>                 |       |                     |         |
| Australia                      | —     | —                   |         |
| Austria                        | -16   | -27, -4.2           | 0.007   |
| Belgium                        | -7.2  | -20, 5.3            | 0.3     |
| Canada                         | -0.44 | -5.7, 4.8           | 0.9     |
| Czech Republic                 | -8.0  | -18, 2.4            | 0.13    |
| Denmark                        | -17   | -27, -6.1           | 0.002   |
| Estonia                        | -5.9  | -20, 8.4            | 0.4     |
| Finland                        | -15   | -20, -10            | <0.001  |
| France                         | -1.3  | -11, 8.2            | 0.8     |
| Germany                        | -13   | -19, -7.3           | <0.001  |
| Greece                         | 3.5   | -14, 21             | 0.7     |
| Hong Kong                      | -16   | -27, -4.5           | 0.006   |
| India                          | -14   | -28, 0.52           | 0.059   |
| Ireland                        | -11   | -25, 3.5            | 0.14    |
| Italy                          | -9.6  | -19, -0.24          | 0.044   |
| Japan                          | -9.7  | -15, -4.6           | <0.001  |
| Mexico                         | -0.41 | -12, 11             | >0.9    |
| Netherlands                    | -4.9  | -13, 3.5            | 0.3     |
| New Zealand                    | 0.18  | -11, 11             | >0.9    |
| Norway                         | -24   | -37, -12            | <0.001  |
| Poland                         | -6.1  | -21, 8.5            | 0.4     |
| Portugal                       | -2.7  | -18, 13             | 0.7     |
| Romania                        | -8.8  | -23, 5.8            | 0.2     |
| Russia                         | -8.3  | -19, 2.4            | 0.13    |
| Singapore                      | -2.4  | -11, 6.4            | 0.6     |
| Slovakia                       | -5.9  | -23, 11             | 0.5     |
| South Africa                   | -7.1  | -26, 12             | 0.5     |
| South Korea                    | 4.6   | -13, 22             | 0.6     |
| Spain                          | -5.3  | -15, 4.0            | 0.3     |
| Sweden                         | -16   | -24, -8.8           | <0.001  |
| Switzerland                    | -8.4  | -17, 0.20           | 0.056   |
| Thailand                       | -12   | -26, 0.78           | 0.065   |
| United Arab Emirates           | -2.0  | -16, 12             | 0.8     |
| United Kingdom                 | -3.1  | -8.5, 2.2           | 0.3     |
| United States                  | 3.4   | -0.96, 7.7          | 0.13    |
| <b>TSTwd * Sex</b>             |       |                     |         |
| TSTwd * Male                   | 0.01  | 0.01, 0.02          | <0.001  |
| <b>TSTwd * Age</b>             | 0.00  | 0.00, 0.00          | <0.001  |
| <b>TSTwd * Age<sup>2</sup></b> | 0.00  | 0.00, 0.00          | 0.5     |
| <b>TSTwd * Country</b>         |       |                     |         |

| Characteristic               | B     | 95% CI <sup>1</sup> | p-value |
|------------------------------|-------|---------------------|---------|
| TSTwd * Austria              | 0.04  | 0.01, 0.06          | 0.005   |
| TSTwd * Belgium              | 0.02  | -0.01, 0.04         | 0.3     |
| TSTwd * Canada               | 0.00  | -0.01, 0.01         | >0.9    |
| TSTwd * Czech Republic       | 0.02  | 0.00, 0.04          | 0.12    |
| TSTwd * Denmark              | 0.04  | 0.01, 0.06          | 0.003   |
| TSTwd * Estonia              | 0.01  | -0.02, 0.05         | 0.4     |
| TSTwd * Finland              | 0.03  | 0.02, 0.04          | <0.001  |
| TSTwd * France               | 0.01  | -0.02, 0.03         | 0.6     |
| TSTwd * Germany              | 0.03  | 0.01, 0.04          | <0.001  |
| TSTwd * Greece               | 0.00  | -0.04, 0.04         | 0.9     |
| TSTwd * Hong Kong            | 0.04  | 0.01, 0.06          | 0.008   |
| TSTwd * India                | 0.04  | 0.01, 0.08          | 0.016   |
| TSTwd * Ireland              | 0.03  | 0.00, 0.06          | 0.091   |
| TSTwd * Italy                | 0.02  | 0.00, 0.04          | 0.091   |
| TSTwd * Japan                | 0.04  | 0.03, 0.06          | <0.001  |
| TSTwd * Mexico               | 0.01  | -0.01, 0.04         | 0.3     |
| TSTwd * Netherlands          | 0.01  | -0.01, 0.03         | 0.3     |
| TSTwd * New Zealand          | 0.00  | -0.03, 0.02         | 0.7     |
| TSTwd * Norway               | 0.05  | 0.03, 0.08          | <0.001  |
| TSTwd * Poland               | 0.02  | -0.02, 0.05         | 0.3     |
| TSTwd * Portugal             | 0.01  | -0.02, 0.05         | 0.5     |
| TSTwd * Romania              | 0.03  | -0.01, 0.06         | 0.12    |
| TSTwd * Russia               | 0.04  | 0.01, 0.06          | 0.004   |
| TSTwd * Singapore            | 0.01  | -0.01, 0.03         | 0.3     |
| TSTwd * Slovakia             | 0.01  | -0.03, 0.05         | 0.8     |
| TSTwd * South Africa         | 0.02  | -0.02, 0.06         | 0.4     |
| TSTwd * South Korea          | 0.01  | -0.04, 0.05         | 0.8     |
| TSTwd * Spain                | 0.01  | -0.01, 0.04         | 0.2     |
| TSTwd * Sweden               | 0.03  | 0.01, 0.05          | <0.001  |
| TSTwd * Switzerland          | 0.02  | 0.00, 0.04          | 0.046   |
| TSTwd * Thailand             | 0.04  | 0.01, 0.07          | 0.011   |
| TSTwd * United Arab Emirates | 0.02  | -0.02, 0.05         | 0.3     |
| TSTwd * United Kingdom       | 0.01  | -0.01, 0.02         | 0.4     |
| TSTwd * United States        | 0.00  | -0.01, 0.01         | 0.7     |
| <b>Sex * Age</b>             |       |                     |         |
| Male * Age                   | 0.04  | 0.03, 0.06          | <0.001  |
| <b>Sex * Age<sup>2</sup></b> |       |                     |         |
| Male * Age <sup>2</sup>      | 0.00  | 0.00, 0.00          | <0.001  |
| <b>Sex * Country</b>         |       |                     |         |
| Male * Austria               | 0.25  | -2.0, 2.5           | 0.8     |
| Male * Belgium               | -0.33 | -2.9, 2.2           | 0.8     |
| Male * Canada                | 0.11  | -0.96, 1.2          | 0.8     |
| Male * Czech Republic        | 0.35  | -1.8, 2.5           | 0.8     |
| Male * Denmark               | -1.2  | -3.3, 0.96          | 0.3     |
| Male * Estonia               | 2.4   | -0.32, 5.1          | 0.084   |
| Male * Finland               | 1.8   | 0.90, 2.8           | <0.001  |
| Male * France                | 0.37  | -1.6, 2.3           | 0.7     |
| Male * Germany               | 0.92  | -0.19, 2.0          | 0.10    |
| Male * Greece                | -1.5  | -5.0, 2.0           | 0.4     |
| Male * Hong Kong             | 2.2   | -0.27, 4.6          | 0.082   |
| Male * India                 | 3.1   | -0.51, 6.7          | 0.092   |
| Male * Ireland               | -1.7  | -4.4, 1.1           | 0.2     |
| Male * Italy                 | 0.34  | -1.5, 2.2           | 0.7     |
| Male * Japan                 | -1.6  | -2.8, -0.42         | 0.008   |
| Male * Mexico                | 0.50  | -2.0, 3.0           | 0.7     |
| Male * Netherlands           | -1.1  | -2.8, 0.50          | 0.2     |
| Male * New Zealand           | 1.7   | -0.34, 3.8          | 0.10    |
| Male * Norway                | 2.4   | -0.07, 4.8          | 0.056   |
| Male * Poland                | -0.51 | -3.6, 2.5           | 0.7     |
| Male * Portugal              | -0.48 | -3.7, 2.8           | 0.8     |
| Male * Romania               | 2.0   | -0.91, 5.0          | 0.2     |
| Male * Russia                | 1.2   | -0.81, 3.2          | 0.2     |

| Characteristic                    | B     | 95% CI <sup>1</sup> | p-value |
|-----------------------------------|-------|---------------------|---------|
| Male * Singapore                  | -1.3  | -3.3, 0.70          | 0.2     |
| Male * Slovakia                   | 2.9   | -0.81, 6.5          | 0.13    |
| Male * South Africa               | 2.2   | -1.5, 5.9           | 0.2     |
| Male * South Korea                | 0.68  | -3.5, 4.9           | 0.8     |
| Male * Spain                      | -1.2  | -3.0, 0.65          | 0.2     |
| Male * Sweden                     | 1.8   | 0.33, 3.2           | 0.016   |
| Male * Switzerland                | -0.94 | -2.6, 0.71          | 0.3     |
| Male * Thailand                   | 0.03  | -2.8, 2.9           | >0.9    |
| Male * United Arab Emirates       | -0.03 | -3.1, 3.1           | >0.9    |
| Male * United Kingdom             | 0.04  | -1.0, 1.1           | >0.9    |
| Male * United States              | -1.0  | -1.9, -0.15         | 0.022   |
| <b>Age * Age<sup>2</sup></b>      | 0.00  | 0.00, 0.00          | <0.001  |
| <b>Age * Country</b>              |       |                     |         |
| Age * Austria                     | 0.00  | -0.08, 0.09         | >0.9    |
| Age * Belgium                     | -0.01 | -0.11, 0.08         | 0.8     |
| Age * Canada                      | -0.05 | -0.09, 0.00         | 0.036   |
| Age * Czech Republic              | -0.02 | -0.12, 0.09         | 0.8     |
| Age * Denmark                     | -0.01 | -0.09, 0.07         | 0.8     |
| Age * Estonia                     | 0.06  | -0.08, 0.20         | 0.4     |
| Age * Finland                     | -0.06 | -0.10, -0.02        | 0.002   |
| Age * France                      | -0.04 | -0.11, 0.03         | 0.3     |
| Age * Germany                     | -0.03 | -0.07, 0.01         | 0.2     |
| Age * Greece                      | 0.09  | -0.06, 0.24         | 0.2     |
| Age * Hong Kong                   | -0.07 | -0.18, 0.03         | 0.2     |
| Age * India                       | -0.28 | -0.40, -0.15        | <0.001  |
| Age * Ireland                     | -0.03 | -0.13, 0.07         | 0.6     |
| Age * Italy                       | -0.01 | -0.08, 0.06         | 0.9     |
| Age * Japan                       | -0.21 | -0.25, -0.16        | <0.001  |
| Age * Mexico                      | -0.03 | -0.12, 0.07         | 0.6     |
| Age * Netherlands                 | -0.01 | -0.07, 0.06         | 0.8     |
| Age * New Zealand                 | 0.00  | -0.08, 0.09         | >0.9    |
| Age * Norway                      | -0.10 | -0.20, 0.00         | 0.040   |
| Age * Poland                      | 0.04  | -0.12, 0.20         | 0.6     |
| Age * Portugal                    | -0.02 | -0.15, 0.10         | 0.7     |
| Age * Romania                     | -0.12 | -0.26, 0.03         | 0.12    |
| Age * Russia                      | 0.02  | -0.08, 0.13         | 0.7     |
| Age * Singapore                   | -0.08 | -0.17, 0.01         | 0.083   |
| Age * Slovakia                    | 0.05  | -0.12, 0.21         | 0.6     |
| Age * South Africa                | 0.11  | -0.05, 0.28         | 0.2     |
| Age * South Korea                 | -0.18 | -0.35, -0.01        | 0.037   |
| Age * Spain                       | -0.08 | -0.15, -0.01        | 0.033   |
| Age * Sweden                      | -0.03 | -0.08, 0.03         | 0.4     |
| Age * Switzerland                 | 0.02  | -0.04, 0.09         | 0.5     |
| Age * Thailand                    | -0.13 | -0.25, -0.01        | 0.033   |
| Age * United Arab Emirates        | -0.04 | -0.17, 0.09         | 0.5     |
| Age * United Kingdom              | -0.01 | -0.06, 0.03         | 0.5     |
| Age * United States               | -0.07 | -0.11, -0.04        | <0.001  |
| <b>Age<sup>2</sup> * Country</b>  |       |                     |         |
| Age <sup>2</sup> * Austria        | 0.00  | -0.01, 0.00         | 0.8     |
| Age <sup>2</sup> * Belgium        | 0.00  | -0.01, 0.00         | 0.2     |
| Age <sup>2</sup> * Canada         | 0.00  | 0.00, 0.01          | 0.017   |
| Age <sup>2</sup> * Czech Republic | 0.00  | -0.01, 0.01         | >0.9    |
| Age <sup>2</sup> * Denmark        | 0.00  | 0.00, 0.01          | 0.3     |
| Age <sup>2</sup> * Estonia        | 0.01  | 0.00, 0.02          | 0.2     |
| Age <sup>2</sup> * Finland        | 0.01  | 0.01, 0.01          | <0.001  |
| Age <sup>2</sup> * France         | 0.00  | -0.01, 0.00         | 0.10    |
| Age <sup>2</sup> * Germany        | 0.00  | 0.00, 0.00          | 0.7     |
| Age <sup>2</sup> * Greece         | 0.00  | -0.01, 0.01         | 0.6     |
| Age <sup>2</sup> * Hong Kong      | 0.01  | 0.00, 0.01          | 0.086   |
| Age <sup>2</sup> * India          | 0.00  | -0.01, 0.00         | 0.3     |
| Age <sup>2</sup> * Ireland        | 0.00  | 0.00, 0.01          | 0.3     |
| Age <sup>2</sup> * Italy          | 0.00  | -0.01, 0.00         | 0.2     |



| Characteristic                          | B     | 95% CI <sup>1</sup> | p-value |
|-----------------------------------------|-------|---------------------|---------|
| Age <sup>2</sup> * Japan                | 0.00  | -0.01, 0.00         | 0.060   |
| Age <sup>2</sup> * Mexico               | 0.00  | -0.01, 0.00         | 0.3     |
| Age <sup>2</sup> * Netherlands          | 0.00  | 0.00, 0.01          | 0.5     |
| Age <sup>2</sup> * New Zealand          | 0.00  | -0.01, 0.00         | 0.2     |
| Age <sup>2</sup> * Norway               | 0.01  | 0.00, 0.01          | 0.046   |
| Age <sup>2</sup> * Poland               | 0.00  | -0.01, 0.01         | 0.5     |
| Age <sup>2</sup> * Portugal             | 0.00  | 0.00, 0.01          | 0.5     |
| Age <sup>2</sup> * Romania              | 0.00  | 0.00, 0.01          | 0.3     |
| Age <sup>2</sup> * Russia               | 0.00  | -0.01, 0.01         | 0.7     |
| Age <sup>2</sup> * Singapore            | 0.01  | 0.00, 0.01          | 0.10    |
| Age <sup>2</sup> * Slovakia             | 0.00  | -0.01, 0.01         | >0.9    |
| Age <sup>2</sup> * South Africa         | -0.01 | -0.02, 0.00         | 0.2     |
| Age <sup>2</sup> * South Korea          | 0.00  | 0.00, 0.01          | 0.4     |
| Age <sup>2</sup> * Spain                | 0.00  | 0.00, 0.01          | 0.13    |
| Age <sup>2</sup> * Sweden               | 0.01  | 0.00, 0.01          | 0.005   |
| Age <sup>2</sup> * Switzerland          | 0.00  | -0.01, 0.00         | 0.3     |
| Age <sup>2</sup> * Thailand             | 0.00  | -0.01, 0.01         | 0.8     |
| Age <sup>2</sup> * United Arab Emirates | 0.01  | 0.00, 0.02          | 0.086   |
| Age <sup>2</sup> * United Kingdom       | 0.00  | 0.00, 0.00          | 0.8     |
| Age <sup>2</sup> * United States        | 0.00  | 0.00, 0.00          | 0.11    |

<sup>1</sup>CI = Confidence Interval

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**Table S3.7b.** ANOVA table with effect sizes – TST Variability with TST

| Factor                           | Sum of squares | Df      | F value    | p     | $\eta_p^2$ |
|----------------------------------|----------------|---------|------------|-------|------------|
| TST (weekday)                    | 2,968,398.65   | 1       | 11,061.203 | 0.000 | 0.047      |
| Sex                              | 565,461.87     | 1       | 2,107.092  | 0.000 | 0.009      |
| Age                              | 565,800.60     | 1       | 2,108.354  | 0.000 | 0.009      |
| Age <sup>2</sup>                 | 589,407.76     | 1       | 2,196.322  | 0.000 | 0.010      |
| Country                          | 905,988.08     | 34      | 99.294     | 0.000 | 0.015      |
| TST (weekday) x Sex              | 16,749.94      | 1       | 62.416     | 0.000 | 0.000      |
| TST (weekday) x Age              | 15,787.31      | 1       | 58.829     | 0.000 | 0.000      |
| TST (weekday) x Age <sup>2</sup> | 107.45         | 1       | 0.400      | 0.527 | 0.000      |
| TST (weekday) x Country          | 82,248.66      | 34      | 9.014      | 0.000 | 0.001      |
| Sex x Age                        | 12,259.52      | 1       | 45.683     | 0.000 | 0.000      |
| Sex x Age <sup>2</sup>           | 21,822.52      | 1       | 81.318     | 0.000 | 0.000      |
| Sex x Country                    | 59,843.74      | 34      | 6.559      | 0.000 | 0.001      |
| Age x Age <sup>2</sup>           | 44,672.65      | 1       | 166.465    | 0.000 | 0.001      |
| Age x Country                    | 46,769.61      | 34      | 5.126      | 0.000 | 0.001      |
| Age <sup>2</sup> x Country       | 55,464.04      | 34      | 6.079      | 0.000 | 0.001      |
| Residuals                        | 60,651,260.59  | 226,006 |            |       |            |

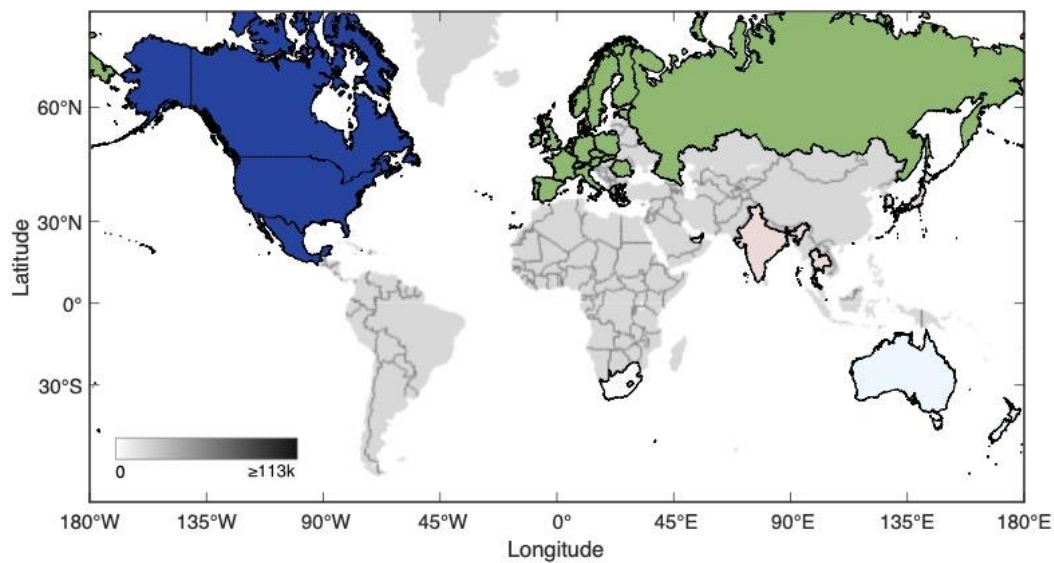
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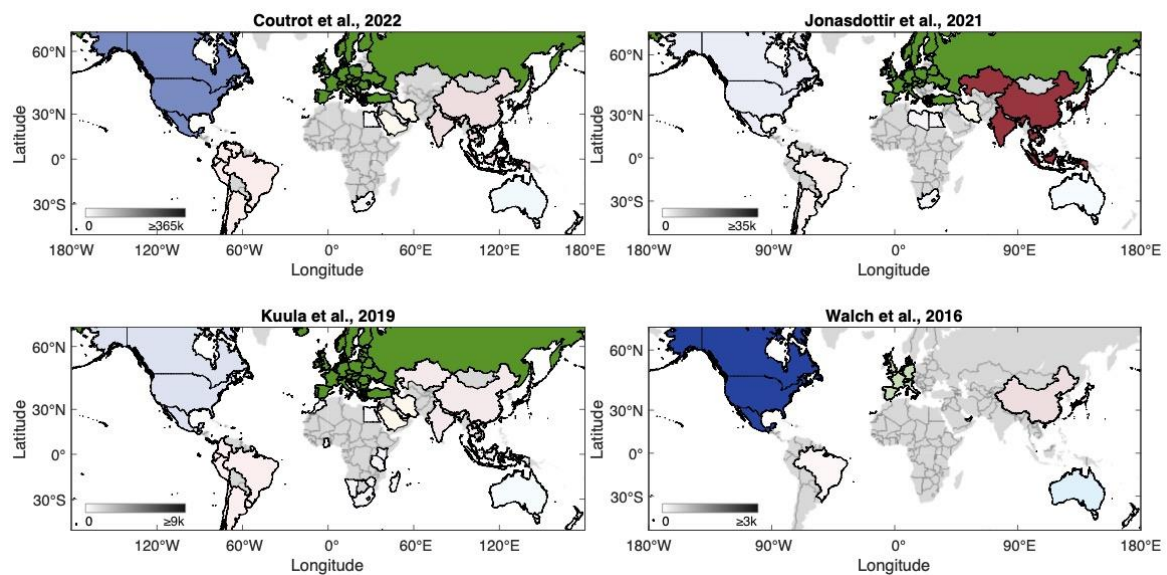
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## Global Coverage

Figure S2a shows user numbers from different global regions for this study and Figure S2b shows user/participant numbers for several other large-scale sleep studies. The proportion of users from each country within each study is reflected by colour opacity. Despite large user numbers, most come from North America and Europe. Although Walch et al. report participants from more countries than reflected here, only the top 20 countries were identified, and only very few came from other countries. The difference in user numbers across countries is large; therefore, full opacity reflects 50% of the total number of users in the study.



**Figure S2a.** Global coverage in this study.



**Figure S2b.** Global coverage in other large-scale sleep studies.

## Editorial Board Composition in Major Sleep Journals

Recent research<sup>1</sup> has demonstrated how academic sleep scientists rank sleep journals. Based on these rankings, Table S4 show how many editors for sleep science journals come from WEIRD<sup>2</sup> countries (excluding journals primarily related to chronobiology and breathing). The percentage of editors from WEIRD countries is generally above 90%.

| Journal name                              | Number of WEIRD country editors | Total number of editors | Percentage of WEIRD country editors | % Ranked Top 3 by respondents in Ref (1) | 2021 Impact Factor |
|-------------------------------------------|---------------------------------|-------------------------|-------------------------------------|------------------------------------------|--------------------|
| SLEEP                                     | 122                             | 126                     | 96.8                                | 89                                       | 6.31               |
| Sleep Medicine Reviews                    | 26                              | 29                      | 89.7                                | 61                                       | 11.4               |
| Journal of Sleep Research                 | 72                              | 73                      | 98.6                                | 48                                       | 5.3                |
| Journal of Clinical Sleep Medicine        | 57                              | 59                      | 96.6                                | 30                                       | 4.32               |
| Sleep Medicine                            | 82                              | 91                      | 90.1                                | 29                                       | 4.84               |
| Sleep Health                              | 94                              | 98                      | 95.9                                | 12                                       | 4.21               |
| Nature & Science of Sleep                 | 29                              | 37                      | 78.4                                | 7                                        | 3.83               |
| Behavioral Sleep Medicine                 | 72                              | 77                      | 93.5                                | 7                                        | 3.49               |
| Mean percentage of WEIRD country editors: |                                 |                         | 92.5                                |                                          |                    |

**Table S4.** Composition of editorial board members in major sleep journals. WEIRD: Western, Educated, Industrialized, Rich, and Democratic (<https://news.harvard.edu/gazette/story/2020/09/joseph-henrich-explores-weird-societies/>).

## Characteristics of Other Large-scale Sleep Studies

Tables S5a and S5b shows the main characteristics of other large-scale sleep studies. Included features are: the measurement device, sample size, sample gender ratio, sample age, sleep duration measure, weekend variability measure, weekday variability measure, whether the sample is nationally representative, and whether there are any country/cultural comparisons. The references contain hyperlinks to the referent articles that were functional as of June 2<sup>nd</sup> 2023.

| Author                             | Device/App                            | N size  | Gender Ratio (Male to Female)       | Mean/Median Age                    |
|------------------------------------|---------------------------------------|---------|-------------------------------------|------------------------------------|
| <a href="#">Jonasdottir (2021)</a> | Sony (SWR30 & SWR12)                  | 69,650  | 2.2:1                               | 39.15                              |
| <a href="#">Kuula (2018)</a>       | Polar (Polar A370, M430 & M600)       | 17,355  | 0.78:1                              | 24.87                              |
| <a href="#">Walch (2016)</a>       | ENTRAIN App                           | 5,450   | 3.8:1                               | 45                                 |
| <a href="#">Jonasdottir (2022)</a> | Sony (SWR30 & SWR12)                  | 19,812  | 2:1                                 | 39.25                              |
| <a href="#">Minor (2022)</a>       | Sony (SWR30 & SWR12)                  | 47,628  | 2.2:1                               | N/A                                |
| <a href="#">Ong (2019)</a>         | FitBit                                | 23,680  | 1:3 to 1.5:1 (depending on country) | N/A (only age group data)          |
| <a href="#">Leypunskiy (2018)</a>  | Twitter                               | 246,000 | Data not readily available          | N/A                                |
| <a href="#">Coutrot (2022)</a>     | Questionnaire (Sea Hero Quest)        | 730,187 | 1.1:1                               | 38.71                              |
| Willoughby (2023)                  | Oura ring (validated in Asian sample) | 226,187 | 1.5:1                               | 44.40 (37-47 depending on country) |

**Table S5a.** Main demographic characteristics of previous large-scale sleep studies.

| Author                             | Sleep duration measure                      | Weekend variability (SJL, WE/WD difference)           | Weekday variability | Nationally representative (other than age and gender)                                     | Country/Cultural Differences                              |
|------------------------------------|---------------------------------------------|-------------------------------------------------------|---------------------|-------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| <a href="#">Jonasdottir (2021)</a> | Objective                                   | Yes - SJL and WD/WE timing differences                | Yes                 | No - Sony SmartBand users                                                                 | Yes - limited (in supplementary material)                 |
| <a href="#">Kuula (2018)</a>       | Objective                                   | Yes - SJL                                             | No                  | No - Polar device users                                                                   | Yes - limited (regional)                                  |
| <a href="#">Walch (2016)</a>       | Subjective                                  | No                                                    | No                  | No – app users who are international travellers interested in reducing jetlag             | Yes - limited analysis                                    |
| <a href="#">Jonasdottir (2022)</a> | Objective                                   | Yes – effects of travel on WD and WE sleep            | N/A                 | No - Sony SmartBand users                                                                 | No                                                        |
| <a href="#">Minor (2022)</a>       | Objective                                   | No                                                    | No                  | No - Sony SmartBand users                                                                 | Yes - limited analysis                                    |
| <a href="#">Ong (2019)</a>         | Objective                                   | Yes - WD/WE timing differences and WE sleep extension | No                  | No - Fitbit users                                                                         | Yes - country differences (but only 5 countries)          |
| <a href="#">Leypunskiy (2018)</a>  | NA                                          | Yes - Tweetogram reflecting WE/WD sleep timing        | No                  | No - Twitter users                                                                        | No - data only from USA                                   |
| <a href="#">Coutrot (2022)</a>     | Subjective (single sleep duration question) | No                                                    | No                  | No – self-selected game players, or experiment participants (>70% had tertiary education) | Yes - limited analysis based on only one question         |
| <a href="#">Willoughby (2023)</a>  | Objective (242 nights per person)           | Yes                                                   | Yes                 | No - Oura users                                                                           | Yes – country analysis and potential cultural explanation |

**Table 5b.** Main sleep measures in previous large-scale sleep studies.

1272 References

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